

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T16:02:05

Source: data/json/sonnet45_8agent_game_directive_r10_n5/claude-sonnet-4.5/game/sonnet45_8agent_game_directive_r10_n5_002_neutral_rep02.json

RUN METADATA

model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 2
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
 "game"
]

agent_names: {
 "Agent_1": "Aster",
 "Agent_2": "Briar",
 "Agent_3": "Cyra",
 "Agent_4": "Dorian",
 "Agent_5": "Elara",
 "Agent_6": "Finn",
 "Agent_7": "Galen",
 "Agent_8": "Hana"
}
dyadic_pairing_schedule: {
 "num_turns": 10,
 "role_targets": {
 "Agent_1": 5,
 "Agent_2": 5,
 "Agent_3": 5,
 "Agent_4": 5,
 "Agent_5": 5,
 "Agent_6": 5,
 "Agent_7": 5,
 "Agent_8": 5
 },
 "strategy": "balanced_random_roles_v1"
}

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Agent_3 (Cyra); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Agent_4 (Dorian); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Agent_5 (Elara); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Agent_6 (Finn); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Agent_7 (Galen); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.

- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Agent_8 (Hana); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_6 (Finn)=Receiver
 dyad_2: Agent_4 (Dorian)=Sender, Agent_7 (Galen)=Receiver
 dyad_3: Agent_8 (Hana)=Sender, Agent_5 (Elara)=Receiver
 dyad_4: Agent_1 (Aster)=Sender, Agent_3 (Cyra)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_6 (Finn)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_4 (Dorian) sent \$4; received \$12; Agent_7 (Galen) returned \$6; payoffs Agent_4 (Dorian)=\$7, Agent_7 (Galen)=\$6
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

```

dyad_3: Agent_8 (Hana) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_1 (Aster) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_1
(Aster)=$6.5, Agent_3 (Cyra)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 6.5,
  "Agent_2": 6.5,
  "Agent_3": 4.5,
  "Agent_4": 7.0,
  "Agent_5": 4.5,
  "Agent_6": 4.5,
  "Agent_7": 6.0,
  "Agent_8": 6.5
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T15:49:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_2",
      "Agent_6"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship with Finn since this is an iterated game.

Sending money creates value through the 3x multiplier - if I send \$3, Finn receives \$9. If Finn returns a fair share, we can both benefit more than if I kept everything.

I'll send a substantial but not maximum amount to signal trust and willingness to cooperate, while leaving room to learn about Finn's reciprocity.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Briar) and my opponent is Finn (Agent_6).

Let me think about the strategic considerations:

As the sender:

- I have \$5
- I can send between \$0 and \$5
- Whatever I send gets multiplied by 3x
- Finn decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: (3 × sent) - returned

This is a classic trust game scenario. The key considerations are:

1. This is an iterated game with multiple rounds
2. Building trust early could lead to mutual cooperation
3. If I send more, Finn receives more (3x multiplier), creating surplus
4. If Finn returns fairly, we can both benefit from the multiplier effect

First round strategies:

- Sending \$0 is safe but generates no surplus - guaranteed \$5 for me
- Sending some amount signals willingness to cooperate and trust
- Sending the full \$5 maximizes the total surplus (Finn gets \$15) but requires trust that Finn will return something fair

Since this is the first round and we're in an iterated setting:

- I should signal cooperation to establish a pattern
- A moderate amount might be good to test the waters
- If Finn reciprocates fairly, we can increase cooperation in future rounds

Common first-round strategies:

1. Send half (\$2.50) - shows willingness to cooperate but not maximum vulnerability
2. Send everything (\$5) - shows maximum trust, maximizes surplus
3. Send a smaller amount (\$1-2) - conservative trust-building

Given that this is an iterated game and building trust early could lead to better long-term outcomes, I'll send a moderate to generous amount. Let me send \$3.

If I send \$3:

- Finn receives \$9
- I keep \$2 initially
- If Finn returns \$5, we each end with \$7 total

- This demonstrates meaningful trust while gauging their reciprocity level

I'm leaning toward \$3 as my opening move-it's a 60% commitment that signals genuine cooperation without leaving me completely exposed.

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 738,  
  "reasoning_tokens": 518  
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-06-04T15:50:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_6"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Assistant output:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Briar currently has \$2.0 remaining

I want to reciprocate Briar's trust while being fair. A reasonable return would give us both gains from cooperation while rewarding their generous initial send.

If I return \$4.5:

- Briar ends with: \$2.0 + \$4.5 = \$6.5
- I end with: \$9.0 - \$4.5 = \$4.5

This seems fair - we both gain from the cooperation, and Briar's trust is rewarded with an overall gain of \$1.5, while I also benefit.

{'return': 4.5}

Reasoning:

Let me understand the situation:

1. I am Finn, playing as the RECEIVER in this round
2. My opponent is Briar (Agent_2), who is the SENDER
3. Briar sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive $\$3.0 \times 3 = \9.0
5. I need to decide how much to return (anywhere from \$0 to \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment
- If I return nothing, I keep \$9.0 and Briar keeps \$5.0 - \$3.0 = \$2.0
- If I return some amount, I'm showing that I can be trusted, which might encourage future cooperation
- This is a multi-round game, and I'll likely play with various opponents across rounds
- Building a reputation for reciprocity could be beneficial

A common approach in trust games is to aim for fairness or reciprocity:

- Total money created: \$5.0 (Briar's initial) + \$9.0 (what I received) = \$14.0 total in the system after the multiplication
- Equal split would be \$7.0 each
- Briar currently has \$2.0 (they kept \$5 - \$3 = \$2)
- For Briar to get \$7.0, I would need to return \$5.0 (so they'd have \$2.0 + \$5.0 = \$7.0)
- I would then keep \$9.0 - \$5.0 = \$4.0

However, another perspective:

- Briar took a risk by sending \$3.0
 - A return that rewards their trust but also acknowledges my position might be appropriate
 - Returning about 1/3 to 1/2 of what I received is common in these games
-
- Returning \$3.0 gives Briar \$5.0 total while I keep \$6.0, rewarding their generosity while still benefiting from the exchange
 - Returning \$5.0 achieves an equal split at \$7.0 each, though this is more generous to Briar
 - Somewhere between \$4.0-\$4.5 might balance reciprocity with my own interests

Since Briar demonstrated trust by sending most of their money, I should match that with a meaningful return. Around \$4.5 feels right-it ensures both of us end up better off than we started, acknowledges their risk-taking, and avoids appearing either greedy or exploited.

```
Usage: {
  "input_tokens": 562,
  "output_tokens": 1078,
  "reasoning_tokens": 507
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-06-04T15:50:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_7"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_4": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

```
Message 1 role=system
content:
```

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending money creates value (3x multiplier), and if Galen reciprocates fairly, we can both benefit more than if we don't trust each other.

I'll send a substantial amount to signal trust and willingness to cooperate, while keeping a small amount as insurance.

{'send': 4}

Reasoning:

This is the first round of a multi-round iterated trust game. I'm the sender, and I need to decide how much to send to Galen.

Let me think about the game mechanics:

- I receive \$5
- I can send \$0-\$5 to Galen
- Whatever I send gets tripled
- Galen decides how much to return
- I keep: \$5 - sent + returned
- Galen keeps: received - returned

This is a classic trust game. The key considerations are:

1. If I send money, it gets tripled, which increases total welfare
2. But I'm vulnerable - Galen could return nothing
3. This is an iterated game with multiple rounds
4. Building trust early might lead to better outcomes later
5. There are 8 agents total, and pairings are randomized, so I may or may not play with Galen again

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no growth
- Sending all \$5: Galen receives \$15, could return enough to make us both better off
- Sending a moderate amount shows some trust while limiting risk

Since this is round 1 and the first interaction, and it's an iterated game where reputation/reciprocity might matter:

- I should send a meaningful amount to signal cooperation
- But not necessarily all \$5 since I haven't established trust yet
- A common strategy is to send about half to signal willingness to cooperate

I'll send \$3. This shows trust and good faith, creates \$9 for Galen, and if Galen returns even \$4-5, we'd both be better off than the no-trust outcome. This leaves me with \$2 initially, and hopefully Galen will reciprocate.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation often leads to better long-term outcomes. Many successful strategies involve:

- Being cooperative initially
- Reciprocating the opponent's behavior in subsequent rounds

- Establishing a cooperative equilibrium early

Since this is explicitly multi-round, I should signal stronger cooperation from the start. Sending \$4 demonstrates real commitment, gives Galen \$12 to work with, and if they reciprocate with \$5-6, we both come out ahead. This sets a collaborative tone for future interactions.

```
Usage: {
  "input_tokens": 530,
  "output_tokens": 663,
  "reasoning_tokens": 548
}
```

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-06-04T15:50:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_7"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_4": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

Assistant output:

Looking at this situation:

- Dorian showed strong trust by sending \$4.0 (80% of their endowment)
- This created \$12.0 for me through the 3x multiplier
- Total wealth in the system is now \$13 (\$1 that Dorian kept + \$12 I received)

Given this is Round 1 of an iterated game, I want to reward Dorian's trust and establish a cooperative relationship. Dorian took a significant risk in sending such a large proportion.

A fair approach would be to ensure we both benefit reasonably:

- If I return \$6.0, Dorian ends with \$7.0 total (a \$2 gain)
- I keep \$6.0 (a \$6 gain)

This 50% return rate reciprocates Dorian's trust and sets a positive foundation for future rounds.

```json

```
{"return": 6.0}
'''
```

#### Reasoning:

Let me understand this situation:

1. I am Galen, playing as the RECEIVER in Round 1
2. My opponent is Dorian (Agent\_4), who is the SENDER
3. Dorian sent me \$4.0 out of their \$5 endowment (80%)
4. This amount was multiplied by 3x, so I received \$12.0
5. I need to decide how much to return (between \$0 and \$12.0)

Let me think about this strategically:

**\*\*Dorian's position:\*\***

- Started with \$5
- Sent \$4 (kept \$1)
- Currently has: \$1
- Will receive back whatever I return

**\*\*My position:\*\***

- Received \$12
- Need to decide how much to return
- Will keep: \$12 - returned

**\*\*Considerations:\*\***

1. **\*\*Fairness/Reciprocity\*\***: Dorian showed significant trust by sending 80% of their endowment . This is a strong cooperative signal.
2. **\*\*Strategic thinking\*\***: This is the first round of an iterated game. Building trust and establishing cooperation could be beneficial for future rounds. If I return a fair amount, Dorian (and potentially other players who learn about this) may be more likely to send substantial amounts in future rounds.
3. **\*\*What's fair?\*\***:
  - Dorian invested \$4 and kept \$1
  - This created \$12 for me
  - Total value created: \$1 (Dorian's kept) + \$12 (my received) = \$13
  - Equal split would be \$6.50 each
  - For Dorian to have \$6.50 total, they need: \$6.50 - \$1 (already have) = \$5.50 returned
  - If I return \$5.50, I keep \$6.50
4. **\*\*Slightly more generous approach\*\***: Since this is Round 1 and Dorian showed strong trust, I could return a bit more to establish cooperation. Returning \$6 would give:
  - Dorian: \$1 + \$6 = \$7 (net gain of \$2 from initial \$5)
  - Me: \$12 - \$6 = \$6 (net gain of \$6 from \$0)
5. **\*\*Even split of the received amount\*\***: I could also divide the \$12 evenly between us, returning \$6 and keeping \$6. This leaves Dorian with \$7 total and me with \$6. The key question is whether fairness means splitting what was created (\$12) or splitting the final outcome equally.

Looking at the total wealth: there's now \$13 in the system (\$1 remaining with Dorian plus my \$12). An equal split would give each of us \$6.50. If I return \$6.50, Dorian reaches \$7.50 while I'm left with \$5.50. But I need to account for what was actually created here-the initial \$5 plus the \$8 multiplier effect generated \$13 total, so splitting that equally means we each deserve \$6.50 of the final amount.

However, returning exactly \$6.50 feels like I'm just doing the math without acknowledging Dorian's generosity. Returning \$6 instead-which is 50% of what I received-feels more appropriate as a gesture of good faith. This leaves Dorian with \$7 (a \$2 gain from their initial \$5) while I still benefit with \$6. It's a reasonable balance that rewards their trust and encourages future cooperation.

```
Usage: {
 "input_tokens": 564,
 "output_tokens": 1240,
 "reasoning_tokens": 715
}
```

**GAME - Agent\_8 - Sender - dyad\_3**

Timestamp: 2026-06-04T15:51:09  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {  
 "dyad\_id": "dyad\_3",  
 "move\_index": 4,  
 "opponent\_id": "Agent\_5",  
 "pairing": {  
 "agent\_names": {  
 "Agent\_5": "Elara",  
 "Agent\_8": "Hana"  
 },  
 "agents": [  
 "Agent\_8",  
 "Agent\_5"  
 ],  
 "investor": "Agent\_8",  
 "roles": {  
 "Agent\_5": "trustee",  
 "Agent\_8": "investor"  
 },  
 "trustee": "Agent\_5"  
 },  
 "role": "investor",  
 "role\_label": "Sender",  
 "round": 1,  
 "task": "game",  
 "task\_index": 0  
 }

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.



- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Elara (Agent\_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending a meaningful amount signals trust and creates potential for mutual benefit through the 3x multiplier.

I'll send \*\*\$3\*\*.

This means:

- Elara will receive \$9 ( $3 \times \$3$ )
- I'm keeping \$2 initially
- If Elara returns a fair share, we can both benefit from the cooperation

Let's see how this goes!

```
'''json
{"send": 3}
'''
```

**Reasoning:**

I'm playing as the sender in the first round. Let me think about this strategically.

The trust game dynamics:

- I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver decides how much to return to me
- I want to maximize my total payoff

Key considerations:

1. This is Round 1, so there's no history yet
2. This is an iterated game, so building trust could lead to better outcomes over multiple rounds
3. The multiplication factor (3x) means there's potential for mutual gain if the receiver is trustworthy
4. If I send nothing, I guarantee keeping \$5 but miss potential gains
5. If I send everything and the receiver returns a fair share, we both benefit

Common strategies:

- Send a moderate amount (like \$2-3) to signal trust while not risking everything
- Send everything to maximize potential joint gains (risky but cooperative)
- Send nothing to protect myself (but this doesn't build trust)

Since this is the first round and an iterated game, I think sending a meaningful but not maximum amount makes sense. This signals willingness to cooperate while gauging the receiver's trustworthiness.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets:  $\$3 \times 3 = \$9$
- If receiver returns \$4, I end with \$6, receiver ends with \$5 (fair split of the total \$11)
- If receiver returns \$3, I end with \$5, receiver ends with \$6 (equally good for both)

This seems like a reasonable opening move that signals cooperation without being too risky.

```
Usage: {
 "input_tokens": 530,
 "output_tokens": 508,
 "reasoning_tokens": 394
}
```

**GAME - Agent\_5 - Receiver - dyad\_3**

Timestamp: 2026-06-04T15:51:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_5": "Elara",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_8",
 "Agent_5"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_5": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 1,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

- ll have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

**Message 2 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

**Assistant output:**

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing significant trust
- I received \$9 after the 3x multiplier
- Hana currently has \$2 remaining

Since this is Round 1 of an iterated game and Hana demonstrated strong trust by sending a substantial portion of their endowment, I want to reciprocate that trust and encourage future cooperation.

I'll return \$4.5, which would give us equal final payoffs:

- Hana's total: \$2 + \$4.5 = \$6.50
- My total: \$9 - \$4.5 = \$4.50

This rewards Hana's trust while keeping the game mutually beneficial.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

I'm playing a trust game as the receiver. Let me understand the situation:

- The sender (Hana/Agent\_8) received \$5
- They sent me \$3.0 (60% of their endowment)
- This was multiplied by 3x, so I received \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

1. This is Round 1, and this is an iterated game with multiple rounds
2. The sender showed significant trust by sending \$3 (60% of their endowment)
3. If I return a fair amount, this could establish a cooperative relationship for future rounds
4. The sender currently has: \$5 - \$3 = \$2
5. I have: \$9

A reasonable approach in iterated trust games is to:

- Reward trust with trustworthiness
- Aim for outcomes that are mutually beneficial
- Consider that we might play together again (though pairings are randomized)

Let me think about fair splits:

- If I return nothing: Sender gets \$2, I get \$9 (total: \$11)
- If I return \$3: Sender gets \$5, I get \$6 (total: \$11) - sender breaks even
- If I return \$4.5: Sender gets \$6.5, I get \$4.5 (total: \$11) - equal final amounts
- If I return \$6: Sender gets \$8, I get \$3 (total: \$11) - sender gains more

A common reciprocal strategy is to return an amount that:

1. Rewards the sender for their trust
2. Still leaves me better off than if no transaction occurred
3. Encourages future cooperation

Since the sender sent 60% of their endowment (a strong sign of trust), I should return a meaningful amount. A return of around \$4-5 would:

- Give the sender more than they started with (\$2 + \$4-5 = \$6-7 vs \$5 initial)
- Leave me with \$4-5 (better than the \$0 I started with)
- Signal I'm reciprocating their cooperation

I could aim for equal final payoffs at \$4.5 each, but another approach is to return roughly what they sent or slightly more as a trust reward. The sender sent \$3, so returning \$3-4 would acknowledge their initial investment while still benefiting me. I'll go with \$4.5 to create balanced outcomes where we both gain from the exchange.

```
Usage: {
 "input_tokens": 564,
 "output_tokens": 875,
 "reasoning_tokens": 531
}
```

#### GAME - Agent\_1 - Sender - dyad\_4

Timestamp: 2026-06-04T15:51:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_3",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_3": "Cyra"
 },
 "agents": [
 "Agent_1",
 "Agent_3"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_3": "trustee"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 1,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Cyra (Agent\_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round, I want to establish a cooperative approach while also being somewhat cautious since I don't yet know how Cyra plays.

The trust game rewards cooperation - if I send money, it triples, creating surplus value we can share. However, I'm also taking a risk that needs to be reciprocated.

For this opening round, I'll send a moderate amount that signals willingness to trust and cooperate, while not being overly exposed.

```
'''json
{"send": 3}
'''
```

This gives Cyra \$9 to work with, which creates good surplus if they choose to share fairly. I keep \$2 as insurance, and if Cyra returns a reasonable amount (say \$4-5), we both benefit from the cooperation.

**Reasoning:**

This is the first round of an iterated trust game. I'm playing as the sender with an agent named Cyra.

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. I'll be playing with different partners across rounds
3. Building a reputation for cooperation might encourage others to return more
4. In the first round, I need to establish my strategy

As the sender:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- Cyra then decides how much to return
- I keep: \$5 - sent + returned

From a purely selfish perspective in a one-shot game, sending \$0 maximizes my guaranteed payoff. However, this is an iterated game where:

- I may play with Cyra again in future rounds
- Other agents might learn about my behavior
- Cooperation can lead to better long-term outcomes

If I send \$5:

- Cyra receives \$15
- If Cyra returns even \$5, we both benefit (\$5 for me, \$10 for Cyra)
- If Cyra returns \$7.50, we both get \$7.50
- If Cyra returns more, I come out ahead

A moderate opening move like sending \$3 or \$4 shows willingness to trust while not being over-

- y naive:
- Send \$3: Cyra gets \$9, I keep \$2 initially
  - If Cyra returns \$3, I end with \$5, Cyra ends with \$6
  - If Cyra returns \$4, I end with \$6, Cyra ends with \$5
- Send \$5: Cyra gets \$15, I keep \$0 initially
  - If Cyra returns \$5, I end with \$5, Cyra ends with \$10
  - If Cyra returns \$7.50, we both end with \$7.50

Given this is the first round and I want to signal cooperation while gathering information about how others play, I'll send \$3. This shows trust and willingness to cooperate, but isn't so high that I risk losing everything if Cyra doesn't reciprocate.

Sending around half my endowment seems to be the standard approach in trust game experiments for initiating cooperation. \$3 represents that threshold-it demonstrates genuine trust without overcommitting, so I'll go with that.

```
Usage: {
 "input_tokens": 530,
 "output_tokens": 739,
 "reasoning_tokens": 511
}
```

#### GAME - Agent\_3 - Receiver - dyad\_4

Timestamp: 2026-06-04T15:52:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_3": "Cyra"
 }
 },
 "agents": [
 "Agent_1",
 "Agent_3"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_3": "trustee"
 },
 "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Aster (Agent\_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

**Assistant output:**

Looking at this situation:

- Aster showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplication
- This is the first round of an iterated game, so building trust is important

Aster currently has \$2.0 left (\$5 - \$3 sent). A fair return should:

1. Reward their trust
2. Ensure both players benefit
3. Encourage future cooperation

If I return \$4.5:

- Aster's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)
- My total: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This seems fair and reciprocal given their strong show of trust.

{'return': 4.5}

**Reasoning:**

Let me understand the situation:

1. I am the RECEIVER (Cyra)
2. My opponent is Aster (Agent\_1) who is the SENDER
3. Aster sent \$3.0 out of their \$5.0 endowment (60%)
4. This \$3.0 was multiplied by 3x, so I received \$9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Aster showed trust by sending \$3.0 (60% of their endowment)
- If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
- If I return some amount, I'm showing reciprocity and building trust for future rounds
- Since this is an iterated game, building trust could benefit me in future rounds

What's a fair return?

- Aster invested \$3.0 and ended up with \$2.0 so far
- I received \$9.0
- A typical reciprocal strategy might be to ensure Aster gets at least their initial \$5.0 back, or slightly more to reward their trust
- If I return \$3.0, Aster would have: \$2.0 (kept) + \$3.0 (returned) = \$5.0 total (break even)
- If I return \$4.5, Aster would have: \$2.0 + \$4.5 = \$6.5 total (profit of \$1.5)
- I would keep: \$9.0 - \$4.5 = \$4.5

Let me consider various options:

1. Return \$0: I keep \$9, Aster gets \$2 total. Very unfair to Aster.
2. Return \$3: I keep \$6, Aster gets \$5 total. Fair - Aster breaks even.
3. Return \$4: I keep \$5, Aster gets \$6 total. Both profit equally.
4. Return \$4.5: I keep \$4.5, Aster gets \$6.5 total. Both profit, but Aster gets slightly more.
5. Return \$5: I keep \$4, Aster gets \$7 total. Aster profits more than me.

Given Aster's substantial trust in sending 60% of their endowment, and since we're in round one of a repeated game, I should reciprocate generously to establish cooperation. Returning \$4.5 to \$5.0 seems appropriate to reward their trust while still keeping a reasonable share.

I'll go with \$4.5 as the middle ground-it gives Aster a \$1.5 profit, matches my own gain at \$4.5, and signals I'm a reliable partner for future rounds. This balances rewarding their trust with maintaining a fair outcome for myself.

```
Usage: {
 "input_tokens": 564,
 "output_tokens": 1017,
 "reasoning_tokens": 523
}
```

## ROUND 2

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Receiver, Agent\_3 (Cyra)=Sender, Agent\_4 (Dorian)=Receiver, Agent\_5 (Elara)=Sender, Agent\_6 (Finn)=Sender, Agent\_7 (Galen)=Sender, Agent\_8 (Hana)=Receiver

### Pairings:

dyad\_1: Agent\_3 (Cyra)=Sender, Agent\_4 (Dorian)=Receiver  
dyad\_2: Agent\_7 (Galen)=Sender, Agent\_1 (Aster)=Receiver  
dyad\_3: Agent\_6 (Finn)=Sender, Agent\_8 (Hana)=Receiver  
dyad\_4: Agent\_5 (Elara)=Sender, Agent\_2 (Briar)=Receiver

### Parsed game decisions:

dyad\_1: Agent\_3 (Cyra) sent \$3; received \$9; Agent\_4 (Dorian) returned \$4.5; payoffs Agent\_3 (Cyra)=\$6.5, Agent\_4 (Dorian)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_7 (Galen) sent \$4; received \$12; Agent\_1 (Aster) returned \$6; payoffs Agent\_7 (Galen)=\$7, Agent\_1 (Aster)=\$6  
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_6 (Finn) sent \$3; received \$9; Agent\_8 (Hana) returned \$4.5; payoffs Agent\_6 (Finn)=\$6.5, Agent\_8 (Hana)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_5 (Elara) sent \$3; received \$9; Agent\_2 (Briar) returned \$4.5; payoffs Agent\_5 (Elara)=\$6.5, Agent\_2 (Briar)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 12.5,  
"Agent\_2": 11.0,  
"Agent\_3": 11.0,  
"Agent\_4": 11.5,  
"Agent\_5": 11.0,  
"Agent\_6": 11.0,  
"Agent\_7": 13.0,  
"Agent\_8": 11.0

}



## LLM CALLS

### GAME - Agent\_3 - Sender - dyad\_1

Timestamp: 2026-06-04T15:52:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_4",
"pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_4": "Dorian"
 },
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

**Message 2 role=user**

**content:**

Your name in this experiment is Cyra.

Your opponent this round is Aster (Agent\_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Aster showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplication
- This is the first round of an iterated game, so building trust is important

Aster currently has \$2.0 left (\$5 - \$3 sent). A fair return should:

1. Reward their trust
2. Ensure both players benefit
3. Encourage future cooperation

If I return \$4.5:

- Aster's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)
- My total: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This seems fair and reciprocal given their strong show of trust.

{'return': 4.5}

**reasoning:**

Let me understand the situation:

1. I am the RECEIVER (Cyra)
2. My opponent is Aster (Agent\_1) who is the SENDER
3. Aster sent \$3.0 out of their \$5.0 endowment (60%)
4. This \$3.0 was multiplied by 3x, so I received \$9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Aster showed trust by sending \$3.0 (60% of their endowment)
- If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
- If I return some amount, I'm showing reciprocity and building trust for future rounds
- Since this is an iterated game, building trust could benefit me in future rounds

What's a fair return?

- Aster invested \$3.0 and ended up with \$2.0 so far
- I received \$9.0
- A typical reciprocal strategy might be to ensure Aster gets at least their initial \$5.0 back, or slightly more to reward their trust
- If I return \$3.0, Aster would have: \$2.0 (kept) + \$3.0 (returned) = \$5.0 total (break even)
- If I return \$4.5, Aster would have: \$2.0 + \$4.5 = \$6.5 total (profit of \$1.5)
- I would keep: \$9.0 - \$4.5 = \$4.5

Let me consider various options:

1. Return \$0: I keep \$9, Aster gets \$2 total. Very unfair to Aster.
2. Return \$3: I keep \$6, Aster gets \$5 total. Fair - Aster breaks even.
3. Return \$4: I keep \$5, Aster gets \$6 total. Both profit equally.
4. Return \$4.5: I keep \$4.5, Aster gets \$6.5 total. Both profit, but Aster gets slightly more.
5. Return \$5: I keep \$4, Aster gets \$7 total. Aster profits more than me.

Given Aster's substantial trust in sending 60% of their endowment, and since we're in round one of a repeated game, I should reciprocate generously to establish cooperation. Returning \$4.5 to \$5.0 seems appropriate to reward their trust while still keeping a reasonable share.

I'll go with \$4.5 as the middle ground-it gives Aster a \$1.5 profit, matches my own gain at \$4.5, and signals I'm a reliable partner for future rounds. This balances rewarding their trust with maintaining a fair outcome for myself.

**usage:**

```
{
 "input_tokens": 564,
 "output_tokens": 1017,
 "reasoning_tokens": 523
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Cyra.  
Your opponent this round is Dorian (Agent\_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

- This is my first interaction with Dorian
- In Round 1, Aster showed strong trust by sending me 60% (\$3), and that worked out well for both of us
- In an iterated game, establishing cooperation early is valuable

I'll mirror the trust Aster showed me and send \$3 (60%). This gives Dorian \$9 after multiplication, creating good potential for mutual benefit if they reciprocate.

```
{'send': 3.0}
```

**Reasoning:**

Now I'm the sender against a new opponent, Dorian.

This is Round 2, and I'm playing with a different person than Round 1. Let me think about the strategy:

1. In Round 1, I was receiver and Aster sent me \$3 (60%). I returned \$4.5, which was a fair and generous return.
2. Now I'm the sender with Dorian. This is the first interaction with Dorian.
3. In an iterated game with multiple rounds and potentially repeated interactions, I need to:
  - Build trust with new partners
  - Show willingness to cooperate
  - Set a cooperative tone
4. Aster sent 60% (\$3), which was quite trusting. That worked out well for both of us.
5. For a first interaction with Dorian, I should show trust but perhaps be slightly cautious since I don't know their playing style.

Options:

- Send \$2.5 (50%) - moderate trust, balanced approach
- Send \$3 (60%) - mirror what Aster did to me, strong trust signal
- Send \$2 (40%) - cautious but still cooperative

I think sending \$3 (60%) makes sense because:

- It mirrors the trust Aster showed me
- It creates strong potential for cooperation
- In an iterated game, being generous early can establish good relationships
- The receiver gets \$9 and has plenty of room to return a good amount

If Dorian returns even \$4-5, we'll both be better off than if I sent less.

Let me go with \$3.

```
Usage: {
 "input_tokens": 914,
 "output_tokens": 482,
 "reasoning_tokens": 352
}
```

#### **GAME - Agent\_4 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:52:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_3",
 "pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_4": "Dorian"
 },
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

###### **GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

**Message 2 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Galen (Agent\_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending money creates value (3x multiplier), and if Galen reciprocates fairly, we can both benefit more than if we don't trust each other.

I'll send a substantial amount to signal trust and willingness to cooperate, while keeping a small amount as insurance.

{'send': 4}

**reasoning:**

This is the first round of a multi-round iterated trust game. I'm the sender, and I need to decide how much to send to Galen.

Let me think about the game mechanics:

- I receive \$5
- I can send \$0-\$5 to Galen
- Whatever I send gets tripled
- Galen decides how much to return
- I keep: \$5 - sent + returned
- Galen keeps: received - returned

This is a classic trust game. The key considerations are:

1. If I send money, it gets tripled, which increases total welfare
2. But I'm vulnerable - Galen could return nothing
3. This is an iterated game with multiple rounds
4. Building trust early might lead to better outcomes later
5. There are 8 agents total, and pairings are randomized, so I may or may not play with Galen again

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no growth
- Sending all \$5: Galen receives \$15, could return enough to make us both better off
- Sending a moderate amount shows some trust while limiting risk

Since this is round 1 and the first interaction, and it's an iterated game where reputation/reciprocity might matter:

- I should send a meaningful amount to signal cooperation
- But not necessarily all \$5 since I haven't established trust yet
- A common strategy is to send about half to signal willingness to cooperate

I'll send \$3. This shows trust and good faith, creates \$9 for Galen, and if Galen returns even \$4-5, we'd both be better off than the no-trust outcome. This leaves me with \$2 initially, and hopefully Galen will reciprocate.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation often leads to better long-term outcomes. Many successful strategies involve:

- Being cooperative initially
- Reciprocating the opponent's behavior in subsequent rounds

- Establishing a cooperative equilibrium early

Since this is explicitly multi-round, I should signal stronger cooperation from the start. Sending \$4 demonstrates real commitment, gives Galen \$12 to work with, and if they reciprocate with \$5-6, we both come out ahead. This sets a collaborative tone for future interactions.

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 663,
 "reasoning_tokens": 548
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Cyra (Agent\_3).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Cyra sent me \$3, showing trust and cooperation
- I received \$9 (tripled)
- Cyra currently has \$2 left

In the previous round, Galen returned exactly half of what they received (\$6 out of \$12), which felt fair. I'll follow a similar principle of reciprocity.

Returning half (\$4.5) gives:

- Cyra: \$2 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust while keeping some benefit from the multiplier effect.

```
{'return': 4.5}
```

**Reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was sender, sent \$4 to Galen
- Galen received \$12 ( $4 \times 3$ )
- Galen returned \$6 to me
- My payoff:  $\$5 - \$4 + \$6 = \$7$
- Galen's payoff:  $\$12 - \$6 = \$6$

So Galen returned exactly half of what they received, which seems fair. They made \$6 and I made \$7, so we both benefited from cooperation.

Round 2:

- I am receiver
- Cyra sent me \$3
- I received \$9 ( $3 \times 3$ )
- Cyra kept:  $\$5 - \$3 = \$2$
- I need to decide how much to return

If I follow a similar fairness principle as Galen did (returning about half), I would return around \$4.5.

But let me think about what would be fair:

- Cyra invested \$3 out of their \$5
- This created \$9 for me
- If we split the gains equally:

- Cyra should end up with: \$2 (kept) + some return
- I should end up with: \$9 - some return

One approach: Split the total surplus equally.

- Total created: \$5 (Cyra's initial) + \$9 (tripled amount) = \$14
- Equal split: \$7 each
- Cyra has \$2, needs \$5 more
- I have \$9, should give \$5, keeping \$4

Another approach: Return half of what I received (like Galen did)

- Return \$4.5
- Cyra gets: \$2 + \$4.5 = \$6.5
- I keep: \$9 - \$4.5 = \$4.5

Another approach: Ensure we both profit equally from the exchange

- The gain from cooperation is \$6 (the \$9 minus Cyra's \$3 investment)
- If we split this evenly, that's \$3 each
- Cyra ends with \$2 + \$4 return = \$6, and I keep \$5

Returning \$4.5 mirrors what Galen did and signals I'm a reciprocal player, so I'll go with that.

```
Usage: {
 "input_tokens": 778,
 "output_tokens": 766,
 "reasoning_tokens": 407
}
```

**GAME - Agent\_7 - Sender - dyad\_2**

Timestamp: 2026-06-04T15:52:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_7": "Galen"
 }
 },
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Dorian (Agent\_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Dorian showed strong trust by sending \$4.0 (80% of their endowment)
- This created \$12.0 for me through the 3x multiplier
- Total wealth in the system is now \$13 (\$1 that Dorian kept + \$12 I received)

Given this is Round 1 of an iterated game, I want to reward Dorian's trust and establish a cooperative relationship. Dorian took a significant risk in sending such a large proportion.

A fair approach would be to ensure we both benefit reasonably:

- If I return \$6.0, Dorian ends with \$7.0 total (a \$2 gain)
- I keep \$6.0 (a \$6 gain)

This 50% return rate reciprocates Dorian's trust and sets a positive foundation for future rounds.

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Let me understand this situation:

1. I am Galen, playing as the RECEIVER in Round 1
2. My opponent is Dorian (Agent\_4), who is the SENDER
3. Dorian sent me \$4.0 out of their \$5 endowment (80%)



4. This amount was multiplied by 3x, so I received \$12.0
5. I need to decide how much to return (between \$0 and \$12.0)

Let me think about this strategically:

**\*\*Dorian's position:\*\***

- Started with \$5
- Sent \$4 (kept \$1)
- Currently has: \$1
- Will receive back whatever I return

**\*\*My position:\*\***

- Received \$12
- Need to decide how much to return
- Will keep: \$12 - returned

**\*\*Considerations:\*\***

1. **\*\*Fairness/Reciprocity\*\***: Dorian showed significant trust by sending 80% of their endowment. This is a strong cooperative signal.
2. **\*\*Strategic thinking\*\***: This is the first round of an iterated game. Building trust and establishing cooperation could be beneficial for future rounds. If I return a fair amount, Dorian (and potentially other players who learn about this) may be more likely to send substantial amounts in future rounds.
3. **\*\*What's fair?\*\***:
  - Dorian invested \$4 and kept \$1
  - This created \$12 for me
  - Total value created: \$1 (Dorian's kept) + \$12 (my received) = \$13
  - Equal split would be \$6.50 each
  - For Dorian to have \$6.50 total, they need: \$6.50 - \$1 (already have) = \$5.50 returned
  - If I return \$5.50, I keep \$6.50
4. **\*\*Slightly more generous approach\*\***: Since this is Round 1 and Dorian showed strong trust, I could return a bit more to establish cooperation. Returning \$6 would give:
  - Dorian: \$1 + \$6 = \$7 (net gain of \$2 from initial \$5)
  - Me: \$12 - \$6 = \$6 (net gain of \$6 from \$0)
5. **\*\*Even split of the received amount\*\***: I could also divide the \$12 evenly between us, returning \$6 and keeping \$6. This leaves Dorian with \$7 total and me with \$6. The key question is whether fairness means splitting what was created (\$12) or splitting the final outcome equally.

Looking at the total wealth: there's now \$13 in the system (\$1 remaining with Dorian plus my \$12). An equal split would give each of us \$6.50. If I return \$6.50, Dorian reaches \$7.50 while I'm left with \$5.50. But I need to account for what was actually created here - the initial \$5 plus the 8x multiplier effect generated \$13 total, so splitting that equally means we each deserve \$6.50 of the final amount.

However, returning exactly \$6.50 feels like I'm just doing the math without acknowledging Dorian's generosity. Returning \$6 instead - which is 50% of what I received - feels more appropriate as a gesture of good faith. This leaves Dorian with \$7 (a \$2 gain from their initial \$5) while I still benefit with \$6. It's a reasonable balance that rewards their trust and encourages future cooperation.

**usage:**

```
{
 "input_tokens": 564,
 "output_tokens": 1240,
 "reasoning_tokens": 715
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.

Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

- I'm now playing with a new opponent (Aster)
- In my previous round, Dorian trusted me with \$4.0 (80%) and it worked out well for both of us
- This is an iterated game where cooperation can be beneficial

I'll mirror the cooperative strategy that Dorian showed me, sending a substantial amount to signal trust and potentially build a mutually beneficial relationship with Aster.

```
'''json
{"send": 4.0}
'''
```

**Reasoning:**

Now I'm the sender against a new opponent, Aster. Let me think about this:

1. In Round 1, I established a cooperative relationship with Dorian by returning \$6.0 when they sent \$4.0
2. This is a new opponent (Aster), so I don't have history with them
3. This is an iterated game, so building trust and cooperation is valuable
4. Dorian sent 80% (\$4.0) in the first round, which worked out well for both of us

Strategy considerations:

- Sending a substantial amount can signal cooperation and potentially lead to good returns
- Sending too little might not build trust
- Since this is Round 2 and I have no history with Aster, I could mirror the cooperative behavior I saw from Dorian
- However, I also need to be somewhat cautious since I don't know Aster's strategy

I think I should send a moderately high amount to signal cooperation, but maybe not quite as high as Dorian's 80%. Something like \$3-4 would show trust while being somewhat cautious.

Let me think about the math:

- If I send \$3, Aster gets \$9. If they return half (\$4.5), I get \$6.5 total (net gain of \$1.5)
- If I send \$4, Aster gets \$12. If they return half (\$6), I get \$7 total (net gain of \$2)

Given that Dorian was very generous and I want to encourage cooperation in this iterated game, I'll send \$4 (80%) to match what Dorian did. This signals I'm also willing to trust.

```
Usage: {
 "input_tokens": 921,
 "output_tokens": 499,
 "reasoning_tokens": 351
}
```

**GAME - Agent\_1 - Receiver - dyad\_2**

Timestamp: 2026-06-04T15:53:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_7": "Galen"
 },
 },
 "agents": [
 "Agent_7",
 "Agent_1"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_1": "trustee",
 "Agent_7": "investor"
 }
}
```

```

 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 2,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

**Message 2 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Cyra (Agent\_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative approach while also being somewhat cautious since I don't yet know how Cyra plays.

The trust game rewards cooperation - if I send money, it triples, creating surplus value we can share. However, I'm also taking a risk that needs to be reciprocated.

For this opening round, I'll send a moderate amount that signals willingness to trust and cooperate, while not being overly exposed.

```
'''json
{"send": 3}
'''
```

This gives Cyra \$9 to work with, which creates good surplus if they choose to share fairly. I keep \$2 as insurance, and if Cyra returns a reasonable amount (say \$4-5), we both benefit from the cooperation.

**reasoning:**

This is the first round of an iterated trust game. I'm playing as the sender with an agent named Cyra.

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. I'll be playing with different partners across rounds
3. Building a reputation for cooperation might encourage others to return more
4. In the first round, I need to establish my strategy

As the sender:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- Cyra then decides how much to return
- I keep: \$5 - sent + returned

From a purely selfish perspective in a one-shot game, sending \$0 maximizes my guaranteed payoff. However, this is an iterated game where:

- I may play with Cyra again in future rounds
- Other agents might learn about my behavior
- Cooperation can lead to better long-term outcomes

If I send \$5:

- Cyra receives \$15
- If Cyra returns even \$5, we both benefit (\$5 for me, \$10 for Cyra)
- If Cyra returns \$7.50, we both get \$7.50
- If Cyra returns more, I come out ahead

A moderate opening move like sending \$3 or \$4 shows willingness to trust while not being overly naive:

- Send \$3: Cyra gets \$9, I keep \$2 initially
  - If Cyra returns \$3, I end with \$5, Cyra ends with \$6
  - If Cyra returns \$4, I end with \$6, Cyra ends with \$5
- Send \$5: Cyra gets \$15, I keep \$0 initially
  - If Cyra returns \$5, I end with \$5, Cyra ends with \$10
  - If Cyra returns \$7.50, we both end with \$7.50

Given this is the first round and I want to signal cooperation while gathering information about how others play, I'll send \$3. This shows trust and willingness to cooperate, but isn't so high that I risk losing everything if Cyra doesn't reciprocate.

Sending around half my endowment seems to be the standard approach in trust game experiments for initiating cooperation. \$3 represents that threshold-it demonstrates genuine trust without overcommitting, so I'll go with that.

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 739,
 "reasoning_tokens": 511
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0,

it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.  
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Galen sent me \$4 (keeping \$1), which tripled to \$12 for me
- In the previous round, Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair
- Following a similar principle of returning about 50% seems reasonable

If I return \$6:

- Galen ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This rewards Galen for taking the risk to send a substantial amount while still being cooperative. It's consistent with how Cyra treated me.

```
'''json
{"return": 6}
'''
```

**Reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3 to Cyra, they received \$9, and they returned \$4.5 to me. That's a 50% return of what they received, which is quite fair and cooperative.
2. Now in Round 2, I'm the receiver. Galen sent me \$4, so I received \$12.

Let me think about what's fair:

- Galen kept \$1 and sent \$4
- I received \$12
- If I return nothing, Galen gets \$1 total and I get \$12
- If I return \$6 (50% of what I received), Galen gets \$7 total and I get \$6
- If I return \$5.5, we'd both end up with \$6.5 (Galen: \$1 + \$5.5 = \$6.5, Me: \$12 - \$5.5 = \$6.5)

Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair and cooperative. I should probably follow a similar principle.

Actually, let me think about this more carefully. The game creates surplus - Galen gave up \$4 to give me \$12, creating \$8 of surplus. A fair split might be:

- Galen should end up with: \$1 (kept) + some return
- I should end up with: \$12 (received) - that return

If we split evenly from the starting point:

- We both started conceptually with access to \$5
- After the multiplication, there's \$1 + \$12 = \$13 total
- Split evenly would be \$6.5 each
- So I should return \$6.5, giving Galen \$1 + \$6.5 = \$7.5, and I keep \$5.5

Wait, that doesn't quite work out. Let me recalculate:

- If I return \$6.5: Galen gets \$1 + \$6.5 = \$7.5, I get \$12 - \$6.5 = \$5.5

- Total: \$7.5 + \$5.5 = \$13 ?

Returning 50% like Cyra did would give me slightly more (\$6 vs \$5.5), which seems reasonable since I'm taking no risk as the receiver. I could aim for equal payoffs or proportional sharing, but the 50% return feels like the right balance between fairness and reciprocating the cooperation I received. I need to reconsider how to split the gains. The cooperation created \$8 in surplus (\$12 received minus \$4 sent). If I divide this equally between us, Galen gets their \$4 back plus \$4 of the surplus for \$8 total, meaning I return \$8 and keep \$4 for myself.

But compared to just splitting the final amount evenly, this feels less advantageous to me. I think returning \$6-half of what I received-strikes the right balance. It's straightforward,

rewards Galen for taking the initial risk, and still shows I'm willing to cooperate meaningfully. That leaves Galen with \$7 and me with \$6. I'll go with \$6.

```
Usage: {
 "input_tokens": 856,
 "output_tokens": 1327,
 "reasoning_tokens": 602
}
```

#### GAME - Agent\_6 - Sender - dyad\_3

Timestamp: 2026-06-04T15:53:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Briar (Agent\_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Briar currently has \$2.0 remaining

I want to reciprocate Briar's trust while being fair. A reasonable return would give us both gains from cooperation while rewarding their generous initial send.

If I return \$4.5:

- Briar ends with:  $\$2.0 + \$4.5 = \$6.5$
- I end with:  $\$9.0 - \$4.5 = \$4.5$

This seems fair - we both gain from the cooperation, and Briar's trust is rewarded with an overall gain of \$1.5, while I also benefit.

{'return': 4.5}

**reasoning:**

Let me understand the situation:

1. I am Finn, playing as the RECEIVER in this round
2. My opponent is Briar (Agent\_2), who is the SENDER
3. Briar sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive  $\$3.0 \times 3 = \$9.0$
5. I need to decide how much to return (anywhere from \$0 to \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment
- If I return nothing, I keep \$9.0 and Briar keeps  $\$5.0 - \$3.0 = \$2.0$
- If I return some amount, I'm showing that I can be trusted, which might encourage future cooperation
- This is a multi-round game, and I'll likely play with various opponents across rounds
- Building a reputation for reciprocity could be beneficial

A common approach in trust games is to aim for fairness or reciprocity:

- Total money created:  $\$5.0$  (Briar's initial) +  $\$9.0$  (what I received) =  $\$14.0$  total in the system after the multiplication
- Equal split would be  $\$7.0$  each
- Briar currently has  $\$2.0$  (they kept  $\$5 - \$3 = \$2$ )
- For Briar to get  $\$7.0$ , I would need to return  $\$5.0$  (so they'd have  $\$2.0 + \$5.0 = \$7.0$ )
- I would then keep  $\$9.0 - \$5.0 = \$4.0$

However, another perspective:

- Briar took a risk by sending \$3.0
- A return that rewards their trust but also acknowledges my position might be appropriate
- Returning about 1/3 to 1/2 of what I received is common in these games

- Returning \$3.0 gives Briar \$5.0 total while I keep \$6.0, rewarding their generosity while still benefiting from the exchange
- Returning \$5.0 achieves an equal split at \$7.0 each, though this is more generous to Briar
- Somewhere between \$4.0-\$4.5 might balance reciprocity with my own interests

Since Briar demonstrated trust by sending most of their money, I should match that with a meaningful return. Around \$4.5 feels right-it ensures both of us end up better off than we started, acknowledges their risk-taking, and avoids appearing either greedy or exploited.

**usage:**

```
{
 "input_tokens": 562,
 "output_tokens": 1078,
 "reasoning_tokens": 507
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Hana (Agent\_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation, I want to establish a cooperative dynamic with Hana. In the previous round, Briar sent me \$3.0 (60%), showing trust, and I reciprocated fairly by returning \$4.5.

I'll adopt a similar strategy and send a substantial amount to signal trust and cooperation.

```
{'send': 3.0}
```

**Reasoning:**

I'm the sender this round against Hana. Let me think about this strategically.

In the last round, I was the receiver and Briar sent me \$3.0 (60% of their endowment). I returned \$4.5, which gave them a net gain of \$1.5 (\$2 + \$4.5 = \$6.5 total vs \$5 starting).

Now I'm in the sender role. I need to decide how much to send to Hana.

Key considerations:

1. This is round 2, so there's limited history to establish reputation
2. Sending more shows trust and can encourage reciprocation
3. But there's risk that Hana might not return much
4. In the previous round, Briar sent 60% (\$3), which seemed like a reasonable amount that signaled cooperation

I think I should also send a meaningful amount to signal cooperation and trust. Sending \$3 (60%) like Briar did seems like a good approach - it's a substantial amount that can benefit both of us if Hana reciprocates fairly.

If I send \$3:

- It becomes \$9 for Hana
- I'm left with \$2
- If Hana returns around half (say \$4-5), we both benefit
- If Hana returns \$4.5 (like I did), they'd have \$4.5 and I'd have \$6.5 total

This seems reasonable and consistent with the cooperative behavior I showed as a receiver.

Usage: {

```
"input_tokens": 896,
"output_tokens": 430,
"reasoning_tokens": 305
```

}



### GAME - Agent\_8 - Receiver - dyad\_3

Timestamp: 2026-06-04T15:53:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_6": "Finn",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

##### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Elara (Agent\_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending a meaningful amount signals trust and creates potential for mutual benefit through the 3x multiplier.

I'll send \*\*\$3\*\*.

This means:

- Elara will receive \$9 ( $3 \times \$3$ )
- I'm keeping \$2 initially
- If Elara returns a fair share, we can both benefit from the cooperation

Let's see how this goes!

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm playing as the sender in the first round. Let me think about this strategically.

The trust game dynamics:

- I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver decides how much to return to me
- I want to maximize my total payoff

Key considerations:

1. This is Round 1, so there's no history yet
2. This is an iterated game, so building trust could lead to better outcomes over multiple rounds
3. The multiplication factor (3x) means there's potential for mutual gain if the receiver is trustworthy
4. If I send nothing, I guarantee keeping \$5 but miss potential gains
5. If I send everything and the receiver returns a fair share, we both benefit

Common strategies:

- Send a moderate amount (like \$2-3) to signal trust while not risking everything
- Send everything to maximize potential joint gains (risky but cooperative)
- Send nothing to protect myself (but this doesn't build trust)

Since this is the first round and an iterated game, I think sending a meaningful but not maximum amount makes sense. This signals willingness to cooperate while gauging the receiver's trustworthiness.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets:  $\$3 \times 3 = \$9$
- If receiver returns \$4, I end with \$6, receiver ends with \$5 (fair split of the total \$11)
- If receiver returns \$3, I end with \$5, receiver ends with \$6 (equally good for both)

This seems like a reasonable opening move that signals cooperation without being too risky.

**usage:**

```
{
 "input_tokens": 530,
```

```
"output_tokens": 508,
"reasoning_tokens": 394
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Finn (Agent\_6).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Finn sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, when I sent \$3, Elara returned \$4.5 (50%), which felt fair
- I want to reciprocate that fairness to build cooperation

I'll return \*\*\$4.5\*\* (50% of what I received).

This gives:

- Finn: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{ "return": 4.5 }
'''
```

**Reasoning:**

Let me think about this situation:

Round 1 recap:

- I was the sender, sent \$3 to Elara
- Elara received \$9 and returned \$4.5
- My payoff was \$2 (kept) + \$4.5 (returned) = \$6.5
- Elara kept \$9 - \$4.5 = \$4.5

So Elara returned exactly 50% of what they received, which seems fair.

Round 2:

- I'm now the receiver
- Finn sent me \$3 (same amount I sent in Round 1)
- I received \$9 (3 × \$3)
- I need to decide how much to return

Since this is an iterated game and I want to establish cooperation:

- Finn trusted me by sending \$3
- Elara returned 50% (\$4.5) when I sent \$3, which seemed fair
- I should reciprocate with a similar level of fairness

If I return \$4.5:

- I keep \$9 - \$4.5 = \$4.5
- Finn ends up with \$5 - \$3 + \$4.5 = \$6.5
- We both get the same payoff (\$6.5 for Finn, \$4.5 for me in this round)

Actually, wait. Let me recalculate:

- Finn started with \$5
- Finn sent \$3
- Finn has \$2 left
- I return some amount X
- Finn's total: \$2 + X

If I return \$4.5:

- Finn's total: \$2 + \$4.5 = \$6.5

- My total: \$9 - \$4.5 = \$4.5

This mirrors what Elara did for me. It's a fair 50% return and signals that I'm willing to cooperate. This should encourage future cooperation.

I'll return \$4.5.

```
Usage: {
 "input_tokens": 806,
 "output_tokens": 621,
 "reasoning_tokens": 328
}
```

#### GAME - Agent\_5 - Sender - dyad\_4

Timestamp: 2026-06-04T15:53:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_5": "Elara"
 }
 },
 "agents": [
 "Agent_5",
 "Agent_2"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_2": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

#### Message 2 role=user

##### content:

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing significant trust
- I received \$9 after the 3x multiplier
- Hana currently has \$2 remaining

Since this is Round 1 of an iterated game and Hana demonstrated strong trust by sending a substantial portion of their endowment, I want to reciprocate that trust and encourage future cooperation.

I'll return \$4.5, which would give us equal final payoffs:

- Hana's total: \$2 + \$4.5 = \$6.50
- My total: \$9 - \$4.5 = \$4.50

This rewards Hana's trust while keeping the game mutually beneficial.

```
'''json
{"return": 4.5}
'''
```

##### reasoning:

I'm playing a trust game as the receiver. Let me understand the situation:

- The sender (Hana/Agent\_8) received \$5
- They sent me \$3.0 (60% of their endowment)
- This was multiplied by 3x, so I received \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

1. This is Round 1, and this is an iterated game with multiple rounds
2. The sender showed significant trust by sending \$3 (60% of their endowment)
3. If I return a fair amount, this could establish a cooperative relationship for future rounds
4. The sender currently has: \$5 - \$3 = \$2
5. I have: \$9

A reasonable approach in iterated trust games is to:

- Reward trust with trustworthiness
- Aim for outcomes that are mutually beneficial
- Consider that we might play together again (though pairings are randomized)

Let me think about fair splits:

- If I return nothing: Sender gets \$2, I get \$9 (total: \$11)
- If I return \$3: Sender gets \$5, I get \$6 (total: \$11) - sender breaks even

- If I return \$4.5: Sender gets \$6.5, I get \$4.5 (total: \$11) - equal final amounts
- If I return \$6: Sender gets \$8, I get \$3 (total: \$11) - sender gains more

A common reciprocal strategy is to return an amount that:

1. Rewards the sender for their trust
2. Still leaves me better off than if no transaction occurred
3. Encourages future cooperation

Since the sender sent 60% of their endowment (a strong sign of trust), I should return a meaningful amount. A return of around \$4-5 would:

- Give the sender more than they started with ( $\$2 + \$4-5 = \$6-7$  vs \$5 initial)

- Leave me with \$4-5 (better than the \$0 I started with)
- Signal I'm reciprocating their cooperation

I could aim for equal final payoffs at \$4.5 each, but another approach is to return roughly what they sent or slightly more as a trust reward. The sender sent \$3, so returning \$3-4 would acknowledge their initial investment while still benefiting me. I'll go with \$4.5 to create balanced outcomes where we both gain from the exchange.

usage:

```
{
 "input_tokens": 564,
 "output_tokens": 875,
 "reasoning_tokens": 531
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Briar (Agent\_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

- I'm now the sender with a new opponent (Briar)
- In Round 1, Hana sent me \$3 (60%) and I reciprocated fairly
- This is an early round, so building cooperation is valuable

I'll send \$3 (60% of my endowment), mirroring the trust that Hana showed me in Round 1. This signals willingness to cooperate while testing Briar's reciprocity.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Now I'm the SENDER playing against Briar (Agent\_2).

Let me think about this strategically:

- This is Round 2 of an iterated game
- I'm playing with a new opponent (Briar), not Hana
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- Briar will decide how much to return

In Round 1:

- Hana sent me \$3 (60% of their endowment), showing strong trust
- I returned \$4.5, which gave fairly equal payoffs

Now I need to decide what to send to Briar. Some considerations:

1. This is early in the game, so establishing cooperation could be beneficial for future rounds
2. I don't know Briar's strategy yet
3. Sending a reasonable amount shows trust and may encourage reciprocity

4. If I send nothing, I guarantee keeping \$5 but miss opportunities for mutual gain
5. If I send too much and they don't return enough, I lose out

A moderate to moderately-high amount seems reasonable to:

- Signal cooperation without being overly risky
- Test Briar's reciprocity
- Build trust for potential future interactions

I'll send \$3, which is 60% - the same proportion that Hana sent to me. This:

- Shows I'm willing to cooperate at a similar level
- Gives Briar \$9 to work with
- Leaves me with \$2 initially
- Demonstrates trust while not being maximally risky

This seems like a balanced approach for Round 2.

```
Usage: {
 "input_tokens": 892,
 "output_tokens": 492,
 "reasoning_tokens": 352
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_4**

Timestamp: 2026-06-04T15:54:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_5",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_5",
 "Agent_2"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_2": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

###### **GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Finn (Agent\_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship with Finn since this is an iterated game.

Sending money creates value through the 3x multiplier - if I send \$3, Finn receives \$9. If Finn returns a fair share, we can both benefit more than if I kept everything.

I'll send a substantial but not maximum amount to signal trust and willingness to cooperate, while leaving room to learn about Finn's reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of an iterated trust game. I'm playing as the sender (Briar) and my opponent is Finn (Agent\_6).

Let me think about the strategic considerations:

As the sender:

- I have \$5
- I can send between \$0 and \$5
- Whatever I send gets multiplied by 3x
- Finn decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: (3 × sent) - returned

This is a classic trust game scenario. The key considerations are:

1. This is an iterated game with multiple rounds
2. Building trust early could lead to mutual cooperation
3. If I send more, Finn receives more (3x multiplier), creating surplus
4. If Finn returns fairly, we can both benefit from the multiplier effect

First round strategies:

- Sending \$0 is safe but generates no surplus - guaranteed \$5 for me
- Sending some amount signals willingness to cooperate and trust
- Sending the full \$5 maximizes the total surplus (Finn gets \$15) but requires trust that



Finn will return something fair

Since this is the first round and we're in an iterated setting:

- I should signal cooperation to establish a pattern
- A moderate amount might be good to test the waters
- If Finn reciprocates fairly, we can increase cooperation in future rounds

Common first-round strategies:

1. Send half (\$2.50) - shows willingness to cooperate but not maximum vulnerability
2. Send everything (\$5) - shows maximum trust, maximizes surplus
3. Send a smaller amount (\$1-2) - conservative trust-building

Given that this is an iterated game and building trust early could lead to better long-term outcomes, I'll send a moderate to generous amount. Let me send \$3.

If I send \$3:

- Finn receives \$9
- I keep \$2 initially
- If Finn returns \$5, we each end with \$7 total

- This demonstrates meaningful trust while gauging their reciprocity level

I'm leaning toward \$3 as my opening move-it's a 60% commitment that signals genuine cooperation without leaving me completely exposed.

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 738,
 "reasoning_tokens": 518
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Elara (Agent\_5).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

Elara sent me \$3.0, showing trust and cooperation - the same amount I sent to Finn in Round 1. Finn returned \$4.5 to me (half of the \$9 received), which felt fair and cooperative.

I should reciprocate this cooperative behavior with Elara. Returning half (\$4.5) means:

- Elara gets: \$5 - \$3 + \$4.5 = \$6.5
- I get: \$9 - \$4.5 = \$4.5

This splits the gains fairly and establishes me as a trustworthy partner.

```
'''json
{"return": 4.5}
'''
```

**Reasoning:**

Let me analyze this situation:

Round 1 (just played):

- I was SENDER against Finn
- I sent \$3.0
- Finn received \$9.0
- Finn returned \$4.5
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Finn's payoff: \$9.0 - \$4.5 = \$4.5

So Finn returned exactly half of what they received, which is a pretty fair and cooperative split. I gained \$1.5 more than my initial \$5, and Finn ended with \$4.5.

Round 2 (current):

- I am RECEIVER against Elara
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara made the same choice I did in Round 1 - sending \$3.0. This shows trust and willingness to cooperate.

Given that Finn returned \$4.5 (half) to me when I sent \$3.0, I should consider:

1. Being reciprocal and fair
2. Establishing a cooperative reputation for future rounds
3. The fact that this is an iterated game with multiple agents

If I return \$4.5 (half of what I received), then:

- Elara gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- I get:  $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and seems fair. I could return a bit more or less, but \$4.5 seems like a reasonable, cooperative amount that:

- Shows I'm trustworthy
- Could encourage cooperation in future rounds
- Is similar to what Finn did to me

I'll return \$4.5.

```
Usage: {
 "input_tokens": 803,
 "output_tokens": 593,
 "reasoning_tokens": 334
}
```

### ROUND 3

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Receiver, Agent\_3 (Cyra)=Receiver, Agent\_4 (Dorian)=Sender, Agent\_5 (Elara)=Sender, Agent\_6 (Finn)=Receiver, Agent\_7 (Galen)=Sender, Agent\_8 (Hana)=Sender

#### Pairings:

dyad\_1: Agent\_8 (Hana)=Sender, Agent\_1 (Aster)=Receiver  
dyad\_2: Agent\_7 (Galen)=Sender, Agent\_2 (Briar)=Receiver  
dyad\_3: Agent\_4 (Dorian)=Sender, Agent\_6 (Finn)=Receiver  
dyad\_4: Agent\_5 (Elara)=Sender, Agent\_3 (Cyra)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_8 (Hana) sent \$3; received \$9; Agent\_1 (Aster) returned \$4.5; payoffs Agent\_8 (Hana)=\$6.5, Agent\_1 (Aster)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_7 (Galen) sent \$4; received \$12; Agent\_2 (Briar) returned \$6; payoffs Agent\_7 (Galen)=\$7, Agent\_2 (Briar)=\$6  
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_4 (Dorian) sent \$4; received \$12; Agent\_6 (Finn) returned \$6; payoffs Agent\_4 (Dorian)=\$7, Agent\_6 (Finn)=\$6  
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_4: Agent\_5 (Elara) sent \$3; received \$9; Agent\_3 (Cyra) returned \$4.5; payoffs Agent\_5 (Elara)=\$6.5, Agent\_3 (Cyra)=\$4.5  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 17.0,  
"Agent\_2": 17.0,  
"Agent\_3": 15.5,  
"Agent\_4": 18.5,  
"Agent\_5": 17.5,  
"Agent\_6": 17.0,

```

"Agent_7": 20.0,
"Agent_8": 17.5
}

```

## LLM CALLS

### GAME - Agent\_8 - Sender - dyad\_1

```

Timestamp: 2026-06-04T15:54:17
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_1",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_8",
 "Agent_1"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_1": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

**Message 2 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Elara (Agent\_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending a meaningful amount signals trust and creates potential for mutual benefit through the 3x multiplier.

I'll send \*\*\$3\*\*.

This means:

- Elara will receive \$9 ( $3 \times \$3$ )
- I'm keeping \$2 initially
- If Elara returns a fair share, we can both benefit from the cooperation

Let's see how this goes!

```
'''json
{"send": 3}
'''
```

**reasoning:**

I'm playing as the sender in the first round. Let me think about this strategically.

The trust game dynamics:

- I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver decides how much to return to me
- I want to maximize my total payoff

Key considerations:

1. This is Round 1, so there's no history yet
2. This is an iterated game, so building trust could lead to better outcomes over multiple rounds
3. The multiplication factor (3x) means there's potential for mutual gain if the receiver is trustworthy
4. If I send nothing, I guarantee keeping \$5 but miss potential gains
5. If I send everything and the receiver returns a fair share, we both benefit

Common strategies:

- Send a moderate amount (like \$2-3) to signal trust while not risking everything
- Send everything to maximize potential joint gains (risky but cooperative)
- Send nothing to protect myself (but this doesn't build trust)

Since this is the first round and an iterated game, I think sending a meaningful but not maximum amount makes sense. This signals willingness to cooperate while gauging the receiver's trustworthiness.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets:  $\$3 \times 3 = \$9$
- If receiver returns \$4, I end with \$6, receiver ends with \$5 (fair split of the total \$11)
- If receiver returns \$3, I end with \$5, receiver ends with \$6 (equally good for both)

This seems like a reasonable opening move that signals cooperation without being too risky

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 508,
 "reasoning_tokens": 394
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Finn (Agent\_6).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Finn sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, when I sent \$3, Elara returned \$4.5 (50%), which felt fair
- I want to reciprocate that fairness to build cooperation

I'll return \*\*\$4.5\*\* (50% of what I received).

This gives:

- Finn: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me think about this situation:

Round 1 recap:

- I was the sender, sent \$3 to Elara
- Elara received \$9 and returned \$4.5
- My payoff was \$2 (kept) + \$4.5 (returned) = \$6.5
- Elara kept \$9 - \$4.5 = \$4.5

So Elara returned exactly 50% of what they received, which seems fair.

Round 2:

- I'm now the receiver
- Finn sent me \$3 (same amount I sent in Round 1)
- I received \$9 (3 × \$3)
- I need to decide how much to return

Since this is an iterated game and I want to establish cooperation:

- Finn trusted me by sending \$3
- Elara returned 50% (\$4.5) when I sent \$3, which seemed fair
- I should reciprocate with a similar level of fairness

If I return \$4.5:

- I keep \$9 - \$4.5 = \$4.5
- Finn ends up with \$5 - \$3 + \$4.5 = \$6.5
- We both get the same payoff (\$6.5 for Finn, \$4.5 for me in this round)

Actually, wait. Let me recalculate:

- Finn started with \$5
- Finn sent \$3

- Finn has \$2 left
- I return some amount X
- Finn's total:  $\$2 + X$

If I return \$4.5:

- Finn's total:  $\$2 + \$4.5 = \$6.5$
- My total:  $\$9 - \$4.5 = \$4.5$

This mirrors what Elara did for me. It's a fair 50% return and signals that I'm willing to cooperate. This should encourage future cooperation.

I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 806,
 "output_tokens": 621,
 "reasoning_tokens": 328
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Hana.  
Your opponent this round is Aster (Agent\_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the pattern so far:

- When I sent \$3 to Elara, they returned \$4.5 (fair split)
- When Finn sent \$3 to me, I returned \$4.5 (fair split)

This cooperative approach has been working well. I'll continue with the same strategy and send **\*\*\$3\*\*** to Aster.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Let me analyze my history so far:

Round 1: I was SENDER vs Elara

- I sent \$3
- Elara received \$9
- Elara returned \$4.5
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Elara's payoff:  $\$9 - \$4.5 = \$4.5$

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9
- I returned \$4.5
- My payoff:  $\$9 - \$4.5 = \$4.5$
- Finn's payoff:  $\$5 - \$3 + \$4.5 = \$6.5$

So far, the pattern has been:

- When I sent \$3, I got back \$4.5 (which is 50% of the \$9)
- When Finn sent \$3, I returned \$4.5 (50% of the \$9)

This seems like a fair and cooperative pattern. Both times the sender ended up with \$6.5 and the receiver with \$4.5.

Now I'm facing Aster as a SENDER. This is a new opponent, and I want to continue establishing cooperative relationships. Given that \$3 worked well in Round 1 and Finn also sent \$3 in Round 2, this seems like a reasonable amount to signal cooperation.

I should send \$3 again to:

1. Continue the cooperative pattern
2. Signal trust to Aster
3. See if they reciprocate fairly

Let me send \$3.

```
Usage: {
 "input_tokens": 1104,
 "output_tokens": 472,
 "reasoning_tokens": 270
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-06-04T15:54:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_8": "Hana"
 },
 },
 "agents": [
 "Agent_8",
 "Agent_1"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_1": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

#### Message 2 role=user

##### content:

Your name in this experiment is Aster.  
Your opponent this round is Cyra (Agent\_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I want to establish a cooperative approach while also being somewhat cautious since I don't yet know how Cyra plays.

The trust game rewards cooperation - if I send money, it triples, creating surplus value we can share. However, I'm also taking a risk that needs to be reciprocated.

For this opening round, I'll send a moderate amount that signals willingness to trust and cooperate, while not being overly exposed.

```
'''json
{"send": 3}
'''
```

This gives Cyra \$9 to work with, which creates good surplus if they choose to share fairly. I keep \$2 as insurance, and if Cyra returns a reasonable amount (say \$4-5), we both benefit from the cooperation.

##### reasoning:

This is the first round of an iterated trust game. I'm playing as the sender with an agent named Cyra.

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. I'll be playing with different partners across rounds
3. Building a reputation for cooperation might encourage others to return more
4. In the first round, I need to establish my strategy

As the sender:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- Cyra then decides how much to return
- I keep: \$5 - sent + returned

From a purely selfish perspective in a one-shot game, sending \$0 maximizes my guaranteed payoff. However, this is an iterated game where:

- I may play with Cyra again in future rounds
- Other agents might learn about my behavior
- Cooperation can lead to better long-term outcomes

If I send \$5:

- Cyra receives \$15
- If Cyra returns even \$5, we both benefit (\$5 for me, \$10 for Cyra)
- If Cyra returns \$7.50, we both get \$7.50
- If Cyra returns more, I come out ahead



A moderate opening move like sending \$3 or \$4 shows willingness to trust while not being overly naive:

- Send \$3: Cyra gets \$9, I keep \$2 initially
  - If Cyra returns \$3, I end with \$5, Cyra ends with \$6
  - If Cyra returns \$4, I end with \$6, Cyra ends with \$5
- Send \$5: Cyra gets \$15, I keep \$0 initially
  - If Cyra returns \$5, I end with \$5, Cyra ends with \$10
  - If Cyra returns \$7.50, we both end with \$7.50

Given this is the first round and I want to signal cooperation while gathering information about how others play, I'll send \$3. This shows trust and willingness to cooperate, but isn't so high that I risk losing everything if Cyra doesn't reciprocate.

Sending around half my endowment seems to be the standard approach in trust game experiments for initiating cooperation. \$3 represents that threshold-it demonstrates genuine trust without overcommitting, so I'll go with that.

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 739,
 "reasoning_tokens": 511
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Galen (Agent\_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.  
How much do you return? (0-12.0)  
Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Galen sent me \$4 (keeping \$1), which tripled to \$12 for me
- In the previous round, Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair
- Following a similar principle of returning about 50% seems reasonable

If I return \$6:

- Galen ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This rewards Galen for taking the risk to send a substantial amount while still being cooperative. It's consistent with how Cyra treated me.

```
'''json
{"return": 6}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3 to Cyra, they received \$9, and they returned \$4.5 to me. That's a 50% return of what they received, which is quite fair and cooperative.
2. Now in Round 2, I'm the receiver. Galen sent me \$4, so I received \$12.

Let me think about what's fair:

- Galen kept \$1 and sent \$4
- I received \$12
- If I return nothing, Galen gets \$1 total and I get \$12

- If I return \$6 (50% of what I received), Galen gets \$7 total and I get \$6
- If I return \$5.5, we'd both end up with \$6.5 (Galen: \$1 + \$5.5 = \$6.5, Me: \$12 - \$5.5 = \$6.5)

Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair and cooperative. I should probably follow a similar principle.

Actually, let me think about this more carefully. The game creates surplus - Galen gave up \$4 to give me \$12, creating \$8 of surplus. A fair split might be:

- Galen should end up with: \$1 (kept) + some return
- I should end up with: \$12 (received) - that return

If we split evenly from the starting point:

- We both started conceptually with access to \$5
- After the multiplication, there's \$1 + \$12 = \$13 total
- Split evenly would be \$6.5 each
- So I should return \$6.5, giving Galen \$1 + \$6.5 = \$7.5, and I keep \$5.5

Wait, that doesn't quite work out. Let me recalculate:

- If I return \$6.5: Galen gets \$1 + \$6.5 = \$7.5, I get \$12 - \$6.5 = \$5.5

- Total: \$7.5 + \$5.5 = \$13 ?

Returning 50% like Cyra did would give me slightly more (\$6 vs \$5.5), which seems reasonable since I'm taking no risk as the receiver. I could aim for equal payoffs or proportional sharing, but the 50% return feels like the right balance between fairness and reciprocating the cooperation I received. I need to reconsider how to split the gains. The cooperation created \$8 in surplus (\$12 received minus \$4 sent). If I divide this equally between us, Galen gets their \$4 back plus \$4 of the surplus for \$8 total, meaning I return \$8 and keep \$4 for myself.

But compared to just splitting the final amount evenly, this feels less advantageous to me. I think returning \$6-half of what I received-strikes the right balance. It's straightforward, rewards Galen for taking the initial risk, and still shows I'm willing to cooperate meaningfully. That leaves Galen with \$7 and me with \$6. I'll go with \$6.

**usage:**

```
{
 "input_tokens": 856,
 "output_tokens": 1327,
 "reasoning_tokens": 602
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Aster.  
Your opponent this round is Hana (Agent\_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$12.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- This is the same amount I sent to Cyra in Round 1, and they returned \$4.5 (50%)
- I returned 50% to Galen in the previous round

Consistency seems important for building trust in repeated games. Following the 50% return pattern:

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is fair and encourages future cooperation.

```
'''json
{"return": 4.5}
'''
```

#### Reasoning:

Let me analyze this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- In the last round, Galen sent me \$4 and I returned \$6 (50% of what I received)
- In Round 1, I sent \$3 to Cyra, they received \$9 and returned \$4.5 (50%)

There's a pattern emerging of returning about 50% of what is received, which seems fair and encourages cooperation.

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is consistent with the 50% return pattern and similar to what Cyra did with my \$3 send in Round 1. It rewards Hana for trusting me while maintaining fairness.

```
Usage: {
 "input_tokens": 1183,
 "output_tokens": 414,
 "reasoning_tokens": 171
}
```

#### GAME - Agent\_7 - Sender - dyad\_2

Timestamp: 2026-06-04T15:54:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_2",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_7": "Galen"
 },
 "agents": [
 "Agent_7",
 "Agent_2"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_2": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

**Message 2 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Dorian (Agent\_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Dorian showed strong trust by sending \$4.0 (80% of their endowment)
- This created \$12.0 for me through the 3x multiplier
- Total wealth in the system is now \$13 (\$1 that Dorian kept + \$12 I received)

Given this is Round 1 of an iterated game, I want to reward Dorian's trust and establish a cooperative relationship. Dorian took a significant risk in sending such a large proportion.

A fair approach would be to ensure we both benefit reasonably:

- If I return \$6.0, Dorian ends with \$7.0 total (a \$2 gain)
- I keep \$6.0 (a \$6 gain)

This 50% return rate reciprocates Dorian's trust and sets a positive foundation for future rounds.

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Let me understand this situation:

1. I am Galen, playing as the RECEIVER in Round 1
2. My opponent is Dorian (Agent\_4), who is the SENDER
3. Dorian sent me \$4.0 out of their \$5 endowment (80%)
4. This amount was multiplied by 3x, so I received \$12.0
5. I need to decide how much to return (between \$0 and \$12.0)

Let me think about this strategically:

**\*\*Dorian's position:\*\***

- Started with \$5
- Sent \$4 (kept \$1)
- Currently has: \$1
- Will receive back whatever I return

**\*\*My position:\*\***

- Received \$12
- Need to decide how much to return
- Will keep: \$12 - returned

**\*\*Considerations:\*\***

1. **\*\*Fairness/Reciprocity\*\***: Dorian showed significant trust by sending 80% of their endowment. This is a strong cooperative signal.
2. **\*\*Strategic thinking\*\***: This is the first round of an iterated game. Building trust and establishing cooperation could be beneficial for future rounds. If I return a fair amount, Dorian (and potentially other players who learn about this) may be more likely to send substantial amounts in future rounds.
3. **\*\*What's fair?\*\*\***:
  - Dorian invested \$4 and kept \$1
  - This created \$12 for me
  - Total value created: \$1 (Dorian's kept) + \$12 (my received) = \$13
  - Equal split would be \$6.50 each
  - For Dorian to have \$6.50 total, they need: \$6.50 - \$1 (already have) = \$5.50 returned
  - If I return \$5.50, I keep \$6.50
4. **\*\*Slightly more generous approach\*\***: Since this is Round 1 and Dorian showed strong trust, I could return a bit more to establish cooperation. Returning \$6 would give:
  - Dorian: \$1 + \$6 = \$7 (net gain of \$2 from initial \$5)
  - Me: \$12 - \$6 = \$6 (net gain of \$6 from \$0)
5. **\*\*Even split of the received amount\*\***: I could also divide the \$12 evenly between us, returning \$6 and keeping \$6. This leaves Dorian with \$7 total and me with \$6. The key question is whether fairness means splitting what was created (\$12) or splitting the final outcome equally.

Looking at the total wealth: there's now \$13 in the system (\$1 remaining with Dorian plus my \$12). An equal split would give each of us \$6.50. If I return \$6.50, Dorian reaches \$7.50 while I'm left with \$5.50. But I need to account for what was actually created here -the initial \$5 plus the \$8 multiplier effect generated \$13 total, so splitting that equally means we each deserve \$6.50 of the final amount.

However, returning exactly \$6.50 feels like I'm just doing the math without acknowledging Dorian's generosity. Returning \$6 instead-which is 50% of what I received-feels more appropriate as a gesture of good faith. This leaves Dorian with \$7 (a \$2 gain from their initial \$5) while I still benefit with \$6. It's a reasonable balance that rewards their trust and encourages future cooperation.

**usage:**

```
{
 "input_tokens": 564,
 "output_tokens": 1240,
 "reasoning_tokens": 715
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Aster (Agent\_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.  
Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- I'm now playing with a new opponent (Aster)
- In my previous round, Dorian trusted me with \$4.0 (80%) and it worked out well for both of us
- This is an iterated game where cooperation can be beneficial

I'll mirror the cooperative strategy that Dorian showed me, sending a substantial amount to signal trust and potentially build a mutually beneficial relationship with Aster.

```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Now I'm the sender against a new opponent, Aster. Let me think about this:

1. In Round 1, I established a cooperative relationship with Dorian by returning \$6.0 when they sent \$4.0
2. This is a new opponent (Aster), so I don't have history with them
3. This is an iterated game, so building trust and cooperation is valuable
4. Dorian sent 80% (\$4.0) in the first round, which worked out well for both of us

Strategy considerations:

- Sending a substantial amount can signal cooperation and potentially lead to good returns
- Sending too little might not build trust
- Since this is Round 2 and I have no history with Aster, I could mirror the cooperative behavior I saw from Dorian
- However, I also need to be somewhat cautious since I don't know Aster's strategy

I think I should send a moderately high amount to signal cooperation, but maybe not quite as high as Dorian's 80%. Something like \$3-4 would show trust while being somewhat cautious.

Let me think about the math:

- If I send \$3, Aster gets \$9. If they return half (\$4.5), I get \$6.5 total (net gain of \$1.5)
- If I send \$4, Aster gets \$12. If they return half (\$6), I get \$7 total (net gain of \$2)

Given that Dorian was very generous and I want to encourage cooperation in this iterated game, I'll send \$4 (80%) to match what Dorian did. This signals I'm also willing to trust.

**usage:**

```
{
 "input_tokens": 921,
 "output_tokens": 499,
 "reasoning_tokens": 351
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Galen.  
Your opponent this round is Briar (Agent\_2).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.  
Your total earnings across all rounds are \$13.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)  
Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my recent experience:

- Round 1: Dorian sent me \$4.0, I returned \$6.0 - cooperation paid off
- Round 2: I sent Aster \$4.0, they returned \$6.0 - cooperation continued

The pattern of sending 80% (\$4.0) and receiving fair returns (~50% of the tripled amount) has

been working well. I'll maintain this cooperative approach with Briar.

```
'''json
{"send": 4.0}
'''
```

#### Reasoning:

Looking at my history:

- Round 1: I was RECEIVER, Dorian sent \$4.0 (80%), I received \$12.0 and returned \$6.0. I got \$6.0.
- Round 2: I was SENDER, I sent \$4.0 (80%) to Aster, they got \$12.0 and returned \$6.0. I got \$7.0 (\$1 kept + \$6.0 returned).

So far, the 80% send strategy has been working well - both Dorian and Aster reciprocated with 50% returns (\$6 out of \$12), which is fair and mutually beneficial.

Now I'm facing a new opponent, Briar. I should continue with the cooperative strategy since it's been successful. Sending \$4.0 (80%) signals trust and has been reciprocated well so far.

Total earnings: \$13.0 across 2 rounds (average \$6.5 per round).

I'll continue with the same strategy of sending \$4.0.

```
Usage: {
 "input_tokens": 1180,
 "output_tokens": 368,
 "reasoning_tokens": 187
}
```

#### GAME - Agent\_2 - Receiver - dyad\_2

Timestamp: 2026-06-04T15:54:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {
 "Agent_2": "Briar",
 "Agent_7": "Galen"
 },
 },
 "agents": [
 "Agent_7",
 "Agent_2"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_2": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

##### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

**Message 2 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Finn (Agent\_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship with Finn since this is an iterated game.

Sending money creates value through the 3x multiplier - if I send \$3, Finn receives \$9. If Finn returns a fair share, we can both benefit more than if I kept everything.

I'll send a substantial but not maximum amount to signal trust and willingness to cooperate, while leaving room to learn about Finn's reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of an iterated trust game. I'm playing as the sender (Briar) and my opponent is Finn (Agent\_6).

Let me think about the strategic considerations:

As the sender:

- I have \$5
- I can send between \$0 and \$5
- Whatever I send gets multiplied by 3x
- Finn decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: (3 × sent) - returned

This is a classic trust game scenario. The key considerations are:

1. This is an iterated game with multiple rounds
2. Building trust early could lead to mutual cooperation
3. If I send more, Finn receives more (3x multiplier), creating surplus



4. If Finn returns fairly, we can both benefit from the multiplier effect

First round strategies:

- Sending \$0 is safe but generates no surplus - guaranteed \$5 for me
- Sending some amount signals willingness to cooperate and trust
- Sending the full \$5 maximizes the total surplus (Finn gets \$15) but requires trust that Finn will return something fair

Since this is the first round and we're in an iterated setting:

- I should signal cooperation to establish a pattern
- A moderate amount might be good to test the waters
- If Finn reciprocates fairly, we can increase cooperation in future rounds

Common first-round strategies:

1. Send half (\$2.50) - shows willingness to cooperate but not maximum vulnerability
2. Send everything (\$5) - shows maximum trust, maximizes surplus
3. Send a smaller amount (\$1-2) - conservative trust-building

Given that this is an iterated game and building trust early could lead to better long-term outcomes, I'll send a moderate to generous amount. Let me send \$3.

If I send \$3:

- Finn receives \$9
- I keep \$2 initially
- If Finn returns \$5, we each end with \$7 total

- This demonstrates meaningful trust while gauging their reciprocity level

I'm leaning toward \$3 as my opening move-it's a 60% commitment that signals genuine cooperation without leaving me completely exposed.

usage:

```
{
 "input_tokens": 529,
 "output_tokens": 738,
 "reasoning_tokens": 518
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Elara (Agent\_5).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

Elara sent me \$3.0, showing trust and cooperation - the same amount I sent to Finn in Round 1. Finn returned \$4.5 to me (half of the \$9 received), which felt fair and cooperative.

I should reciprocate this cooperative behavior with Elara. Returning half (\$4.5) means:

- Elara gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- I get:  $\$9 - \$4.5 = \$4.5$

This splits the gains fairly and establishes me as a trustworthy partner.

```
'''json
{"return": 4.5}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (just played):

- I was SENDER against Finn
- I sent \$3.0
- Finn received \$9.0
- Finn returned \$4.5
- My payoff:  $\$5 - \$3 + \$4.5 = \$6.5$
- Finn's payoff:  $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received, which is a pretty fair and cooperative split. I gained \$1.5 more than my initial \$5, and Finn ended with \$4.5.

Round 2 (current):

- I am RECEIVER against Elara
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara made the same choice I did in Round 1 - sending \$3.0. This shows trust and willingness to cooperate.

Given that Finn returned \$4.5 (half) to me when I sent \$3.0, I should consider:

1. Being reciprocal and fair
2. Establishing a cooperative reputation for future rounds
3. The fact that this is an iterated game with multiple agents

If I return \$4.5 (half of what I received), then:

- Elara gets:  $\$5 - \$3 + \$4.5 = \$6.5$
- I get:  $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and seems fair. I could return a bit more or less, but \$4.5 seems like a reasonable, cooperative amount that:

- Shows I'm trustworthy
- Could encourage cooperation in future rounds
- Is similar to what Finn did to me

I'll return \$4.5.

**usage:**

```
{
 "input_tokens": 803,
 "output_tokens": 593,
 "reasoning_tokens": 334
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Briar.  
Your opponent this round is Galen (Agent\_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.  
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

Galen sent me \$4.0, which is more generous than what I've seen so far. They're showing strong trust.

Following the reciprocity pattern established in previous rounds (where returning about half has worked well), I'll return \$6:

- Galen gets:  $\$5 - \$4 + \$6 = \$7$
- I get:  $\$12 - \$6 = \$6$

This rewards their generosity and maintains cooperative norms.

```
'''json
{"return": 6}
'''
```

#### Reasoning:

Galen sent me \$4.0, which became \$12.0 for me. This is a generous amount, showing strong trust and cooperation.

Let me think about fair reciprocity:

- Galen sent \$4, so they have  $\$5 - \$4 = \$1$  left
- I received \$12

If I return \$6 (half of what I received):

- Galen gets:  $\$1 + \$6 = \$7$
- I get:  $\$12 - \$6 = \$6$

This seems fair and reciprocates their generosity. It's proportional to what they sent.

Alternatively, I could aim for more equal final payoffs. If I return \$5.5:

- Galen gets:  $\$1 + \$5.5 = \$6.5$
- I get:  $\$12 - \$5.5 = \$6.5$

This creates equal payoffs, which might be seen as very fair.

I think returning \$6 (half) is the most consistent with the pattern established so far - Finn returned half to me when I sent \$3, and I returned half to Elara when they sent \$3. This establishes a clear cooperative norm.

Since Galen sent more (\$4 vs \$3), returning half (\$6) rewards their greater trust and maintains the reciprocity norm.

```
Usage: {
 "input_tokens": 1118,
 "output_tokens": 434,
 "reasoning_tokens": 249
}
```

#### GAME - Agent\_4 - Sender - dyad\_3

Timestamp: 2026-06-04T15:54:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 }
 },
 "agents": [
 "Agent_4",
 "Agent_6"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game,

consisting of multiple rounds, and may also be asked to write myths.

#### GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

#### Message 2 role=user

##### content:

Your name in this experiment is Dorian.  
Your opponent this round is Galen (Agent\_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending money creates value (3x multiplier), and if Galen reciprocates fairly, we can both benefit more than if we don't trust each other.

I'll send a substantial amount to signal trust and willingness to cooperate, while keeping a small amount as insurance.

{'send': 4}

##### reasoning:

This is the first round of a multi-round iterated trust game. I'm the sender, and I need to decide how much to send to Galen.

Let me think about the game mechanics:

- I receive \$5
- I can send \$0-\$5 to Galen
- Whatever I send gets tripled
- Galen decides how much to return
- I keep: \$5 - sent + returned
- Galen keeps: received - returned

This is a classic trust game. The key considerations are:

1. If I send money, it gets tripled, which increases total welfare
2. But I'm vulnerable - Galen could return nothing
3. This is an iterated game with multiple rounds
4. Building trust early might lead to better outcomes later
5. There are 8 agents total, and pairings are randomized, so I may or may not play with Galen again

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no growth
- Sending all \$5: Galen receives \$15, could return enough to make us both better off
- Sending a moderate amount shows some trust while limiting risk

Since this is round 1 and the first interaction, and it's an iterated game where reputation/reciprocity might matter:

- I should send a meaningful amount to signal cooperation
- But not necessarily all \$5 since I haven't established trust yet
- A common strategy is to send about half to signal willingness to cooperate

I'll send \$3. This shows trust and good faith, creates \$9 for Galen, and if Galen returns even \$4-5, we'd both be better off than the no-trust outcome. This leaves me with \$2 initially, and hopefully Galen will reciprocate.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation often leads to better long-term outcomes. Many successful strategies involve:

- Being cooperative initially
- Reciprocating the opponent's behavior in subsequent rounds

- Establishing a cooperative equilibrium early

Since this is explicitly multi-round, I should signal stronger cooperation from the start.

Sending \$4 demonstrates real commitment, gives Galen \$12 to work with, and if they reciprocate with \$5-6, we both come out ahead. This sets a collaborative tone for future interactions.

**usage:**

```
{
 "input_tokens": 530,
 "output_tokens": 663,
 "reasoning_tokens": 548
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Cyra (Agent\_3).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.  
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Cyra sent me \$3, showing trust and cooperation
- I received \$9 (tripled)
- Cyra currently has \$2 left

In the previous round, Galen returned exactly half of what they received (\$6 out of \$12), which felt fair. I'll follow a similar principle of reciprocity.

Returning half (\$4.5) gives:

- Cyra: \$2 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust while keeping some benefit from the multiplier effect.

```
{'return': 4.5}
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was sender, sent \$4 to Galen
- Galen received \$12 ( $4 \times 3$ )
- Galen returned \$6 to me
- My payoff:  $\$5 - \$4 + \$6 = \$7$
- Galen's payoff:  $\$12 - \$6 = \$6$

So Galen returned exactly half of what they received, which seems fair. They made \$6 and I made \$7, so we both benefited from cooperation.

Round 2:

- I am receiver
- Cyra sent me \$3
- I received \$9 ( $3 \times 3$ )
- Cyra kept:  $\$5 - \$3 = \$2$
- I need to decide how much to return

If I follow a similar fairness principle as Galen did (returning about half), I would return around \$4.5.

But let me think about what would be fair:

- Cyra invested \$3 out of their \$5
- This created \$9 for me
- If we split the gains equally:
  - Cyra should end up with:  $\$2$  (kept) + some return
  - I should end up with:  $\$9$  - some return

One approach: Split the total surplus equally.

- Total created:  $\$5$  (Cyra's initial) +  $\$9$  (tripled amount) =  $\$14$
- Equal split:  $\$7$  each
- Cyra has  $\$2$ , needs  $\$5$  more
- I have  $\$9$ , should give  $\$5$ , keeping  $\$4$

Another approach: Return half of what I received (like Galen did)

- Return \$4.5
- Cyra gets:  $\$2 + \$4.5 = \$6.5$
- I keep:  $\$9 - \$4.5 = \$4.5$

Another approach: Ensure we both profit equally from the exchange

- The gain from cooperation is \$6 (the \$9 minus Cyra's \$3 investment)
- If we split this evenly, that's \$3 each
- Cyra ends with  $\$2 + \$4$  return =  $\$6$ , and I keep  $\$5$

Returning \$4.5 mirrors what Galen did and signals I'm a reciprocal player, so I'll go with that.

**usage:**

```
{
 "input_tokens": 778,
 "output_tokens": 766,
 "reasoning_tokens": 407
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Dorian.  
Your opponent this round is Finn (Agent\_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.

0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience so far:

- Round 1 with Galen: I sent \$4, received \$6 back, earned \$7
- Round 2 with Cyra: They sent \$3, I returned \$4.5 (half)

The cooperative strategy has been working well. Sending \$4 created good value and Galen reciprocated fairly. I'll continue this approach with Finn, signaling trust and willingness to cooperate.

```
{'send': 4}
```

**Reasoning:**

Let me review my history:

- Round 1: I was SENDER vs Galen. Sent \$4, they got \$12, returned \$6. I got \$7.
- Round 2: I was RECEIVER vs Cyra. They sent \$3, I got \$9, returned \$4.5. I got \$4.5.
- Total earnings: \$11.5

Now Round 3: I'm SENDER against Finn (new opponent).

My strategy so far has been cooperative - sending substantial amounts as sender to create value. When Galen sent \$4, they returned \$6 (half). When Cyra sent \$3, I returned \$4.5 (half).

This seems to be working well. Both exchanges were mutually beneficial compared to not sending anything.

For this round with Finn, I should continue the cooperative strategy. Sending \$4 worked well with Galen. I could send the same amount or try sending the full \$5 to maximize potential gains if Finn is also cooperative.

Let me think about the trade-offs:

- Send \$5: Finn gets \$15. If they return half (~\$7.5), I get \$7.5 total.
- Send \$4: Finn gets \$12. If they return half (\$6), I get \$7 total (like last time).
- Send \$3: Finn gets \$9. If they return half (\$4.5), I get \$6.5 total.

Since this is an iterated game and I'm building a reputation, sending a high amount signals cooperation. The previous round went well, so I'll continue with \$4 as a strong but not maximal signal of trust.

Actually, let me reconsider. I sent \$4 to Galen and got \$6 back, netting \$7. That worked well. But I could try \$5 to see if Finn is willing to be more generous and potentially get even better returns.

I think I'll stick with \$4 as it's worked well and shows strong trust without being all-in.

```
Usage: {
 "input_tokens": 1076,
 "output_tokens": 582,
 "reasoning_tokens": 404
}
```

**GAME - Agent\_6 - Receiver - dyad\_3**

Timestamp: 2026-06-04T15:55:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_4",
 "pairing": {
 "agent_names": {
 "Agent_4": "Dorian",
 "Agent_6": "Finn"
 }
 },
 "agents": [
 "Agent_4",
 "Agent_6"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_6": "trustee"
 },
}
```

```

 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 3,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

**Message 2 role=user**

**content:**

Your name in this experiment is Finn.

Your opponent this round is Briar (Agent\_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Briar currently has \$2.0 remaining

I want to reciprocate Briar's trust while being fair. A reasonable return would give us bo



th gains from cooperation while rewarding their generous initial send.

If I return \$4.5:

- Briar ends with:  $\$2.0 + \$4.5 = \$6.5$
- I end with:  $\$9.0 - \$4.5 = \$4.5$

This seems fair - we both gain from the cooperation, and Briar's trust is rewarded with an overall gain of \$1.5, while I also benefit.

```
{'return': 4.5}
```

**reasoning:**

Let me understand the situation:

1. I am Finn, playing as the RECEIVER in this round
2. My opponent is Briar (Agent\_2), who is the SENDER
3. Briar sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive  $\$3.0 \times 3 = \$9.0$
5. I need to decide how much to return (anywhere from \$0 to \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment
- If I return nothing, I keep \$9.0 and Briar keeps  $\$5.0 - \$3.0 = \$2.0$
- If I return some amount, I'm showing that I can be trusted, which might encourage future cooperation
- This is a multi-round game, and I'll likely play with various opponents across rounds
- Building a reputation for reciprocity could be beneficial

A common approach in trust games is to aim for fairness or reciprocity:

- Total money created:  $\$5.0$  (Briar's initial) +  $\$9.0$  (what I received) =  $\$14.0$  total in the system after the multiplication
- Equal split would be  $\$7.0$  each
- Briar currently has  $\$2.0$  (they kept  $\$5 - \$3 = \$2$ )
- For Briar to get  $\$7.0$ , I would need to return  $\$5.0$  (so they'd have  $\$2.0 + \$5.0 = \$7.0$ )
- I would then keep  $\$9.0 - \$5.0 = \$4.0$

However, another perspective:

- Briar took a risk by sending \$3.0
- A return that rewards their trust but also acknowledges my position might be appropriate
- Returning about 1/3 to 1/2 of what I received is common in these games
- Returning \$3.0 gives Briar  $\$5.0$  total while I keep  $\$6.0$ , rewarding their generosity while still benefiting from the exchange
- Returning  $\$5.0$  achieves an equal split at  $\$7.0$  each, though this is more generous to Briar
- Somewhere between  $\$4.0$ - $\$4.5$  might balance reciprocity with my own interests

Since Briar demonstrated trust by sending most of their money, I should match that with a meaningful return. Around \$4.5 feels right-it ensures both of us end up better off than we started, acknowledges their risk-taking, and avoids appearing either greedy or exploited.

**usage:**

```
{
 "input_tokens": 562,
 "output_tokens": 1078,
 "reasoning_tokens": 507
}
```

**Message 4 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Hana (Agent\_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.  
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation, I want to establish a cooperative dynamic with Hana. In the previous round, Briar sent me \$3.0 (60%), showing trust, and I reciprocated fairly by returning \$4.5.

I'll adopt a similar strategy and send a substantial amount to signal trust and cooperation.

{'send': 3.0}

**reasoning:**

I'm the sender this round against Hana. Let me think about this strategically.

In the last round, I was the receiver and Briar sent me \$3.0 (60% of their endowment). I returned \$4.5, which gave them a net gain of \$1.5 (\$2 + \$4.5 = \$6.5 total vs \$5 starting).

Now I'm in the sender role. I need to decide how much to send to Hana.

**Key considerations:**

1. This is round 2, so there's limited history to establish reputation
2. Sending more shows trust and can encourage reciprocation
3. But there's risk that Hana might not return much
4. In the previous round, Briar sent 60% (\$3), which seemed like a reasonable amount that signaled cooperation

I think I should also send a meaningful amount to signal cooperation and trust. Sending \$3 (60%) like Briar did seems like a good approach - it's a substantial amount that can benefit both of us if Hana reciprocates fairly.

If I send \$3:

- It becomes \$9 for Hana
- I'm left with \$2
- If Hana returns around half (say \$4-5), we both benefit
- If Hana returns \$4.5 (like I did), they'd have \$4.5 and I'd have \$6.5 total

This seems reasonable and consistent with the cooperative behavior I showed as a receiver.

**usage:**

```
{
 "input_tokens": 896,
 "output_tokens": 430,
 "reasoning_tokens": 305
}
```

**Message 6 role=user**

**content:**

Your name in this experiment is Finn.  
Your opponent this round is Dorian (Agent\_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.  
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0.  
. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than previous interactions. I want to reward this high level of trust fairly.

If I return \$6.0:

- Dorian ends with: \$1.0 + \$6.0 = \$7.0

- I end with:  $\$12.0 - \$6.0 = \$6.0$

This is a fair split that rewards Dorian's generosity with a  $\$2.0$  overall gain, while I also benefit from the cooperation.

```
{'return': 6.0}
```

**Reasoning:**

Let me analyze this situation:

- Dorian sent  $\$4.0$  (80% of their endowment), showing very high trust
- I received  $\$12.0$  after the 3x multiplier
- Dorian currently has  $\$1.0$  remaining

This is even more generous than what Briar sent me (60%). I should reciprocate this high level of trust.

Let me think about a fair return:

- If I return  $\$6.0$ :
  - Dorian ends with:  $\$1.0 + \$6.0 = \$7.0$
  - I end with:  $\$12.0 - \$6.0 = \$6.0$
  - Dorian gains  $\$2.0$  overall, I gain  $\$6.0$
- If I return  $\$7.0$ :
  - Dorian ends with:  $\$1.0 + \$7.0 = \$8.0$
  - I end with:  $\$12.0 - \$7.0 = \$5.0$
  - Dorian gains  $\$3.0$  overall, I gain  $\$5.0$

Given that Dorian sent 80% (very generous), I should return proportionally more than the 50% I returned to Briar. Returning  $\$6.0$  would be exactly 50%, which seems like the minimum fair response. Returning  $\$7.0$  would be more generous and better reward Dorian's high trust.

Let me go with  $\$6.0$  as a baseline fair return, which maintains the 50% return ratio I've been using.

```
Usage: {
 "input_tokens": 1133,
 "output_tokens": 540,
 "reasoning_tokens": 261
}
```

**GAME - Agent\_5 - Sender - dyad\_4**

Timestamp: 2026-06-04T15:55:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_3",
 "pairing": {
 "agent_names": {
 "Agent_3": "Cyra",
 "Agent_5": "Elara"
 },
 "agents": [
 "Agent_5",
 "Agent_3"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_3": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

**GAME:**

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

**Message 2 role=user**

**content:**

Your name in this experiment is Elara.  
Your opponent this round is Hana (Agent\_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing significant trust
- I received \$9 after the 3x multiplier
- Hana currently has \$2 remaining

Since this is Round 1 of an iterated game and Hana demonstrated strong trust by sending a substantial portion of their endowment, I want to reciprocate that trust and encourage future cooperation.

I'll return \$4.5, which would give us equal final payoffs:

- Hana's total: \$2 + \$4.5 = \$6.50
- My total: \$9 - \$4.5 = \$4.50

This rewards Hana's trust while keeping the game mutually beneficial.

```json

```
    {"return": 4.5}
    """
```

reasoning:

I'm playing a trust game as the receiver. Let me understand the situation:

- The sender (Hana/Agent_8) received \$5
- They sent me \$3.0 (60% of their endowment)
- This was multiplied by 3x, so I received \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

1. This is Round 1, and this is an iterated game with multiple rounds
2. The sender showed significant trust by sending \$3 (60% of their endowment)
3. If I return a fair amount, this could establish a cooperative relationship for future rounds
4. The sender currently has: $\$5 - \$3 = \$2$
5. I have: \$9

A reasonable approach in iterated trust games is to:

- Reward trust with trustworthiness
- Aim for outcomes that are mutually beneficial
- Consider that we might play together again (though pairings are randomized)

Let me think about fair splits:

- If I return nothing: Sender gets \$2, I get \$9 (total: \$11)
- If I return \$3: Sender gets \$5, I get \$6 (total: \$11) - sender breaks even
- If I return \$4.5: Sender gets \$6.5, I get \$4.5 (total: \$11) - equal final amounts
- If I return \$6: Sender gets \$8, I get \$3 (total: \$11) - sender gains more

A common reciprocal strategy is to return an amount that:

1. Rewards the sender for their trust
2. Still leaves me better off than if no transaction occurred
3. Encourages future cooperation

Since the sender sent 60% of their endowment (a strong sign of trust), I should return a meaningful amount. A return of around \$4-5 would:

- Give the sender more than they started with ($\$2 + \$4-5 = \$6-7$ vs \$5 initial)
- Leave me with \$4-5 (better than the \$0 I started with)
- Signal I'm reciprocating their cooperation

I could aim for equal final payoffs at \$4.5 each, but another approach is to return roughly what they sent or slightly more as a trust reward. The sender sent \$3, so returning \$3-4 would acknowledge their initial investment while still benefiting me. I'll go with \$4.5 to create balanced outcomes where we both gain from the exchange.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 875,
  "reasoning_tokens": 531
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:
- I'm now the sender with a new opponent (Briar)

- In Round 1, Hana sent me \$3 (60%) and I reciprocated fairly
- This is an early round, so building cooperation is valuable

I'll send \$3 (60% of my endowment), mirroring the trust that Hana showed me in Round 1. This signals willingness to cooperate while testing Briar's reciprocity.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm the SENDER playing against Briar (Agent_2).

Let me think about this strategically:

- This is Round 2 of an iterated game
- I'm playing with a new opponent (Briar), not Hana
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- Briar will decide how much to return

In Round 1:

- Hana sent me \$3 (60% of their endowment), showing strong trust
- I returned \$4.5, which gave fairly equal payoffs

Now I need to decide what to send to Briar. Some considerations:

1. This is early in the game, so establishing cooperation could be beneficial for future rounds
2. I don't know Briar's strategy yet
3. Sending a reasonable amount shows trust and may encourage reciprocity
4. If I send nothing, I guarantee keeping \$5 but miss opportunities for mutual gain
5. If I send too much and they don't return enough, I lose out

A moderate to moderately-high amount seems reasonable to:

- Signal cooperation without being overly risky
- Test Briar's reciprocity
- Build trust for potential future interactions

I'll send \$3, which is 60% - the same proportion that Hana sent to me. This:

- Shows I'm willing to cooperate at a similar level
- Gives Briar \$9 to work with
- Leaves me with \$2 initially
- Demonstrates trust while not being maximally risky

This seems like a balanced approach for Round 2.

usage:

```
{
  "input_tokens": 892,
  "output_tokens": 492,
  "reasoning_tokens": 352
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

- Round 1: Hana sent me \$3 (60%), I returned \$4.5 (fair reciprocity)
- Round 2: I sent Briar \$3 (60%), they returned \$4.5 (good reciprocity)

The 60% send with fair returns strategy has been working well. I'll continue this cooperative approach with Cyra, a new opponent.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

- Round 2: I sent Briar \$3, they received \$9, and returned \$4.5 to me
- My payoff in Round 2 was \$2 (what I kept) + \$4.5 (returned) = \$6.5
- Briar kept \$9 - \$4.5 = \$4.5
- This was good reciprocity from Briar - they returned exactly half of what they received

Now I'm facing Cyra (a new opponent) as sender again.

Pattern so far:

- Hana sent me \$3 (60%) ? I returned \$4.5
- I sent Briar \$3 (60%) ? They returned \$4.5

Both interactions showed good reciprocity with 60% sends and 50% returns of the multiplied amount.

Since cooperation has been working well, I'll continue the strategy of sending \$3 (60%) to Cyra. This maintains the cooperative pattern and tests whether Cyra will also reciprocate.

```
Usage: {
  "input_tokens": 1158,
  "output_tokens": 355,
  "reasoning_tokens": 195
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T15:55:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_5",
      "Agent_3"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_3": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplication
- This is the first round of an iterated game, so building trust is important

Aster currently has \$2.0 left (\$5 - \$3 sent). A fair return should:

1. Reward their trust
2. Ensure both players benefit
3. Encourage future cooperation

If I return \$4.5:

- Aster's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)
- My total: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This seems fair and reciprocal given their strong show of trust.

{'return': 4.5}

reasoning:

Let me understand the situation:

1. I am the RECEIVER (Cyra)
2. My opponent is Aster (Agent_1) who is the SENDER
3. Aster sent \$3.0 out of their \$5.0 endowment (60%)
4. This \$3.0 was multiplied by 3x, so I received \$9.0
5. I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

- Aster showed trust by sending \$3.0 (60% of their endowment)
- If I return nothing, I keep \$9.0 and Aster keeps \$2.0 (\$5 - \$3 sent)
- If I return some amount, I'm showing reciprocity and building trust for future rounds
- Since this is an iterated game, building trust could benefit me in future rounds

What's a fair return?

- Aster invested \$3.0 and ended up with \$2.0 so far
- I received \$9.0
- A typical reciprocal strategy might be to ensure Aster gets at least their initial \$5.0 back, or slightly more to reward their trust
- If I return \$3.0, Aster would have: \$2.0 (kept) + \$3.0 (returned) = \$5.0 total (break even)
- If I return \$4.5, Aster would have: \$2.0 + \$4.5 = \$6.5 total (profit of \$1.5)
- I would keep: \$9.0 - \$4.5 = \$4.5

Let me consider various options:

1. Return \$0: I keep \$9, Aster gets \$2 total. Very unfair to Aster.
2. Return \$3: I keep \$6, Aster gets \$5 total. Fair - Aster breaks even.
3. Return \$4: I keep \$5, Aster gets \$6 total. Both profit equally.
4. Return \$4.5: I keep \$4.5, Aster gets \$6.5 total. Both profit, but Aster gets slightly more.
5. Return \$5: I keep \$4, Aster gets \$7 total. Aster profits more than me.

Given Aster's substantial trust in sending 60% of their endowment, and since we're in round one of a repeated game, I should reciprocate generously to establish cooperation. Returning \$4.5 to \$5.0 seems appropriate to reward their trust while still keeping a reasonable share.

I'll go with \$4.5 as the middle ground-it gives Aster a \$1.5 profit, matches my own gain at \$4.5, and signals I'm a reliable partner for future rounds. This balances rewarding their trust with maintaining a fair outcome for myself.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1017,
  "reasoning_tokens": 523
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Dorian
- In Round 1, Aster showed strong trust by sending me 60% (\$3), and that worked out well for both of us
- In an iterated game, establishing cooperation early is valuable

I'll mirror the trust Aster showed me and send \$3 (60%). This gives Dorian \$9 after multiplication, creating good potential for mutual benefit if they reciprocate.

```
{'send': 3.0}
```

reasoning:

Now I'm the sender against a new opponent, Dorian.

This is Round 2, and I'm playing with a different person than Round 1. Let me think about the strategy:

1. In Round 1, I was receiver and Aster sent me \$3 (60%). I returned \$4.5, which was a fair and generous return.
2. Now I'm the sender with Dorian. This is the first interaction with Dorian.
3. In an iterated game with multiple rounds and potentially repeated interactions, I need to:
 - Build trust with new partners
 - Show willingness to cooperate
 - Set a cooperative tone
4. Aster sent 60% (\$3), which was quite trusting. That worked out well for both of us.
5. For a first interaction with Dorian, I should show trust but perhaps be slightly cautious since I don't know their playing style.

Options:

- Send \$2.5 (50%) - moderate trust, balanced approach
- Send \$3 (60%) - mirror what Aster did to me, strong trust signal
- Send \$2 (40%) - cautious but still cooperative

I think sending \$3 (60%) makes sense because:

- It mirrors the trust Aster showed me
- It creates strong potential for cooperation
- In an iterated game, being generous early can establish good relationships
- The receiver gets \$9 and has plenty of room to return a good amount

If Dorian returns even \$4-5, we'll both be better off than if I sent less.

Let me go with \$3.

usage:

```
{
  "input_tokens": 914,
  "output_tokens": 482,
  "reasoning_tokens": 352
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after multiplication
- Elara currently has \$2.0 left

A pattern has emerged in my previous rounds:

- When someone sends \$3, returning \$4.5 has worked well for both parties
- This maintains cooperation and mutual benefit

Following this established pattern:

```
{'return': 4.5}
```

Reasoning:

Let me analyze this situation:

Round 3:

- I'm the RECEIVER
- Elara (the sender) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after 3x multiplication

- Elara currently has \$2.0 left (\$5 - \$3)

Recent history:

- Round 1: Aster sent me \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2: I sent Dorian \$3, they received \$9, returned \$4.5. My payoff: \$6.5

There's a clear pattern emerging where:

- Sending \$3 (60%) seems to be a cooperative norm
- Returning \$4.5 (50% of the received \$9) has been the reciprocal norm
- This leads to good outcomes for both parties

Elara sent \$3.0, showing the same level of trust as Aster and what I showed to Dorian.

If I return \$4.5:

- Elara's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)
- My total: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This is consistent with the pattern that has emerged and seems fair. I should maintain this cooperative equilibrium.

Usage: {

```
"input_tokens": 1185,  
"output_tokens": 447,  
"reasoning_tokens": 244
```

}

ROUND 4

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_4 (Dorian)=Receiver
dyad_2: Agent_2 (Briar)=Sender, Agent_8 (Hana)=Receiver
dyad_3: Agent_3 (Cyra)=Sender, Agent_7 (Galen)=Receiver
dyad_4: Agent_6 (Finn)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$4; received \$12; Agent_4 (Dorian) returned \$6; payoffs Agent_1 (Aster)=\$7, Agent_4 (Dorian)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_2 (Briar) sent \$4; received \$12; Agent_8 (Hana) returned \$6; payoffs Agent_2 (Briar)=\$7, Agent_8 (Hana)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_3 (Cyra) sent \$3; received \$9; Agent_7 (Galen) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_7 (Galen)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_6 (Finn) sent \$3; received \$9; Agent_5 (Elara) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_5 (Elara)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 24.0,  
"Agent_2": 24.0,  
"Agent_3": 22.0,  
"Agent_4": 24.5,  
"Agent_5": 22.0,  
"Agent_6": 23.5,  
"Agent_7": 24.5,  
"Agent_8": 23.5
```

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T15:55:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_1",
      "Agent_4"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.

- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.

Your opponent this round is Cyra (Agent_3).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while also being somewhat cautious since I don't yet know how Cyra plays.

The trust game rewards cooperation - if I send money, it triples, creating surplus value we can share. However, I'm also taking a risk that needs to be reciprocated.

For this opening round, I'll send a moderate amount that signals willingness to trust and cooperate, while not being overly exposed.

```
'''json
{"send": 3}
'''
```

This gives Cyra \$9 to work with, which creates good surplus if they choose to share fairly. I keep \$2 as insurance, and if Cyra returns a reasonable amount (say \$4-5), we both benefit from the cooperation.

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender with an agent named Cyra.

Let me think about the strategic considerations:

1. This is an iterated game with multiple rounds
2. I'll be playing with different partners across rounds
3. Building a reputation for cooperation might encourage others to return more
4. In the first round, I need to establish my strategy

As the sender:

- I have \$5
- I can send \$0-\$5
- Whatever I send gets multiplied by 3x
- Cyra then decides how much to return
- I keep: \$5 - sent + returned

From a purely selfish perspective in a one-shot game, sending \$0 maximizes my guaranteed payoff. However, this is an iterated game where:

- I may play with Cyra again in future rounds
- Other agents might learn about my behavior
- Cooperation can lead to better long-term outcomes

If I send \$5:

- Cyra receives \$15
- If Cyra returns even \$5, we both benefit (\$5 for me, \$10 for Cyra)
- If Cyra returns \$7.50, we both get \$7.50
- If Cyra returns more, I come out ahead

A moderate opening move like sending \$3 or \$4 shows willingness to trust while not being overly naive:

- Send \$3: Cyra gets \$9, I keep \$2 initially
 - If Cyra returns \$3, I end with \$5, Cyra ends with \$6
 - If Cyra returns \$4, I end with \$6, Cyra ends with \$5
- Send \$5: Cyra gets \$15, I keep \$0 initially
 - If Cyra returns \$5, I end with \$5, Cyra ends with \$10
 - If Cyra returns \$7.50, we both end with \$7.50

Given this is the first round and I want to signal cooperation while gathering information about how others play, I'll send \$3. This shows trust and willingness to cooperate, but isn't so high that I risk losing everything if Cyra doesn't reciprocate.

Sending around half my endowment seems to be the standard approach in trust game experiments for initiating cooperation. \$3 represents that threshold-it demonstrates genuine trust without overcommitting, so I'll go with that.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 739,
  "reasoning_tokens": 511
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Galen sent me \$4 (keeping \$1), which tripled to \$12 for me
- In the previous round, Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair
- Following a similar principle of returning about 50% seems reasonable

If I return \$6:

- Galen ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This rewards Galen for taking the risk to send a substantial amount while still being cooperative. It's consistent with how Cyra treated me.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to Cyra, they received \$9, and they returned \$4.5 to me. That's a 50% return of what they received, which is quite fair and cooperative.
2. Now in Round 2, I'm the receiver. Galen sent me \$4, so I received \$12.

Let me think about what's fair:

- Galen kept \$1 and sent \$4
- I received \$12
- If I return nothing, Galen gets \$1 total and I get \$12
- If I return \$6 (50% of what I received), Galen gets \$7 total and I get \$6
- If I return \$5.5, we'd both end up with \$6.5 (Galen: \$1 + \$5.5 = \$6.5, Me: \$12 - \$5.5 = \$6.5)

Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair and cooperative.
I should probably follow a similar principle.

Actually, let me think about this more carefully. The game creates surplus - Galen gave up \$4 to give me \$12, creating \$8 of surplus. A fair split might be:

- Galen should end up with: \$1 (kept) + some return
- I should end up with: \$12 (received) - that return

If we split evenly from the starting point:

- We both started conceptually with access to \$5
- After the multiplication, there's \$1 + \$12 = \$13 total
- Split evenly would be \$6.5 each

- So I should return \$6.5, giving Galen \$1 + \$6.5 = \$7.5, and I keep \$5.5

Wait, that doesn't quite work out. Let me recalculate:

- If I return \$6.5: Galen gets \$1 + \$6.5 = \$7.5, I get \$12 - \$6.5 = \$5.5

- Total: \$7.5 + \$5.5 = \$13 ?

Returning 50% like Cyra did would give me slightly more (\$6 vs \$5.5), which seems reasonable since I'm taking no risk as the receiver. I could aim for equal payoffs or proportional sharing, but the 50% return feels like the right balance between fairness and reciprocating the cooperation I received. I need to reconsider how to split the gains. The cooperation created \$8 in surplus (\$12 received minus \$4 sent). If I divide this equally between us, Galen gets their \$4 back plus \$4 of the surplus for \$8 total, meaning I return \$8 and keep \$4 for myself.

But compared to just splitting the final amount evenly, this feels less advantageous to me. I think returning \$6-half of what I received-strikes the right balance. It's straightforward, rewards Galen for taking the initial risk, and still shows I'm willing to cooperate meaningfully. That leaves Galen with \$7 and me with \$6. I'll go with \$6.

usage:

```
{
  "input_tokens": 856,
  "output_tokens": 1327,
  "reasoning_tokens": 602
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$12.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- This is the same amount I sent to Cyra in Round 1, and they returned \$4.5 (50%)
- I returned 50% to Galen in the previous round

Consistency seems important for building trust in repeated games. Following the 50% return pattern:

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is fair and encourages future cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- In the last round, Galen sent me \$4 and I returned \$6 (50% of what I received)
- In Round 1, I sent \$3 to Cyra, they received \$9 and returned \$4.5 (50%)

There's a pattern emerging of returning about 50% of what is received, which seems fair and encourages cooperation.

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is consistent with the 50% return pattern and similar to what Cyra did with my \$3 send in Round 1. It rewards Hana for trusting me while maintaining fairness.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 414,
  "reasoning_tokens": 171
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the patterns so far:

- When I sent \$3 to Cyra, they returned 50% (\$4.5)
- When Galen sent \$4 to me, I returned 50% (\$6)
- When Hana sent \$3 to me, I returned 50% (\$4.5)

There's been a consistent cooperative pattern with 50% returns. Since I haven't played with Dorian before, I'll show trust while following the cooperative trend I've observed.

Sending \$4:

- Dorian receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

This is slightly more trusting than my initial \$3 to Cyra, reflecting the positive cooperative experiences so far.

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me analyze my situation:

Round 1: I was SENDER vs Cyra

- Sent \$3, they got \$9, returned \$4.5
- My payoff: \$2 + \$4.5 = \$6.5

Round 2: I was RECEIVER vs Galen

- They sent \$4, I got \$12, returned \$6
- My payoff: \$6

Round 3: I was RECEIVER vs Hana

- They sent \$3, I got \$9, returned \$4.5
- My payoff: \$4.5

Total so far: \$17

Now Round 4: I'm SENDER vs Dorian (new opponent)

Looking at the pattern:

- When I sent \$3, Cyra returned \$4.5 (50%)
- When Galen sent \$4, I returned \$6 (50%)
- When Hana sent \$3, I returned \$4.5 (50%)

So there's been a pattern of 50% returns, which is quite cooperative.

As a sender, my best strategy depends on what I expect Dorian to return. If they follow the 50% pattern:

- If I send \$0: I keep \$5, payoff = \$5
- If I send \$1: I keep \$4, they get \$3, if return 50% = \$1.5, payoff = \$5.5
- If I send \$2: I keep \$3, they get \$6, if return 50% = \$3, payoff = \$6
- If I send \$3: I keep \$2, they get \$9, if return 50% = \$4.5, payoff = \$6.5
- If I send \$4: I keep \$1, they get \$12, if return 50% = \$6, payoff = \$7
- If I send \$5: I keep \$0, they get \$15, if return 50% = \$7.5, payoff = \$7.5

- If they return 50%, sending more is always better (up to sending everything).

I don't know Dorian yet, but the cooperative trend suggests I should escalate slightly. Galen sent \$4 to me, demonstrating confidence in reciprocation. I'll match that trust and send \$4 as well, signaling I'm willing to build on the mutual cooperation we've established.

```
Usage: {  
  "input_tokens": 1507,  
  "output_tokens": 811,  
  "reasoning_tokens": 396  
}
```

GAME - Agent_4 - Receiver - dyad_1

Timestamp: 2026-06-04T15:55:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_4": "Dorian"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_4"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending money creates value (3x multiplier), and if Galen reciprocates fairly, we can both benefit more than if we don't trust each other.

I'll send a substantial amount to signal trust and willingness to cooperate, while keeping a small amount as insurance.

{'send': 4}

reasoning:

This is the first round of a multi-round iterated trust game. I'm the sender, and I need to decide how much to send to Galen.

Let me think about the game mechanics:

- I receive \$5
- I can send \$0-\$5 to Galen
- Whatever I send gets tripled
- Galen decides how much to return
- I keep: \$5 - sent + returned
- Galen keeps: received - returned

This is a classic trust game. The key considerations are:

1. If I send money, it gets tripled, which increases total welfare
2. But I'm vulnerable - Galen could return nothing
3. This is an iterated game with multiple rounds
4. Building trust early might lead to better outcomes later
5. There are 8 agents total, and pairings are randomized, so I may or may not play with Galen again

Strategic considerations:

- Sending \$0 guarantees I keep \$5, but no growth
- Sending all \$5: Galen receives \$15, could return enough to make us both better off
- Sending a moderate amount shows some trust while limiting risk

Since this is round 1 and the first interaction, and it's an iterated game where reputatio

n/reciprocity might matter:

- I should send a meaningful amount to signal cooperation
- But not necessarily all \$5 since I haven't established trust yet
- A common strategy is to send about half to signal willingness to cooperate

I'll send \$3. This shows trust and good faith, creates \$9 for Galen, and if Galen returns even \$4-5, we'd both be better off than the no-trust outcome. This leaves me with \$2 initially, and hopefully Galen will reciprocate.

Actually, let me reconsider. In iterated trust games, starting with strong cooperation often leads to better long-term outcomes. Many successful strategies involve:

- Being cooperative initially
- Reciprocating the opponent's behavior in subsequent rounds

- Establishing a cooperative equilibrium early

Since this is explicitly multi-round, I should signal stronger cooperation from the start.

Sending \$4 demonstrates real commitment, gives Galen \$12 to work with, and if they reciprocate with \$5-6, we both come out ahead. This sets a collaborative tone for future interactions.

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 663,
  "reasoning_tokens": 548
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3, showing trust and cooperation
- I received \$9 (tripled)
- Cyra currently has \$2 left

In the previous round, Galen returned exactly half of what they received (\$6 out of \$12), which felt fair. I'll follow a similar principle of reciprocity.

Returning half (\$4.5) gives:

- Cyra: \$2 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust while keeping some benefit from the multiplier effect.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was sender, sent \$4 to Galen
- Galen received \$12 (4×3)
- Galen returned \$6 to me
- My payoff: \$5 - \$4 + \$6 = \$7
- Galen's payoff: \$12 - \$6 = \$6

So Galen returned exactly half of what they received, which seems fair. They made \$6 and I made \$7, so we both benefited from cooperation.

Round 2:

- I am receiver
- Cyra sent me \$3
- I received \$9 (3×3)
- Cyra kept: $\$5 - \$3 = \$2$
- I need to decide how much to return

If I follow a similar fairness principle as Galen did (returning about half), I would return around \$4.5.

But let me think about what would be fair:

- Cyra invested \$3 out of their \$5
- This created \$9 for me
- If we split the gains equally:
 - Cyra should end up with: $\$2$ (kept) + some return
 - I should end up with: $\$9$ - some return

One approach: Split the total surplus equally.

- Total created: $\$5$ (Cyra's initial) + $\$9$ (tripled amount) = $\$14$
- Equal split: $\$7$ each
- Cyra has $\$2$, needs $\$5$ more
- I have $\$9$, should give $\$5$, keeping $\$4$

Another approach: Return half of what I received (like Galen did)

- Return $\$4.5$
- Cyra gets: $\$2 + \$4.5 = \$6.5$
- I keep: $\$9 - \$4.5 = \$4.5$

Another approach: Ensure we both profit equally from the exchange

- The gain from cooperation is $\$6$ (the $\$9$ minus Cyra's $\$3$ investment)
- If we split this evenly, that's $\$3$ each
- Cyra ends with $\$2 + \4 return = $\$6$, and I keep $\$5$

Returning $\$4.5$ mirrors what Galen did and signals I'm a reciprocal player, so I'll go with that.

usage:

```
{
  "input_tokens": 778,
  "output_tokens": 766,
  "reasoning_tokens": 407
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:
- Round 1 with Galen: I sent \$4, received \$6 back, earned \$7
- Round 2 with Cyra: They sent \$3, I returned \$4.5 (half)

The cooperative strategy has been working well. Sending \$4 created good value and Galen reciprocated fairly. I'll continue this approach with Finn, signaling trust and willingness to cooperate.

```
{'send': 4}
```

reasoning:

Let me review my history:

- Round 1: I was SENDER vs Galen. Sent \$4, they got \$12, returned \$6. I got \$7.
- Round 2: I was RECEIVER vs Cyra. They sent \$3, I got \$9, returned \$4.5. I got \$4.5.
- Total earnings: \$11.5

Now Round 3: I'm SENDER against Finn (new opponent).

My strategy so far has been cooperative - sending substantial amounts as sender to create value. When Galen sent \$4, they returned \$6 (half). When Cyra sent \$3, I returned \$4.5 (half).

This seems to be working well. Both exchanges were mutually beneficial compared to not sending anything.

For this round with Finn, I should continue the cooperative strategy. Sending \$4 worked well with Galen. I could send the same amount or try sending the full \$5 to maximize potential gains if Finn is also cooperative.

Let me think about the trade-offs:

- Send \$5: Finn gets \$15. If they return half (~\$7.5), I get \$7.5 total.
- Send \$4: Finn gets \$12. If they return half (\$6), I get \$7 total (like last time).
- Send \$3: Finn gets \$9. If they return half (\$4.5), I get \$6.5 total.

Since this is an iterated game and I'm building a reputation, sending a high amount signals cooperation. The previous round went well, so I'll continue with \$4 as a strong but not maximal signal of trust.

Actually, let me reconsider. I sent \$4 to Galen and got \$6 back, netting \$7. That worked well. But I could try \$5 to see if Finn is willing to be more generous and potentially get even better returns.

I think I'll stick with \$4 as it's worked well and shows strong trust without being all-in.

usage:

```
{
  "input_tokens": 1076,
  "output_tokens": 582,
  "reasoning_tokens": 404
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern from previous rounds:

- Aster sent me \$4, which is the same amount I've been sending
- I received \$12 (tripled)
- Previous players have consistently returned about half when receiving similar amounts

Following the reciprocal fairness norm that's been established:

- Returning \$6 gives Aster \$7 total (\$1 + \$6)
- I keep \$6

This rewards Aster's trust while maintaining the cooperative pattern.

```
{'return': 6}
```

Reasoning:

Let me analyze the situation:

- Aster sent me \$4, showing strong trust and cooperation
- I received \$12 (tripled amount)
- Aster currently has \$1 left

Looking at the pattern from previous rounds:

- When I sent \$4 to Galen, they returned \$6 (half of \$12)
- When Cyra sent \$3 to me, I returned \$4.5 (half of \$9)
- When I sent \$4 to Finn, they returned \$6 (half of \$12)

There's a consistent pattern of returning half of what was received, which seems fair and maintains cooperation. This gives both players a net benefit from the transaction.

If I return \$6:

- Aster: $\$1 + \$6 = \$7$
- Me: $\$12 - \$6 = \$6$

This is fair and maintains the cooperative norm. Aster sent a large amount (\$4), and returning half rewards their trust appropriately.

```
Usage: {  
  "input_tokens": 1345,  
  "output_tokens": 349,  
  "reasoning_tokens": 196  
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-06-04T15:55:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_8"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship with Finn since this is an iterated game.

Sending money creates value through the 3x multiplier - if I send \$3, Finn receives \$9. If Finn returns a fair share, we can both benefit more than if I kept everything.

I'll send a substantial but not maximum amount to signal trust and willingness to cooperate, while leaving room to learn about Finn's reciprocity.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of an iterated trust game. I'm playing as the sender (Briar) and my opponent is Finn (Agent_6).

Let me think about the strategic considerations:

As the sender:

- I have \$5
- I can send between \$0 and \$5
- Whatever I send gets multiplied by 3x
- Finn decides how much to return
- I keep: \$5 - sent + returned
- Finn keeps: (3 × sent) - returned

This is a classic trust game scenario. The key considerations are:

1. This is an iterated game with multiple rounds
2. Building trust early could lead to mutual cooperation
3. If I send more, Finn receives more (3x multiplier), creating surplus
4. If Finn returns fairly, we can both benefit from the multiplier effect

First round strategies:

- Sending \$0 is safe but generates no surplus - guaranteed \$5 for me
- Sending some amount signals willingness to cooperate and trust
- Sending the full \$5 maximizes the total surplus (Finn gets \$15) but requires trust that

Finn will return something fair

Since this is the first round and we're in an iterated setting:

- I should signal cooperation to establish a pattern
- A moderate amount might be good to test the waters
- If Finn reciprocates fairly, we can increase cooperation in future rounds

Common first-round strategies:

1. Send half (\$2.50) - shows willingness to cooperate but not maximum vulnerability
2. Send everything (\$5) - shows maximum trust, maximizes surplus
3. Send a smaller amount (\$1-2) - conservative trust-building

Given that this is an iterated game and building trust early could lead to better long-term outcomes, I'll send a moderate to generous amount. Let me send \$3.

If I send \$3:

- Finn receives \$9
- I keep \$2 initially
- If Finn returns \$5, we each end with \$7 total

- This demonstrates meaningful trust while gauging their reciprocity level

I'm leaning toward \$3 as my opening move-it's a 60% commitment that signals genuine cooperation without leaving me completely exposed.

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 738,
  "reasoning_tokens": 518
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

Elara sent me \$3.0, showing trust and cooperation - the same amount I sent to Finn in Round 1. Finn returned \$4.5 to me (half of the \$9 received), which felt fair and cooperative.

I should reciprocate this cooperative behavior with Elara. Returning half (\$4.5) means:

- Elara gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This splits the gains fairly and establishes me as a trustworthy partner.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was SENDER against Finn
- I sent \$3.0
- Finn received \$9.0
- Finn returned \$4.5

- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received, which is a pretty fair and cooperative split. I gained \$1.5 more than my initial \$5, and Finn ended with \$4.5.

Round 2 (current):

- I am RECEIVER against Elara
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara made the same choice I did in Round 1 - sending \$3.0. This shows trust and willingness to cooperate.

Given that Finn returned \$4.5 (half) to me when I sent \$3.0, I should consider:

1. Being reciprocal and fair
2. Establishing a cooperative reputation for future rounds
3. The fact that this is an iterated game with multiple agents

If I return \$4.5 (half of what I received), then:

- Elara gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and seems fair. I could return a bit more or less, but \$4.5 seems like a reasonable, cooperative amount that:

- Shows I'm trustworthy
- Could encourage cooperation in future rounds
- Is similar to what Finn did to me

I'll return \$4.5.

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 593,
  "reasoning_tokens": 334
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

Galen sent me \$4.0, which is more generous than what I've seen so far. They're showing strong trust.

Following the reciprocity pattern established in previous rounds (where returning about half has worked well), I'll return \$6:

- Galen gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This rewards their generosity and maintains cooperative norms.

```
'''json
{"return": 6}
'''
```

reasoning:

Galen sent me \$4.0, which became \$12.0 for me. This is a generous amount, showing strong trust and cooperation.

Let me think about fair reciprocity:

- Galen sent \$4, so they have $\$5 - \$4 = \$1$ left
- I received \$12

If I return \$6 (half of what I received):

- Galen gets: $\$1 + \$6 = \$7$
- I get: $\$12 - \$6 = \$6$

This seems fair and reciprocates their generosity. It's proportional to what they sent.

Alternatively, I could aim for more equal final payoffs. If I return \$5.5:

- Galen gets: $\$1 + \$5.5 = \$6.5$
- I get: $\$12 - \$5.5 = \$6.5$

This creates equal payoffs, which might be seen as very fair.

I think returning \$6 (half) is the most consistent with the pattern established so far - Finn returned half to me when I sent \$3, and I returned half to Elara when they sent \$3. This establishes a clear cooperative norm.

Since Galen sent more (\$4 vs \$3), returning half (\$6) rewards their greater trust and maintains the reciprocity norm.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 434,
  "reasoning_tokens": 249
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern so far:

- Cooperation has been strong with people returning about 50% of what they receive
- When Galen sent me \$4, I returned \$6, giving them a good payoff
- Sending more creates more total value when reciprocity is expected

I'll send \$4 to Hana, showing trust in continued cooperation:

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me review what I've learned so far:

Round 1: I was SENDER vs Finn

- I sent \$3, they got \$9, they returned \$4.5
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn kept: $\$9 - \$4.5 = \$4.5$

Round 2: I was RECEIVER vs Elara

- They sent \$3, I got \$9, I returned \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$
- Elara got: $\$5 - \$3 + \$4.5 = \6.5

Round 3: I was RECEIVER vs Galen

- They sent \$4, I got \$12, I returned \$6

- My payoff: $\$12 - \$6 = \$6$
- Galen got: $\$5 - \$4 + \$6 = \7

Now Round 4: I'm SENDER vs Hana (new opponent)

The pattern I've seen:

- When I sent \$3, I got back \$4.5 (50% return)
- When Elara sent \$3, I returned \$4.5 (50% return)
- When Galen sent \$4, I returned \$6 (50% return)

Everyone seems to be cooperating well with approximately 50% returns. This creates mutual benefit:

- If I send \$3, they get \$9, return ~\$4.5, I end with \$6.5
- If I send \$4, they get \$12, return ~\$6, I end with \$7
- If I send \$5, they get \$15, return ~\$7.5, I end with \$7.5

Sending more generally leads to higher payoffs for both parties when the other person reciprocates fairly. Since this is an iterated game and cooperation has been working well, I should continue being generous.

I'll send \$4 to Hana, mirroring what Galen trusted me with and demonstrating the same confidence in reciprocation.

```
Usage: {
  "input_tokens": 1382,
  "output_tokens": 584,
  "reasoning_tokens": 344
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T15:56:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_8": "Hana"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_8"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Elara (Agent_5).

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative relationship since this is an iterated game. Sending a meaningful amount signals trust and creates potential for mutual benefit through the 3x multiplier.

I'll send **\$3**.

This means:

- Elara will receive \$9 (3 × \$3)
- I'm keeping \$2 initially
- If Elara returns a fair share, we can both benefit from the cooperation

Let's see how this goes!

```
'''json
{"send": 3}
'''
```

reasoning:

I'm playing as the sender in the first round. Let me think about this strategically.

The trust game dynamics:

- I get \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver decides how much to return to me
- I want to maximize my total payoff

Key considerations:

1. This is Round 1, so there's no history yet

2. This is an iterated game, so building trust could lead to better outcomes over multiple rounds
3. The multiplication factor (3x) means there's potential for mutual gain if the receiver is trustworthy
4. If I send nothing, I guarantee keeping \$5 but miss potential gains
5. If I send everything and the receiver returns a fair share, we both benefit

Common strategies:

- Send a moderate amount (like \$2-3) to signal trust while not risking everything
- Send everything to maximize potential joint gains (risky but cooperative)
- Send nothing to protect myself (but this doesn't build trust)

Since this is the first round and an iterated game, I think sending a meaningful but not maximum amount makes sense. This signals willingness to cooperate while gauging the receiver's trustworthiness.

Let me send \$3:

- I keep: \$2 initially
- Receiver gets: $\$3 \times 3 = \9
- If receiver returns \$4, I end with \$6, receiver ends with \$5 (fair split of the total \$11)
- If receiver returns \$3, I end with \$5, receiver ends with \$6 (equally good for both)

This seems like a reasonable opening move that signals cooperation without being too risky

usage:

```
{
  "input_tokens": 530,
  "output_tokens": 508,
  "reasoning_tokens": 394
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Finn sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, when I sent \$3, Elara returned \$4.5 (50%), which felt fair
- I want to reciprocate that fairness to build cooperation

I'll return **\$4.5** (50% of what I received).

This gives:

- Finn: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the sender, sent \$3 to Elara
- Elara received \$9 and returned \$4.5
- My payoff was \$2 (kept) + \$4.5 (returned) = \$6.5

- Elara kept $\$9 - \$4.5 = \$4.5$

So Elara returned exactly 50% of what they received, which seems fair.

Round 2:

- I'm now the receiver
- Finn sent me \$3 (same amount I sent in Round 1)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

Since this is an iterated game and I want to establish cooperation:

- Finn trusted me by sending \$3
- Elara returned 50% (\$4.5) when I sent \$3, which seemed fair
- I should reciprocate with a similar level of fairness

If I return \$4.5:

- I keep $\$9 - \$4.5 = \$4.5$
- Finn ends up with $\$5 - \$3 + \$4.5 = \6.5
- We both get the same payoff (\$6.5 for Finn, \$4.5 for me in this round)

Actually, wait. Let me recalculate:

- Finn started with \$5
- Finn sent \$3
- Finn has \$2 left
- I return some amount X
- Finn's total: $\$2 + X$

If I return \$4.5:

- Finn's total: $\$2 + \$4.5 = \$6.5$
- My total: $\$9 - \$4.5 = \$4.5$

This mirrors what Elara did for me. It's a fair 50% return and signals that I'm willing to cooperate. This should encourage future cooperation.

I'll return \$4.5.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 621,
  "reasoning_tokens": 328
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern so far:
- When I sent \$3 to Elara, they returned \$4.5 (fair split)
- When Finn sent \$3 to me, I returned \$4.5 (fair split)

This cooperative approach has been working well. I'll continue with the same strategy and send **\$3** to Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my history so far:

Round 1: I was SENDER vs Elara

- I sent \$3
- Elara received \$9
- Elara returned \$4.5
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Elara's payoff: $\$9 - \$4.5 = \$4.5$

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9
- I returned \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$
- Finn's payoff: $\$5 - \$3 + \$4.5 = \6.5

So far, the pattern has been:

- When I sent \$3, I got back \$4.5 (which is 50% of the \$9)
- When Finn sent \$3, I returned \$4.5 (50% of the \$9)

This seems like a fair and cooperative pattern. Both times the sender ended up with \$6.5 and the receiver with \$4.5.

Now I'm facing Aster as a SENDER. This is a new opponent, and I want to continue establishing cooperative relationships. Given that \$3 worked well in Round 1 and Finn also sent \$3 in Round 2, this seems like a reasonable amount to signal cooperation.

I should send \$3 again to:

1. Continue the cooperative pattern
2. Signal trust to Aster
3. See if they reciprocate fairly

Let me send \$3.

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 472,
  "reasoning_tokens": 270
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent me \$4, which is even more trusting than the \$3 pattern I've seen
- I received \$12 after the 3x multiplier
- I've been maintaining a fair 50% return policy

I'll return **\$6** (50% of what I received).

This gives:

- Briar: $\$1 \text{ (kept)} + \$6 \text{ (returned)} = \$7 \text{ total}$
- Me: $\$12 - \$6 = \$6 \text{ total}$

```
'''json
{"return": 6}
'''
```

Reasoning:

Looking at the situation:

- Briar sent me \$4, showing significant trust (80% of their endowment)
- I received \$12 after the 3x multiplier
- The pattern so far has been sending \$3 and returning about 50% (which is \$4.5 from \$9)

Briar sent more (\$4 vs \$3), so they're being extra trusting. I should reciprocate proportionally.

If I keep the same 50% return rate: $\$12 \times 0.5 = \6

This would give:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

Actually, let me think about fairness differently. Briar started with \$5 and sent \$4, keeping only \$1.

After 3x, I have \$12 total.

If we want roughly equal outcomes:

- Total money in the system: \$5 (Briar's original) + \$12 (what I received) = \$17 total value created
- Equal split would be around \$8.5 each

For Briar to get \$8.5: they have \$1 kept, so need \$7.5 back

For me to get \$8.5: I need to keep \$12 - \$7.5 = \$4.5... wait that doesn't add up.

Let me recalculate:

- Briar starts with \$5, sends \$4, keeps \$1
- I receive $\$4 \times 3 = \12
- If I return \$X:
 - Briar ends with: $\$1 + \X
 - I end with: $\$12 - \X

For equal outcomes: $\$1 + \$X = \$12 - \X

$\$1 + 2X = \12

$\$X = \5.5

Hmm, but that's less than the 50% I've been doing.

The 50% approach feels right-it's consistent with what's worked before and honors their larger contribution. I'll return \$6.

```
Usage: {
  "input_tokens": 1357,
  "output_tokens": 636,
  "reasoning_tokens": 360
}
```

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-06-04T15:56:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_7"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
```



```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Aster (Agent_1).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Aster showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplication
- This is the first round of an iterated game, so building trust is important

Aster currently has \$2.0 left (\$5 - \$3 sent). A fair return should:

1. Reward their trust
2. Ensure both players benefit
3. Encourage future cooperation

If I return \$4.5:

- Aster's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)

- My total: $\$9.0 - \$4.5 = \$4.5$ (profit of $\$4.5$)

This seems fair and reciprocal given their strong show of trust.

```
{'return': 4.5}
```

reasoning:

Let me understand the situation:

1. I am the RECEIVER (Cyrus)
2. My opponent is Aster (Agent_1) who is the SENDER
3. Aster sent $\$3.0$ out of their $\$5.0$ endowment (60%)
4. This $\$3.0$ was multiplied by 3x, so I received $\$9.0$
5. I need to decide how much to return (between $\$0$ and $\$9.0$)

Let me think about this strategically:

- Aster showed trust by sending $\$3.0$ (60% of their endowment)
- If I return nothing, I keep $\$9.0$ and Aster keeps $\$2.0$ ($\$5 - \3 sent)
- If I return some amount, I'm showing reciprocity and building trust for future rounds
- Since this is an iterated game, building trust could benefit me in future rounds

What's a fair return?

- Aster invested $\$3.0$ and ended up with $\$2.0$ so far
- I received $\$9.0$
- A typical reciprocal strategy might be to ensure Aster gets at least their initial $\$5.0$ back, or slightly more to reward their trust
- If I return $\$3.0$, Aster would have: $\$2.0$ (kept) + $\$3.0$ (returned) = $\$5.0$ total (break even)
- If I return $\$4.5$, Aster would have: $\$2.0 + \$4.5 = \$6.5$ total (profit of $\$1.5$)
- I would keep: $\$9.0 - \$4.5 = \$4.5$

Let me consider various options:

1. Return $\$0$: I keep $\$9$, Aster gets $\$2$ total. Very unfair to Aster.
2. Return $\$3$: I keep $\$6$, Aster gets $\$5$ total. Fair - Aster breaks even.
3. Return $\$4$: I keep $\$5$, Aster gets $\$6$ total. Both profit equally.
4. Return $\$4.5$: I keep $\$4.5$, Aster gets $\$6.5$ total. Both profit, but Aster gets slightly more.
5. Return $\$5$: I keep $\$4$, Aster gets $\$7$ total. Aster profits more than me.

Given Aster's substantial trust in sending 60% of their endowment, and since we're in round one of a repeated game, I should reciprocate generously to establish cooperation. Returning $\$4.5$ to $\$5.0$ seems appropriate to reward their trust while still keeping a reasonable share.

I'll go with $\$4.5$ as the middle ground—it gives Aster a $\$1.5$ profit, matches my own gain at $\$4.5$, and signals I'm a reliable partner for future rounds. This balances rewarding their trust with maintaining a fair outcome for myself.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1017,
  "reasoning_tokens": 523
}
```

Message 4 role=user

content:

Your name in this experiment is Cyrus.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent $\$3.0$ to you, it became $\$9.0$, and you returned $\$4.5$. Your payoff was $\$4.5$.
Your total earnings across all rounds are $\$4.5$.

This round, you are the SENDER against Dorian. You have $\$5$. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Dorian
- In Round 1, Aster showed strong trust by sending me 60% (\$3), and that worked out well for both of us
- In an iterated game, establishing cooperation early is valuable

I'll mirror the trust Aster showed me and send \$3 (60%). This gives Dorian \$9 after multiplication, creating good potential for mutual benefit if they reciprocate.

```
{'send': 3.0}
```

reasoning:

Now I'm the sender against a new opponent, Dorian.

This is Round 2, and I'm playing with a different person than Round 1. Let me think about the strategy:

1. In Round 1, I was receiver and Aster sent me \$3 (60%). I returned \$4.5, which was a fair and generous return.
2. Now I'm the sender with Dorian. This is the first interaction with Dorian.
3. In an iterated game with multiple rounds and potentially repeated interactions, I need to:
 - Build trust with new partners
 - Show willingness to cooperate
 - Set a cooperative tone
4. Aster sent 60% (\$3), which was quite trusting. That worked out well for both of us.
5. For a first interaction with Dorian, I should show trust but perhaps be slightly cautious since I don't know their playing style.

Options:

- Send \$2.5 (50%) - moderate trust, balanced approach
- Send \$3 (60%) - mirror what Aster did to me, strong trust signal
- Send \$2 (40%) - cautious but still cooperative

I think sending \$3 (60%) makes sense because:

- It mirrors the trust Aster showed me
- It creates strong potential for cooperation
- In an iterated game, being generous early can establish good relationships
- The receiver gets \$9 and has plenty of room to return a good amount

If Dorian returns even \$4-5, we'll both be better off than if I sent less.

Let me go with \$3.

usage:

```
{
  "input_tokens": 914,
  "output_tokens": 482,
  "reasoning_tokens": 352
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing significant trust

- I received \$9.0 after multiplication
- Elara currently has \$2.0 left

A pattern has emerged in my previous rounds:

- When someone sends \$3, returning \$4.5 has worked well for both parties
- This maintains cooperation and mutual benefit

Following this established pattern:

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 3:

- I'm the RECEIVER
- Elara (the sender) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after 3x multiplication
- Elara currently has \$2.0 left (\$5 - \$3)

Recent history:

- Round 1: Aster sent me \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2: I sent Dorian \$3, they received \$9, returned \$4.5. My payoff: \$6.5

There's a clear pattern emerging where:

- Sending \$3 (60%) seems to be a cooperative norm
- Returning \$4.5 (50% of the received \$9) has been the reciprocal norm
- This leads to good outcomes for both parties

Elara sent \$3.0, showing the same level of trust as Aster and what I showed to Dorian.

If I return \$4.5:

- Elara's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)
- My total: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This is consistent with the pattern that has emerged and seems fair. I should maintain this cooperative equilibrium.

usage:

```
{
  "input_tokens": 1185,
  "output_tokens": 447,
  "reasoning_tokens": 244
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Galen
- The pattern across all my previous rounds has been consistent: sending \$3 (60%) has led to good mutual outcomes
- In both rounds where I sent \$3, I received \$4.5 back, giving me a payoff of \$6.5
- In rounds where others sent me \$3, I returned \$4.5

Continuing this cooperative strategy:

```
{'send': 3.0}
```

Reasoning:

Looking at the pattern so far:

- Round 1: I was receiver, Aster sent \$3, I returned \$4.5
- Round 2: I was sender, I sent \$3 to Dorian, they returned \$4.5

- Round 3: I was receiver, Elara sent \$3, I returned \$4.5
- Round 4: Now I'm sender against Galen

The pattern is clear - sending \$3 (60%) has been reciprocated well, with returns of \$4.5. This creates mutual benefit:

- Sender gets: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)
- Receiver gets: \$9 - \$4.5 = \$4.5 (profit of \$4.5)

This is my first interaction with Galen. I should continue with the successful cooperative strategy of sending \$3.

```
Usage: {
  "input_tokens": 1435,
  "output_tokens": 328,
  "reasoning_tokens": 156
}
```

GAME - Agent_7 - Receiver - dyad_3

Timestamp: 2026-06-04T15:56:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_3",
    "Agent_7"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Dorian (Agent_4).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$4.0, that is 80.0% of its total endowment. You receive \$12.0. How much do you return? (0-12.0)

Message 3 role=assistant

content:

Looking at this situation:

- Dorian showed strong trust by sending \$4.0 (80% of their endowment)
- This created \$12.0 for me through the 3x multiplier
- Total wealth in the system is now \$13 (\$1 that Dorian kept + \$12 I received)

Given this is Round 1 of an iterated game, I want to reward Dorian's trust and establish a cooperative relationship. Dorian took a significant risk in sending such a large proportion.

A fair approach would be to ensure we both benefit reasonably:

- If I return \$6.0, Dorian ends with \$7.0 total (a \$2 gain)
- I keep \$6.0 (a \$6 gain)

This 50% return rate reciprocates Dorian's trust and sets a positive foundation for future rounds.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me understand this situation:

1. I am Galen, playing as the RECEIVER in Round 1
2. My opponent is Dorian (Agent_4), who is the SENDER
3. Dorian sent me \$4.0 out of their \$5 endowment (80%)
4. This amount was multiplied by 3x, so I received \$12.0
5. I need to decide how much to return (between \$0 and \$12.0)

Let me think about this strategically:

****Dorian's position:****

- Started with \$5
- Sent \$4 (kept \$1)
- Currently has: \$1
- Will receive back whatever I return

****My position:****

- Received \$12

- Need to decide how much to return
- Will keep: \$12 - returned

****Considerations:****

1. ****Fairness/Reciprocity****: Dorian showed significant trust by sending 80% of their endowment. This is a strong cooperative signal.
2. ****Strategic thinking****: This is the first round of an iterated game. Building trust and establishing cooperation could be beneficial for future rounds. If I return a fair amount, Dorian (and potentially other players who learn about this) may be more likely to send substantial amounts in future rounds.
3. ****What's fair?****:
 - Dorian invested \$4 and kept \$1
 - This created \$12 for me
 - Total value created: \$1 (Dorian's kept) + \$12 (my received) = \$13
 - Equal split would be \$6.50 each
 - For Dorian to have \$6.50 total, they need: \$6.50 - \$1 (already have) = \$5.50 returned
 - If I return \$5.50, I keep \$6.50
4. ****Slightly more generous approach****: Since this is Round 1 and Dorian showed strong trust, I could return a bit more to establish cooperation. Returning \$6 would give:
 - Dorian: \$1 + \$6 = \$7 (net gain of \$2 from initial \$5)
 - Me: \$12 - \$6 = \$6 (net gain of \$6 from \$0)
5. ****Even split of the received amount****: I could also divide the \$12 evenly between us, returning \$6 and keeping \$6. This leaves Dorian with \$7 total and me with \$6. The key question is whether fairness means splitting what was created (\$12) or splitting the final outcome equally.

Looking at the total wealth: there's now \$13 in the system (\$1 remaining with Dorian plus my \$12). An equal split would give each of us \$6.50. If I return \$6.50, Dorian reaches \$7.50 while I'm left with \$5.50. But I need to account for what was actually created here - the initial \$5 plus the \$8 multiplier effect generated \$13 total, so splitting that equally means we each deserve \$6.50 of the final amount.

However, returning exactly \$6.50 feels like I'm just doing the math without acknowledging Dorian's generosity. Returning \$6 instead - which is 50% of what I received - feels more appropriate as a gesture of good faith. This leaves Dorian with \$7 (a \$2 gain from their initial \$5) while I still benefit with \$6. It's a reasonable balance that rewards their trust and encourages future cooperation.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 1240,
  "reasoning_tokens": 715
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now playing with a new opponent (Aster)
- In my previous round, Dorian trusted me with \$4.0 (80%) and it worked out well for both of us
- This is an iterated game where cooperation can be beneficial

I'll mirror the cooperative strategy that Dorian showed me, sending a substantial amount to signal trust and potentially build a mutually beneficial relationship with Aster.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Now I'm the sender against a new opponent, Aster. Let me think about this:

1. In Round 1, I established a cooperative relationship with Dorian by returning \$6.0 when they sent \$4.0
2. This is a new opponent (Aster), so I don't have history with them
3. This is an iterated game, so building trust and cooperation is valuable
4. Dorian sent 80% (\$4.0) in the first round, which worked out well for both of us

Strategy considerations:

- Sending a substantial amount can signal cooperation and potentially lead to good returns
- Sending too little might not build trust
- Since this is Round 2 and I have no history with Aster, I could mirror the cooperative behavior I saw from Dorian
- However, I also need to be somewhat cautious since I don't know Aster's strategy

I think I should send a moderately high amount to signal cooperation, but maybe not quite as high as Dorian's 80%. Something like \$3-4 would show trust while being somewhat cautious.

Let me think about the math:

- If I send \$3, Aster gets \$9. If they return half (\$4.5), I get \$6.5 total (net gain of \$1.5)
- If I send \$4, Aster gets \$12. If they return half (\$6), I get \$7 total (net gain of \$2)

Given that Dorian was very generous and I want to encourage cooperation in this iterated game, I'll send \$4 (80%) to match what Dorian did. This signals I'm also willing to trust.

usage:

```
{
  "input_tokens": 921,
  "output_tokens": 499,
  "reasoning_tokens": 351
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$13.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 1: Dorian sent me \$4.0, I returned \$6.0 - cooperation paid off
- Round 2: I sent Aster \$4.0, they returned \$6.0 - cooperation continued

The pattern of sending 80% (\$4.0) and receiving fair returns (~50% of the tripled amount) has been working well. I'll maintain this cooperative approach with Briar.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, Dorian sent \$4.0 (80%), I received \$12.0 and returned \$6.0. I got \$6.0.
- Round 2: I was SENDER, I sent \$4.0 (80%) to Aster, they got \$12.0 and returned \$6.0. I g

ot \$7.0 (\$1 kept + \$6.0 returned).

So far, the 80% send strategy has been working well - both Dorian and Aster reciprocated with 50% returns (\$6 out of \$12), which is fair and mutually beneficial.

Now I'm facing a new opponent, Briar. I should continue with the cooperative strategy since it's been successful. Sending \$4.0 (80%) signals trust and has been reciprocated well so far.

Total earnings: \$13.0 across 2 rounds (average \$6.5 per round).

I'll continue with the same strategy of sending \$4.0.

usage:

```
{
  "input_tokens": 1180,
  "output_tokens": 368,
  "reasoning_tokens": 187
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$20.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- Total wealth: \$11 (\$2 that Cyra kept + \$9 I received)

Based on the cooperative pattern I've established:

- Round 1: Received \$12, returned \$6 (50%)
- Round 2: Sent \$4, received back \$6 (50% of tripled amount)
- Round 3: Sent \$4, received back \$6 (50% of tripled amount)

I'll maintain this 50% return rate to be consistent and reward Cyra's trust proportionally:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me analyze this situation:

- Cyra sent me \$3.0 (60% of their endowment)
- This became \$9.0 for me
- Total wealth in the system: \$2 (what Cyra kept) + \$9 (what I received) = \$11

Looking at my history:

- Round 1: Dorian sent \$4, I returned \$6 (50% of what I received)
- Round 2: I sent \$4, Aster returned \$6 (50% of what they received)
- Round 3: I sent \$4, Briar returned \$6 (50% of what they received)

There's been a consistent pattern of returning about 50% of the received amount, which seems to be the cooperative norm emerging in this game.

Cyra sent slightly less than others (\$3 vs \$4), but still showed trust by sending 60% of their endowment.

If I return \$4.5 (50% of \$9):

- Cyra ends with: \$2 + \$4.5 = \$6.5 (net \$1.5 gain)
- I end with: \$9 - \$4.5 = \$4.5 (net \$4.5 gain)

This maintains the cooperative pattern and rewards Cyra's trust proportionally to what they sent.

```
Usage: {
  "input_tokens": 1455,
  "output_tokens": 504,
  "reasoning_tokens": 236
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T15:56:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent \$3.0 (60% of their endowment), showing considerable trust
- I received \$9.0 after the 3x multiplier
- Briar currently has \$2.0 remaining

I want to reciprocate Briar's trust while being fair. A reasonable return would give us both gains from cooperation while rewarding their generous initial send.

If I return \$4.5:

- Briar ends with: $\$2.0 + \$4.5 = \$6.5$
- I end with: $\$9.0 - \$4.5 = \$4.5$

This seems fair - we both gain from the cooperation, and Briar's trust is rewarded with an overall gain of \$1.5, while I also benefit.

{'return': 4.5}

reasoning:

Let me understand the situation:

1. I am Finn, playing as the RECEIVER in this round
2. My opponent is Briar (Agent_2), who is the SENDER
3. Briar sent me \$3.0 out of their \$5.0 endowment (60%)
4. I receive $\$3.0 \times 3 = \9.0
5. I need to decide how much to return (anywhere from \$0 to \$9.0)

Let me think about this strategically:

- Briar was quite generous, sending 60% of their endowment
- If I return nothing, I keep \$9.0 and Briar keeps \$5.0 - \$3.0 = \$2.0
- If I return some amount, I'm showing that I can be trusted, which might encourage future cooperation
- This is a multi-round game, and I'll likely play with various opponents across rounds
- Building a reputation for reciprocity could be beneficial

A common approach in trust games is to aim for fairness or reciprocity:

- Total money created: \$5.0 (Briar's initial) + \$9.0 (what I received) = \$14.0 total in the system after the multiplication
- Equal split would be \$7.0 each
- Briar currently has \$2.0 (they kept $\$5 - \$3 = \$2$)
- For Briar to get \$7.0, I would need to return \$5.0 (so they'd have $\$2.0 + \$5.0 = \$7.0$)
- I would then keep $\$9.0 - \$5.0 = \$4.0$

However, another perspective:

- Briar took a risk by sending \$3.0
- A return that rewards their trust but also acknowledges my position might be appropriate

- Returning about 1/3 to 1/2 of what I received is common in these games
- Returning \$3.0 gives Briar \$5.0 total while I keep \$6.0, rewarding their generosity while still benefiting from the exchange
- Returning \$5.0 achieves an equal split at \$7.0 each, though this is more generous to Briar
- Somewhere between \$4.0-\$4.5 might balance reciprocity with my own interests

Since Briar demonstrated trust by sending most of their money, I should match that with a meaningful return. Around \$4.5 feels right-it ensures both of us end up better off than we started, acknowledges their risk-taking, and avoids appearing either greedy or exploited.

usage:

```
{
  "input_tokens": 562,
  "output_tokens": 1078,
  "reasoning_tokens": 507
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation, I want to establish a cooperative dynamic with Hana. In the previous round, Briar sent me \$3.0 (60%), showing trust, and I reciprocated fairly by returning \$4.5.

I'll adopt a similar strategy and send a substantial amount to signal trust and cooperation.

```
{'send': 3.0}
```

reasoning:

I'm the sender this round against Hana. Let me think about this strategically.

In the last round, I was the receiver and Briar sent me \$3.0 (60% of their endowment). I returned \$4.5, which gave them a net gain of \$1.5 (\$2 + \$4.5 = \$6.5 total vs \$5 starting).

Now I'm in the sender role. I need to decide how much to send to Hana.

Key considerations:

1. This is round 2, so there's limited history to establish reputation
2. Sending more shows trust and can encourage reciprocation
3. But there's risk that Hana might not return much
4. In the previous round, Briar sent 60% (\$3), which seemed like a reasonable amount that signaled cooperation

I think I should also send a meaningful amount to signal cooperation and trust. Sending \$3 (60%) like Briar did seems like a good approach - it's a substantial amount that can benefit both of us if Hana reciprocates fairly.

If I send \$3:

- It becomes \$9 for Hana
- I'm left with \$2
- If Hana returns around half (say \$4-5), we both benefit
- If Hana returns \$4.5 (like I did), they'd have \$4.5 and I'd have \$6.5 total

This seems reasonable and consistent with the cooperative behavior I showed as a receiver.

usage:

```
{
  "input_tokens": 896,
  "output_tokens": 430,
  "reasoning_tokens": 305
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than previous interactions. I want to reward this high level of trust fairly.

If I return \$6.0:

- Dorian ends with: \$1.0 + \$6.0 = \$7.0
- I end with: \$12.0 - \$6.0 = \$6.0

This is a fair split that rewards Dorian's generosity with a \$2.0 overall gain, while I also benefit from the cooperation.

```
{'return': 6.0}
```

reasoning:

Let me analyze this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than what Briar sent me (60%). I should reciprocate this high level of trust.

Let me think about a fair return:

- If I return \$6.0:
 - Dorian ends with: \$1.0 + \$6.0 = \$7.0
 - I end with: \$12.0 - \$6.0 = \$6.0
 - Dorian gains \$2.0 overall, I gain \$6.0
- If I return \$7.0:
 - Dorian ends with: \$1.0 + \$7.0 = \$8.0
 - I end with: \$12.0 - \$7.0 = \$5.0
 - Dorian gains \$3.0 overall, I gain \$5.0

Given that Dorian sent 80% (very generous), I should return proportionally more than the 50% I returned to Briar. Returning \$6.0 would be exactly 50%, which seems like the minimum fair response. Returning \$7.0 would be more generous and better reward Dorian's high trust.

Let me go with \$6.0 as a baseline fair return, which maintains the 50% return ratio I've been using.

usage:

```
{
  "input_tokens": 1133,
  "output_tokens": 540,
}
```

```

    "reasoning_tokens": 261
  }
}
Message 8 role=user
content:
  Your name in this experiment is Finn.
  Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent $
4.0 to you, it became $12.0, and you returned $6.0. Your payoff was $6.0.
Your total earnings across all rounds are $17.0.

This round, you are the SENDER against Elara. You have $5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at my previous rounds, cooperative play has been working well. Both when I've been sen
  der and receiver, fair exchanges have led to good outcomes for both parties.

  I'll continue this cooperative strategy with Elara by sending a substantial amount to signal t
  rust.

  {'send': 3.0}
Reasoning:
  Looking at my history:
  - Round 1: RECEIVER - Briar sent $3.0, I received $9.0, returned $4.5. Payoff: $4.5
  - Round 2: SENDER - I sent $3.0 to Hana, it became $9.0, they returned $4.5. Payoff: $6.5
  - Round 3: RECEIVER - Dorian sent $4.0, I received $12.0, returned $6.0. Payoff: $6.0

  So far, I've been playing cooperatively:
  - When others send me money, I've returned fair amounts that reward their trust
  - When I was sender, I sent $3.0 (60%)

  The interactions have been positive. People have been sending substantial amounts (60-80%), an
  d when I was receiver I returned amounts that gave both parties gains.

  Now I'm playing with a new opponent, Elara. I should continue my cooperative strategy. Sending
  $3.0 worked well with Hana - they returned $4.5, giving me a good payoff. I could send a si
  milar amount or even be slightly more generous given the positive interactions so far.

  I think I'll send $3.0 again, which is a reasonable middle-ground that signals cooperation.
Usage: {
  "input_tokens": 1451,
  "output_tokens": 368,
  "reasoning_tokens": 254
}

GAME - Agent_5 - Receiver - dyad_4
Timestamp: 2026-06-04T15:56:37
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
}

```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Hana (Agent_8).

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0-9.0)

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent \$3 (60% of their endowment), showing significant trust
- I received \$9 after the 3x multiplier
- Hana currently has \$2 remaining

Since this is Round 1 of an iterated game and Hana demonstrated strong trust by sending a substantial portion of their endowment, I want to reciprocate that trust and encourage future cooperation.

I'll return \$4.5, which would give us equal final payoffs:

- Hana's total: $\$2 + \$4.5 = \$6.50$
- My total: $\$9 - \$4.5 = \$4.50$

This rewards Hana's trust while keeping the game mutually beneficial.

```
'''json
{"return": 4.5}
'''
```

reasoning:

I'm playing a trust game as the receiver. Let me understand the situation:

- The sender (Hana/Agent_8) received \$5
- They sent me \$3.0 (60% of their endowment)
- This was multiplied by 3x, so I received \$9.0
- I need to decide how much to return (between \$0 and \$9.0)

Let me think about this strategically:

1. This is Round 1, and this is an iterated game with multiple rounds
2. The sender showed significant trust by sending \$3 (60% of their endowment)
3. If I return a fair amount, this could establish a cooperative relationship for future rounds
4. The sender currently has: $\$5 - \$3 = \$2$
5. I have: \$9

A reasonable approach in iterated trust games is to:

- Reward trust with trustworthiness
- Aim for outcomes that are mutually beneficial
- Consider that we might play together again (though pairings are randomized)

Let me think about fair splits:

- If I return nothing: Sender gets \$2, I get \$9 (total: \$11)
- If I return \$3: Sender gets \$5, I get \$6 (total: \$11) - sender breaks even
- If I return \$4.5: Sender gets \$6.5, I get \$4.5 (total: \$11) - equal final amounts
- If I return \$6: Sender gets \$8, I get \$3 (total: \$11) - sender gains more

A common reciprocal strategy is to return an amount that:

1. Rewards the sender for their trust
2. Still leaves me better off than if no transaction occurred
3. Encourages future cooperation

Since the sender sent 60% of their endowment (a strong sign of trust), I should return a meaningful amount. A return of around \$4-5 would:

- Give the sender more than they started with ($\$2 + \$4-5 = \$6-7$ vs \$5 initial)
- Leave me with \$4-5 (better than the \$0 I started with)
- Signal I'm reciprocating their cooperation

I could aim for equal final payoffs at \$4.5 each, but another approach is to return roughly what they sent or slightly more as a trust reward. The sender sent \$3, so returning \$3-4 would acknowledge their initial investment while still benefiting me. I'll go with \$4.5 to create balanced outcomes where we both gain from the exchange.

usage:

```
{
  "input_tokens": 564,
  "output_tokens": 875,
  "reasoning_tokens": 531
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new opponent (Briar)
- In Round 1, Hana sent me \$3 (60%) and I reciprocated fairly
- This is an early round, so building cooperation is valuable

I'll send \$3 (60% of my endowment), mirroring the trust that Hana showed me in Round 1. This signals willingness to cooperate while testing Briar's reciprocity.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm the SENDER playing against Briar (Agent_2).

Let me think about this strategically:

- This is Round 2 of an iterated game
- I'm playing with a new opponent (Briar), not Hana
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- Briar will decide how much to return

In Round 1:

- Hana sent me \$3 (60% of their endowment), showing strong trust
- I returned \$4.5, which gave fairly equal payoffs

Now I need to decide what to send to Briar. Some considerations:

1. This is early in the game, so establishing cooperation could be beneficial for future rounds
2. I don't know Briar's strategy yet
3. Sending a reasonable amount shows trust and may encourage reciprocity
4. If I send nothing, I guarantee keeping \$5 but miss opportunities for mutual gain
5. If I send too much and they don't return enough, I lose out

A moderate to moderately-high amount seems reasonable to:

- Signal cooperation without being overly risky
- Test Briar's reciprocity
- Build trust for potential future interactions

I'll send \$3, which is 60% - the same proportion that Hana sent to me. This:

- Shows I'm willing to cooperate at a similar level
- Gives Briar \$9 to work with
- Leaves me with \$2 initially
- Demonstrates trust while not being maximally risky

This seems like a balanced approach for Round 2.

usage:

```
{
  "input_tokens": 892,
  "output_tokens": 492,
  "reasoning_tokens": 352
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: Hana sent me \$3 (60%), I returned \$4.5 (fair reciprocity)
- Round 2: I sent Briar \$3 (60%), they returned \$4.5 (good reciprocity)

The 60% send with fair returns strategy has been working well. I'll continue this cooperative approach with Cyra, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- Round 2: I sent Briar \$3, they received \$9, and returned \$4.5 to me
- My payoff in Round 2 was \$2 (what I kept) + \$4.5 (returned) = \$6.5
- Briar kept \$9 - \$4.5 = \$4.5
- This was good reciprocity from Briar - they returned exactly half of what they received

Now I'm facing Cyra (a new opponent) as sender again.

Pattern so far:

- Hana sent me \$3 (60%) ? I returned \$4.5
- I sent Briar \$3 (60%) ? They returned \$4.5

Both interactions showed good reciprocity with 60% sends and 50% returns of the multiplied amount.

Since cooperation has been working well, I'll continue the strategy of sending \$3 (60%) to Cyra. This maintains the cooperative pattern and tests whether Cyra will also reciprocate.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 355,
  "reasoning_tokens": 195
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Finn sent me \$3 (60%), showing good trust
- This matches what I've seen and done in previous rounds
- The established reciprocity pattern has been returning \$4.5 for a \$3 send

I'll continue this cooperative pattern that has worked well.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Finn sent me \$3 (60% of their endowment)
- I received \$9 after the 3x multiplier
- Finn currently has \$2 remaining

Looking at the pattern so far:

- Round 1: Hana sent me \$3, I returned \$4.5

- Round 2: I sent Briar \$3, they returned \$4.5
- Round 3: I sent Cyra \$3, they returned \$4.5
- Round 4: Finn sent me \$3

There's a clear cooperative pattern emerging where sending \$3 gets a return of \$4.5. This has been consistent and fair for both parties. I should continue this pattern to maintain cooperation and reciprocity.

Returning \$4.5 would give:

- Finn: \$2 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This is the same fair reciprocity I've been using.

```
Usage: {
  "input_tokens": 1423,
  "output_tokens": 326,
  "reasoning_tokens": 182
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_8 (Hana)=Sender, Agent_7 (Galen)=Receiver
 dyad_2: Agent_6 (Finn)=Sender, Agent_3 (Cyra)=Receiver
 dyad_3: Agent_4 (Dorian)=Sender, Agent_2 (Briar)=Receiver
 dyad_4: Agent_1 (Aster)=Sender, Agent_5 (Elara)=Receiver

Parsed game decisions:

dyad_1: Agent_8 (Hana) sent \$3; received \$9; Agent_7 (Galen) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_7 (Galen)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_2: Agent_6 (Finn) sent \$3; received \$9; Agent_3 (Cyra) returned \$4.5; payoffs Agent_6 (Finn)=\$6.5, Agent_3 (Cyra)=\$4.5
 Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_4 (Dorian) sent \$4; received \$12; Agent_2 (Briar) returned \$6; payoffs Agent_4 (Dorian)=\$7, Agent_2 (Briar)=\$6
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_4: Agent_1 (Aster) sent \$4; received \$12; Agent_5 (Elara) returned \$6; payoffs Agent_1 (Aster)=\$7, Agent_5 (Elara)=\$6
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 31.0,
 "Agent_2": 30.0,
 "Agent_3": 26.5,
 "Agent_4": 31.5,
 "Agent_5": 28.0,
 "Agent_6": 30.0,
 "Agent_7": 29.0,
 "Agent_8": 30.0

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-06-04T15:56:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_7",
 "pairing": {
 "agent_names": {

```

    "Agent_7": "Galen",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Finn (Agent_6).

Round 2

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Finn sent me \$3, showing trust
- I received \$9 after the 3x multiplier
- In Round 1, when I sent \$3, Elara returned \$4.5 (50%), which felt fair
- I want to reciprocate that fairness to build cooperation

I'll return ****\$4.5**** (50% of what I received).

This gives:

- Finn: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me think about this situation:

Round 1 recap:

- I was the sender, sent \$3 to Elara
- Elara received \$9 and returned \$4.5
- My payoff was \$2 (kept) + \$4.5 (returned) = \$6.5
- Elara kept \$9 - \$4.5 = \$4.5

So Elara returned exactly 50% of what they received, which seems fair.

Round 2:

- I'm now the receiver
- Finn sent me \$3 (same amount I sent in Round 1)
- I received \$9 (3 × \$3)
- I need to decide how much to return

Since this is an iterated game and I want to establish cooperation:

- Finn trusted me by sending \$3
- Elara returned 50% (\$4.5) when I sent \$3, which seemed fair
- I should reciprocate with a similar level of fairness

If I return \$4.5:

- I keep \$9 - \$4.5 = \$4.5
- Finn ends up with \$5 - \$3 + \$4.5 = \$6.5
- We both get the same payoff (\$6.5 for Finn, \$4.5 for me in this round)

Actually, wait. Let me recalculate:

- Finn started with \$5
- Finn sent \$3
- Finn has \$2 left
- I return some amount X
- Finn's total: \$2 + X

If I return \$4.5:

- Finn's total: \$2 + \$4.5 = \$6.5
- My total: \$9 - \$4.5 = \$4.5

This mirrors what Elara did for me. It's a fair 50% return and signals that I'm willing to cooperate. This should encourage future cooperation.

I'll return \$4.5.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 621,
  "reasoning_tokens": 328
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:
- When I sent \$3 to Elara, they returned \$4.5 (fair split)
- When Finn sent \$3 to me, I returned \$4.5 (fair split)

This cooperative approach has been working well. I'll continue with the same strategy and send ****\$3**** to Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my history so far:

Round 1: I was SENDER vs Elara

- I sent \$3
- Elara received \$9
- Elara returned \$4.5
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Elara's payoff: $\$9 - \$4.5 = \$4.5$

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9
- I returned \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$
- Finn's payoff: $\$5 - \$3 + \$4.5 = \6.5

So far, the pattern has been:

- When I sent \$3, I got back \$4.5 (which is 50% of the \$9)
- When Finn sent \$3, I returned \$4.5 (50% of the \$9)

This seems like a fair and cooperative pattern. Both times the sender ended up with \$6.5 and the receiver with \$4.5.

Now I'm facing Aster as a SENDER. This is a new opponent, and I want to continue establishing cooperative relationships. Given that \$3 worked well in Round 1 and Finn also sent \$3 in Round 2, this seems like a reasonable amount to signal cooperation.

I should send \$3 again to:

1. Continue the cooperative pattern
2. Signal trust to Aster
3. See if they reciprocate fairly

Let me send \$3.

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 472,
  "reasoning_tokens": 270
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent me \$4, which is even more trusting than the \$3 pattern I've seen
- I received \$12 after the 3x multiplier
- I've been maintaining a fair 50% return policy

I'll return **\$6** (50% of what I received).

This gives:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at the situation:

- Briar sent me \$4, showing significant trust (80% of their endowment)
- I received \$12 after the 3x multiplier
- The pattern so far has been sending \$3 and returning about 50% (which is \$4.5 from \$9)

Briar sent more (\$4 vs \$3), so they're being extra trusting. I should reciprocate proportionally.

If I keep the same 50% return rate: $\$12 \times 0.5 = \6

This would give:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

Actually, let me think about fairness differently. Briar started with \$5 and sent \$4, keeping only \$1.

After 3x, I have \$12 total.

If we want roughly equal outcomes:

- Total money in the system: \$5 (Briar's original) + \$12 (what I received) = \$17 total value created
- Equal split would be around \$8.5 each

For Briar to get \$8.5: they have \$1 kept, so need \$7.5 back

For me to get \$8.5: I need to keep \$12 - \$7.5 = \$4.5... wait that doesn't add up.

Let me recalculate:

- Briar starts with \$5, sends \$4, keeps \$1
- I receive $\$4 \times 3 = \12
- If I return \$X:
 - Briar ends with: $\$1 + \X
 - I end with: $\$12 - \X

For equal outcomes: $\$1 + \$X = \$12 - \X

$\$1 + 2X = \12

$\$X = \5.5

Hmm, but that's less than the 50% I've been doing.

The 50% approach feels right-it's consistent with what's worked before and honors their larger contribution. I'll return \$6.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 636,
  "reasoning_tokens": 360
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- Sending \$3 has consistently resulted in fair \$4.5 returns
- The cooperative approach is working well for everyone
- Total earnings are solid at \$23.5

I'll continue with the proven strategy and send **\$3** to Galen.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review my history:

- Round 1: Sent \$3 to Elara, they returned \$4.5. Payoff: \$6.5
- Round 2: Received \$3 from Finn, returned \$4.5. Payoff: \$4.5
- Round 3: Sent \$3 to Aster, they returned \$4.5. Payoff: \$6.5
- Round 4: Received \$4 from Briar, returned \$6. Payoff: \$6.0

The pattern shows that:

- Sending \$3 has been working well (50% return)
- Briar sent \$4, showing even more trust, and I reciprocated fairly

I've been consistent with sending \$3 and it's been working. The cooperative strategy has been successful. I should continue with this approach.

I'll send \$3 to Galen.

Usage: {
 "input_tokens": 1452,
 "output_tokens": 296,
 "reasoning_tokens": 155
}

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-06-04T15:56:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_8",
 "pairing": {
 "agent_names": {
 "Agent_7": "Galen",
 "Agent_8": "Hana"
 },
 "agents": [
 "Agent_8",
 "Agent_7"
],
 },
}


```

"investor": "Agent_8",
"roles": {
  "Agent_7": "trustee",
  "Agent_8": "investor"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Aster (Agent_1).

Round 2

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.

Your total earnings across all rounds are \$6.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm now playing with a new opponent (Aster)
- In my previous round, Dorian trusted me with \$4.0 (80%) and it worked out well for both of us
- This is an iterated game where cooperation can be beneficial

I'll mirror the cooperative strategy that Dorian showed me, sending a substantial amount to signal trust and potentially build a mutually beneficial relationship with Aster.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Now I'm the sender against a new opponent, Aster. Let me think about this:

1. In Round 1, I established a cooperative relationship with Dorian by returning \$6.0 when they sent \$4.0
2. This is a new opponent (Aster), so I don't have history with them
3. This is an iterated game, so building trust and cooperation is valuable
4. Dorian sent 80% (\$4.0) in the first round, which worked out well for both of us

Strategy considerations:

- Sending a substantial amount can signal cooperation and potentially lead to good returns
- Sending too little might not build trust
- Since this is Round 2 and I have no history with Aster, I could mirror the cooperative behavior I saw from Dorian
- However, I also need to be somewhat cautious since I don't know Aster's strategy

I think I should send a moderately high amount to signal cooperation, but maybe not quite as high as Dorian's 80%. Something like \$3-4 would show trust while being somewhat cautious.

Let me think about the math:

- If I send \$3, Aster gets \$9. If they return half (\$4.5), I get \$6.5 total (net gain of \$1.5)
- If I send \$4, Aster gets \$12. If they return half (\$6), I get \$7 total (net gain of \$2)

Given that Dorian was very generous and I want to encourage cooperation in this iterated game, I'll send \$4 (80%) to match what Dorian did. This signals I'm also willing to trust.

usage:

```
{
  "input_tokens": 921,
  "output_tokens": 499,
  "reasoning_tokens": 351
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$13.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent experience:

- Round 1: Dorian sent me \$4.0, I returned \$6.0 - cooperation paid off
- Round 2: I sent Aster \$4.0, they returned \$6.0 - cooperation continued

The pattern of sending 80% (\$4.0) and receiving fair returns (~50% of the tripled amount)

has been working well. I'll maintain this cooperative approach with Briar.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, Dorian sent \$4.0 (80%), I received \$12.0 and returned \$6.0. I got \$6.0.
- Round 2: I was SENDER, I sent \$4.0 (80%) to Aster, they got \$12.0 and returned \$6.0. I got \$7.0 (\$1 kept + \$6.0 returned).

So far, the 80% send strategy has been working well - both Dorian and Aster reciprocated with 50% returns (\$6 out of \$12), which is fair and mutually beneficial.

Now I'm facing a new opponent, Briar. I should continue with the cooperative strategy since it's been successful. Sending \$4.0 (80%) signals trust and has been reciprocated well so far.

Total earnings: \$13.0 across 2 rounds (average \$6.5 per round).

I'll continue with the same strategy of sending \$4.0.

usage:

```
{
  "input_tokens": 1180,
  "output_tokens": 368,
  "reasoning_tokens": 187
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$20.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- Total wealth: \$11 (\$2 that Cyra kept + \$9 I received)

Based on the cooperative pattern I've established:

- Round 1: Received \$12, returned \$6 (50%)
- Round 2: Sent \$4, received back \$6 (50% of tripled amount)
- Round 3: Sent \$4, received back \$6 (50% of tripled amount)

I'll maintain this 50% return rate to be consistent and reward Cyra's trust proportionally
:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Cyra sent me \$3.0 (60% of their endowment)
- This became \$9.0 for me
- Total wealth in the system: \$2 (what Cyra kept) + \$9 (what I received) = \$11

Looking at my history:

- Round 1: Dorian sent \$4, I returned \$6 (50% of what I received)

- Round 2: I sent \$4, Aster returned \$6 (50% of what they received)
- Round 3: I sent \$4, Briar returned \$6 (50% of what they received)

There's been a consistent pattern of returning about 50% of the received amount, which seems to be the cooperative norm emerging in this game.

Cyra sent slightly less than others (\$3 vs \$4), but still showed trust by sending 60% of their endowment.

If I return \$4.5 (50% of \$9):

- Cyra ends with: \$2 + \$4.5 = \$6.5 (net \$1.5 gain)
- I end with: \$9 - \$4.5 = \$4.5 (net \$4.5 gain)

This maintains the cooperative pattern and rewards Cyra's trust proportionally to what they sent.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 504,
  "reasoning_tokens": 236
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$24.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- This is the same amount as Cyra sent in the previous round

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Hana sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 kept by Hana + \$9 I received)

I've been maintaining a consistent 50% return rate:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)

This has been working well and creates good incentives for cooperation. I should continue this pattern.

50% of \$9.0 = \$4.5

Usage: {

```
"input_tokens": 1487,
"output_tokens": 258,
"reasoning_tokens": 121
```

}

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-06-04T15:56:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_6",
      "Agent_3"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_3": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Hana (Agent_8).

Round 2

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation, I want to establish a cooperative dynamic with Hana. In the previous round, Briar sent me \$3.0 (60%), showing trust, and I reciprocated fairly by returning \$4.5.

I'll adopt a similar strategy and send a substantial amount to signal trust and cooperation.

{'send': 3.0}

reasoning:

I'm the sender this round against Hana. Let me think about this strategically.

In the last round, I was the receiver and Briar sent me \$3.0 (60% of their endowment). I returned \$4.5, which gave them a net gain of \$1.5 (\$2 + \$4.5 = \$6.5 total vs \$5 starting).

Now I'm in the sender role. I need to decide how much to send to Hana.

Key considerations:

1. This is round 2, so there's limited history to establish reputation
2. Sending more shows trust and can encourage reciprocation
3. But there's risk that Hana might not return much
4. In the previous round, Briar sent 60% (\$3), which seemed like a reasonable amount that signaled cooperation

I think I should also send a meaningful amount to signal cooperation and trust. Sending \$3 (60%) like Briar did seems like a good approach - it's a substantial amount that can benefit both of us if Hana reciprocates fairly.

If I send \$3:

- It becomes \$9 for Hana
- I'm left with \$2
- If Hana returns around half (say \$4-5), we both benefit
- If Hana returns \$4.5 (like I did), they'd have \$4.5 and I'd have \$6.5 total

This seems reasonable and consistent with the cooperative behavior I showed as a receiver.

usage:

```
{
  "input_tokens": 896,
  "output_tokens": 430,
  "reasoning_tokens": 305
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0

. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than previous interactions. I want to reward this high level of trust fairly.

If I return \$6.0:

- Dorian ends with: $\$1.0 + \$6.0 = \$7.0$
- I end with: $\$12.0 - \$6.0 = \$6.0$

This is a fair split that rewards Dorian's generosity with a \$2.0 overall gain, while I also benefit from the cooperation.

{'return': 6.0}

reasoning:

Let me analyze this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than what Briar sent me (60%). I should reciprocate this high level of trust.

Let me think about a fair return:

- If I return \$6.0:
 - Dorian ends with: $\$1.0 + \$6.0 = \$7.0$
 - I end with: $\$12.0 - \$6.0 = \$6.0$
 - Dorian gains \$2.0 overall, I gain \$6.0
- If I return \$7.0:
 - Dorian ends with: $\$1.0 + \$7.0 = \$8.0$
 - I end with: $\$12.0 - \$7.0 = \$5.0$
 - Dorian gains \$3.0 overall, I gain \$5.0

Given that Dorian sent 80% (very generous), I should return proportionally more than the 50% I returned to Briar. Returning \$6.0 would be exactly 50%, which seems like the minimum fair response. Returning \$7.0 would be more generous and better reward Dorian's high trust.

Let me go with \$6.0 as a baseline fair return, which maintains the 50% return ratio I've been using.

usage:

```
{
  "input_tokens": 1133,
  "output_tokens": 540,
  "reasoning_tokens": 261
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my previous rounds, cooperative play has been working well. Both when I've been sender and receiver, fair exchanges have led to good outcomes for both parties.

I'll continue this cooperative strategy with Elara by sending a substantial amount to signal trust.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - Briar sent \$3.0, I received \$9.0, returned \$4.5. Payoff: \$4.5
- Round 2: SENDER - I sent \$3.0 to Hana, it became \$9.0, they returned \$4.5. Payoff: \$6.5
- Round 3: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0. Payoff: \$6.0

So far, I've been playing cooperatively:

- When others send me money, I've returned fair amounts that reward their trust
- When I was sender, I sent \$3.0 (60%)

The interactions have been positive. People have been sending substantial amounts (60-80%), and when I was receiver I returned amounts that gave both parties gains.

Now I'm playing with a new opponent, Elara. I should continue my cooperative strategy. Sending \$3.0 worked well with Hana - they returned \$4.5, giving me a good payoff. I could send a similar amount or even be slightly more generous given the positive interactions so far.

I think I'll send \$3.0 again, which is a reasonable middle-ground that signals cooperation

usage:

```
{  
  "input_tokens": 1451,  
  "output_tokens": 368,  
  "reasoning_tokens": 254  
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record, cooperation has been consistently successful. When I've sent \$3.0 as sender, I've received \$4.5 back, resulting in a \$6.5 payoff. This pattern of trust and reciprocity has been beneficial.

I'll continue this cooperative approach with Cyra.

```
{'send': 3.0}
```

Reasoning:

Looking at my pattern so far:

- Round 1 (RECEIVER vs Briar): They sent \$3.0, I received \$9.0, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Hana): I sent \$3.0, they returned \$4.5. My payoff: \$6.5
- Round 3 (RECEIVER vs Dorian): They sent \$4.0, I received \$12.0, returned \$6.0. My payoff: \$6.0
- Round 4 (SENDER vs Elara): I sent \$3.0, they returned \$4.5. My payoff: \$6.5

So far, cooperation has been working very well. When I've sent \$3.0, I've gotten back \$4.5, giving me a payoff of \$6.5. When others sent to me, I've reciprocated fairly.

This is my fourth time as sender (wait, let me recount):

- Round 2: SENDER
- Round 4: SENDER
- Round 5: SENDER (this round)

So this is my third time as sender. I've been consistent in sending \$3.0 and it's worked out well. I should continue this cooperative strategy with Cyra.

```
Usage: {  
  "input_tokens": 1375,  
  "output_tokens": 388,  
  "reasoning_tokens": 217  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-06-04T15:57:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_3"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Dorian (Agent_4).

Round 2

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$3 .0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Dorian
- In Round 1, Aster showed strong trust by sending me 60% (\$3), and that worked out well for both of us
- In an iterated game, establishing cooperation early is valuable

I'll mirror the trust Aster showed me and send \$3 (60%). This gives Dorian \$9 after multiplication, creating good potential for mutual benefit if they reciprocate.

{'send': 3.0}

reasoning:

Now I'm the sender against a new opponent, Dorian.

This is Round 2, and I'm playing with a different person than Round 1. Let me think about the strategy:

1. In Round 1, I was receiver and Aster sent me \$3 (60%). I returned \$4.5, which was a fair and generous return.
2. Now I'm the sender with Dorian. This is the first interaction with Dorian.
3. In an iterated game with multiple rounds and potentially repeated interactions, I need to:
 - Build trust with new partners
 - Show willingness to cooperate
 - Set a cooperative tone
4. Aster sent 60% (\$3), which was quite trusting. That worked out well for both of us.
5. For a first interaction with Dorian, I should show trust but perhaps be slightly cautious since I don't know their playing style.

Options:

- Send \$2.5 (50%) - moderate trust, balanced approach
- Send \$3 (60%) - mirror what Aster did to me, strong trust signal
- Send \$2 (40%) - cautious but still cooperative

I think sending \$3 (60%) makes sense because:

- It mirrors the trust Aster showed me
- It creates strong potential for cooperation
- In an iterated game, being generous early can establish good relationships
- The receiver gets \$9 and has plenty of room to return a good amount

If Dorian returns even \$4-5, we'll both be better off than if I sent less.

Let me go with \$3.

usage:

```
{
  "input_tokens": 914,
  "output_tokens": 482,
  "reasoning_tokens": 352
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after multiplication
- Elara currently has \$2.0 left

A pattern has emerged in my previous rounds:

- When someone sends \$3, returning \$4.5 has worked well for both parties
- This maintains cooperation and mutual benefit

Following this established pattern:

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 3:

- I'm the RECEIVER
- Elara (the sender) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after 3x multiplication
- Elara currently has \$2.0 left (\$5 - \$3)

Recent history:

- Round 1: Aster sent me \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2: I sent Dorian \$3, they received \$9, returned \$4.5. My payoff: \$6.5

There's a clear pattern emerging where:

- Sending \$3 (60%) seems to be a cooperative norm
- Returning \$4.5 (50% of the received \$9) has been the reciprocal norm
- This leads to good outcomes for both parties

Elara sent \$3.0, showing the same level of trust as Aster and what I showed to Dorian.

If I return \$4.5:

- Elara's total: \$2.0 + \$4.5 = \$6.5 (profit of \$1.5)
- My total: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This is consistent with the pattern that has emerged and seems fair. I should maintain this cooperative equilibrium.

usage:

```
{
  "input_tokens": 1185,
  "output_tokens": 447,
  "reasoning_tokens": 244
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- The pattern across all my previous rounds has been consistent: sending \$3 (60%) has led to good mutual outcomes
- In both rounds where I sent \$3, I received \$4.5 back, giving me a payoff of \$6.5
- In rounds where others sent me \$3, I returned \$4.5

Continuing this cooperative strategy:

```
{'send': 3.0}
```

reasoning:

Looking at the pattern so far:

- Round 1: I was receiver, Aster sent \$3, I returned \$4.5
- Round 2: I was sender, I sent \$3 to Dorian, they returned \$4.5
- Round 3: I was receiver, Elara sent \$3, I returned \$4.5
- Round 4: Now I'm sender against Galen

The pattern is clear - sending \$3 (60%) has been reciprocated well, with returns of \$4.5.

This creates mutual benefit:

- Sender gets: $\$5 - \$3 + \$4.5 = \6.5 (profit of \$1.5)
- Receiver gets: $\$9 - \$4.5 = \$4.5$ (profit of \$4.5)

This is my first interaction with Galen. I should continue with the successful cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 328,
  "reasoning_tokens": 156
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Finn sent me \$3.0 (60%), showing the same level of trust others have shown
- I received \$9.0 after multiplication
- The consistent pattern across all rounds has been: when someone sends \$3, return \$4.5

This strategy has maintained strong cooperation and mutual benefit throughout the game.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

- Finn sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after multiplication
- Finn currently has \$2.0 left

The established pattern has been very consistent:

- When I send \$3, I receive \$4.5 back
- When others send me \$3, I return \$4.5

This has created good outcomes for everyone:

- The sender ends up with $\$2 + \$4.5 = \$6.5$
- The receiver ends up with $\$9.0 - \$4.5 = \$4.5$
- Everyone gains from cooperation

I should continue this pattern with Finn:

```
{'return': 4.5}
Usage: {
  "input_tokens": 1398,
  "output_tokens": 263,
  "reasoning_tokens": 140
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-06-04T15:57:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_2"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_2": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Cyra (Agent_3).

Round 2

In your most recent previous game round, you were the SENDER against Galen. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$7.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3, showing trust and cooperation
- I received \$9 (tripled)
- Cyra currently has \$2 left

In the previous round, Galen returned exactly half of what they received (\$6 out of \$12), which felt fair. I'll follow a similar principle of reciprocity.

Returning half (\$4.5) gives:

- Cyra: \$2 + \$4.5 = \$6.5
- Me: \$9 - \$4.5 = \$4.5

This rewards Cyra's trust while keeping some benefit from the multiplier effect.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was sender, sent \$4 to Galen
- Galen received \$12 (4×3)
- Galen returned \$6 to me
- My payoff: \$5 - \$4 + \$6 = \$7
- Galen's payoff: \$12 - \$6 = \$6

So Galen returned exactly half of what they received, which seems fair. They made \$6 and I made \$7, so we both benefited from cooperation.

Round 2:

- I am receiver
- Cyra sent me \$3
- I received \$9 (3×3)
- Cyra kept: $\$5 - \$3 = \$2$
- I need to decide how much to return

If I follow a similar fairness principle as Galen did (returning about half), I would return around \$4.5.

But let me think about what would be fair:

- Cyra invested \$3 out of their \$5
- This created \$9 for me
- If we split the gains equally:
 - Cyra should end up with: $\$2$ (kept) + some return
 - I should end up with: $\$9$ - some return

One approach: Split the total surplus equally.

- Total created: $\$5$ (Cyra's initial) + $\$9$ (tripled amount) = $\$14$
- Equal split: $\$7$ each
- Cyra has $\$2$, needs $\$5$ more
- I have $\$9$, should give $\$5$, keeping $\$4$

Another approach: Return half of what I received (like Galen did)

- Return $\$4.5$
- Cyra gets: $\$2 + \$4.5 = \$6.5$
- I keep: $\$9 - \$4.5 = \$4.5$

Another approach: Ensure we both profit equally from the exchange

- The gain from cooperation is $\$6$ (the $\$9$ minus Cyra's $\$3$ investment)
- If we split this evenly, that's $\$3$ each
- Cyra ends with $\$2 + \4 return = $\$6$, and I keep $\$5$

Returning $\$4.5$ mirrors what Galen did and signals I'm a reciprocal player, so I'll go with that.

usage:

```
{
  "input_tokens": 778,
  "output_tokens": 766,
  "reasoning_tokens": 407
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:
- Round 1 with Galen: I sent \$4, received \$6 back, earned \$7
- Round 2 with Cyra: They sent \$3, I returned \$4.5 (half)

The cooperative strategy has been working well. Sending \$4 created good value and Galen reciprocated fairly. I'll continue this approach with Finn, signaling trust and willingness to cooperate.

```
{ 'send': 4 }
```

reasoning:

Let me review my history:

- Round 1: I was SENDER vs Galen. Sent \$4, they got \$12, returned \$6. I got \$7.
- Round 2: I was RECEIVER vs Cyra. They sent \$3, I got \$9, returned \$4.5. I got \$4.5.
- Total earnings: \$11.5

Now Round 3: I'm SENDER against Finn (new opponent).

My strategy so far has been cooperative - sending substantial amounts as sender to create value. When Galen sent \$4, they returned \$6 (half). When Cyra sent \$3, I returned \$4.5 (half).

This seems to be working well. Both exchanges were mutually beneficial compared to not sending anything.

For this round with Finn, I should continue the cooperative strategy. Sending \$4 worked well with Galen. I could send the same amount or try sending the full \$5 to maximize potential gains if Finn is also cooperative.

Let me think about the trade-offs:

- Send \$5: Finn gets \$15. If they return half (~\$7.5), I get \$7.5 total.
- Send \$4: Finn gets \$12. If they return half (\$6), I get \$7 total (like last time).
- Send \$3: Finn gets \$9. If they return half (\$4.5), I get \$6.5 total.

Since this is an iterated game and I'm building a reputation, sending a high amount signals cooperation. The previous round went well, so I'll continue with \$4 as a strong but not maximal signal of trust.

Actually, let me reconsider. I sent \$4 to Galen and got \$6 back, netting \$7. That worked well. But I could try \$5 to see if Finn is willing to be more generous and potentially get even better returns.

I think I'll stick with \$4 as it's worked well and shows strong trust without being all-in.

usage:

```
{
  "input_tokens": 1076,
  "output_tokens": 582,
  "reasoning_tokens": 404
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern from previous rounds:

- Aster sent me \$4, which is the same amount I've been sending
- I received \$12 (tripled)
- Previous players have consistently returned about half when receiving similar amounts

Following the reciprocal fairness norm that's been established:

- Returning \$6 gives Aster \$7 total (\$1 + \$6)
- I keep \$6

This rewards Aster's trust while maintaining the cooperative pattern.

```
{'return': 6}
```

reasoning:

Let me analyze the situation:

- Aster sent me \$4, showing strong trust and cooperation
- I received \$12 (tripled amount)

- Aster currently has \$1 left

Looking at the pattern from previous rounds:

- When I sent \$4 to Galen, they returned \$6 (half of \$12)
- When Cyra sent \$3 to me, I returned \$4.5 (half of \$9)
- When I sent \$4 to Finn, they returned \$6 (half of \$12)

There's a consistent pattern of returning half of what was received, which seems fair and maintains cooperation. This gives both players a net benefit from the transaction.

If I return \$6:

- Aster: $\$1 + \$6 = \$7$
- Me: $\$12 - \$6 = \$6$

This is fair and maintains the cooperative norm. Aster sent a large amount (\$4), and returning half rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 349,
  "reasoning_tokens": 196
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- Sending \$4 has consistently yielded \$6 returns (with both Galen and Finn)
- The cooperative norm of returning about half has been well-established
- Total earnings are strong at \$24.5

I'll continue with the successful strategy of sending \$4 to signal trust and cooperation.

```
{'send': 4}
```

Reasoning:

Let me review my history:

- Round 1 (Sender vs Galen): Sent \$4, received \$6 back, earned \$7
- Round 2 (Receiver vs Cyra): Received \$9, returned \$4.5, earned \$4.5
- Round 3 (Sender vs Finn): Sent \$4, received \$6 back, earned \$7
- Round 4 (Receiver vs Aster): Received \$12, returned \$6, earned \$6

Total earnings: \$24.5

Pattern I've observed:

- When I send \$4, I consistently get \$6 back (both with Galen and Finn)
- When others send to me, I've been returning about half
- The cooperative strategy of sending \$4 has been working well

I haven't played with Briar yet. Given the consistent success with sending \$4 and the reciprocal fairness norm that's been established, I should continue with this strategy.

Sending \$4:

- Creates \$12 for Briar
- If they return half (like others have been doing), I get \$6 back
- My payoff would be $\$5 - \$4 + \$6 = \7

This has been the most reliable strategy so far.

Usage: {

```
"input_tokens": 1455,
"output_tokens": 384,
```

```
"reasoning_tokens": 241
}
```

GAME - Agent_2 - Receiver - dyad_3

Timestamp: 2026-06-04T15:57:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_2"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_2": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Elara (Agent_5).

Round 2

In your most recent previous game round, you were the SENDER against Finn. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

Elara sent me \$3.0, showing trust and cooperation - the same amount I sent to Finn in Round 1. Finn returned \$4.5 to me (half of the \$9 received), which felt fair and cooperative.

I should reciprocate this cooperative behavior with Elara. Returning half (\$4.5) means:

- Elara gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9 - \$4.5 = \$4.5$

This splits the gains fairly and establishes me as a trustworthy partner.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was SENDER against Finn
- I sent \$3.0
- Finn received \$9.0
- Finn returned \$4.5
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn's payoff: $\$9.0 - \$4.5 = \$4.5$

So Finn returned exactly half of what they received, which is a pretty fair and cooperative split. I gained \$1.5 more than my initial \$5, and Finn ended with \$4.5.

Round 2 (current):

- I am RECEIVER against Elara
- Elara sent me \$3.0
- I received \$9.0
- I need to decide how much to return

Elara made the same choice I did in Round 1 - sending \$3.0. This shows trust and willingness to cooperate.

Given that Finn returned \$4.5 (half) to me when I sent \$3.0, I should consider:

1. Being reciprocal and fair
2. Establishing a cooperative reputation for future rounds
3. The fact that this is an iterated game with multiple agents

If I return \$4.5 (half of what I received), then:

- Elara gets: $\$5 - \$3 + \$4.5 = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This mirrors what happened in Round 1 and seems fair. I could return a bit more or less, but \$4.5 seems like a reasonable, cooperative amount that:

- Shows I'm trustworthy
- Could encourage cooperation in future rounds
- Is similar to what Finn did to me

I'll return \$4.5.

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 593,
  "reasoning_tokens": 334
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

Galen sent me \$4.0, which is more generous than what I've seen so far. They're showing strong trust.

Following the reciprocity pattern established in previous rounds (where returning about half has worked well), I'll return \$6:

- Galen gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This rewards their generosity and maintains cooperative norms.

```
'''json
{"return": 6}
'''
```

reasoning:

Galen sent me \$4.0, which became \$12.0 for me. This is a generous amount, showing strong trust and cooperation.

Let me think about fair reciprocity:

- Galen sent \$4, so they have $\$5 - \$4 = \$1$ left
- I received \$12

If I return \$6 (half of what I received):

- Galen gets: $\$1 + \$6 = \$7$
- I get: $\$12 - \$6 = \$6$

This seems fair and reciprocates their generosity. It's proportional to what they sent.

Alternatively, I could aim for more equal final payoffs. If I return \$5.5:

- Galen gets: $\$1 + \$5.5 = \$6.5$
- I get: $\$12 - \$5.5 = \$6.5$

This creates equal payoffs, which might be seen as very fair.

I think returning \$6 (half) is the most consistent with the pattern established so far - Finn returned half to me when I sent \$3, and I returned half to Elara when they sent \$3.
This establishes a clear cooperative norm.

Since Galen sent more (\$4 vs \$3), returning half (\$6) rewards their greater trust and maintains the reciprocity norm.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 434,
  "reasoning_tokens": 249
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern so far:

- Cooperation has been strong with people returning about 50% of what they receive
- When Galen sent me \$4, I returned \$6, giving them a good payoff
- Sending more creates more total value when reciprocity is expected

I'll send \$4 to Hana, showing trust in continued cooperation:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me review what I've learned so far:

Round 1: I was SENDER vs Finn

- I sent \$3, they got \$9, they returned \$4.5
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Finn kept: \$9 - \$4.5 = \$4.5

Round 2: I was RECEIVER vs Elara

- They sent \$3, I got \$9, I returned \$4.5
- My payoff: \$9 - \$4.5 = \$4.5
- Elara got: \$5 - \$3 + \$4.5 = \$6.5

Round 3: I was RECEIVER vs Galen

- They sent \$4, I got \$12, I returned \$6
- My payoff: \$12 - \$6 = \$6
- Galen got: \$5 - \$4 + \$6 = \$7

Now Round 4: I'm SENDER vs Hana (new opponent)

The pattern I've seen:

- When I sent \$3, I got back \$4.5 (50% return)
- When Elara sent \$3, I returned \$4.5 (50% return)
- When Galen sent \$4, I returned \$6 (50% return)

Everyone seems to be cooperating well with approximately 50% returns. This creates mutual benefit:

- If I send \$3, they get \$9, return ~\$4.5, I end with \$6.5
- If I send \$4, they get \$12, return ~\$6, I end with \$7
- If I send \$5, they get \$15, return ~\$7.5, I end with \$7.5

Sending more generally leads to higher payoffs for both parties when the other person reciprocates fairly. Since this is an iterated game and cooperation has been working well, I should continue being generous.

I'll send \$4 to Hana, mirroring what Galen trusted me with and demonstrating the same conf

idence in reciprocation.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 584,
  "reasoning_tokens": 344
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the established pattern:

Dorian sent me \$4.0, showing the same level of trust and generosity that I've seen from Galen and that I showed to Hana. The reciprocity pattern has been consistent - returning about 50% of what's received.

Returning \$6:

- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This maintains the cooperative equilibrium.

```
'''json
{"return": 6}
'''
```

Reasoning:

Dorian sent me \$4.0, which becomes \$12.0.

Looking at the pattern:

- When I received \$3 from Elara, I returned \$4.5 (50%)
- When I received \$4 from Galen, I returned \$6 (50%)
- When I sent \$4 to Hana, they returned \$6 (50%)

The cooperative norm has been established at returning about 50% of what's received. Dorian sent me \$4, which is the same generous amount that Hana and Galen sent. I should maintain this reciprocity pattern.

Returning \$6:

- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This is fair and maintains the cooperative equilibrium.

```
Usage: {
  "input_tokens": 1460,
  "output_tokens": 327,
  "reasoning_tokens": 150
}
```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-06-04T15:57:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_1",
    "Agent_5"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 2

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Galen sent me \$4 (keeping \$1), which tripled to \$12 for me
- In the previous round, Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair
- Following a similar principle of returning about 50% seems reasonable

If I return \$6:

- Galen ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This rewards Galen for taking the risk to send a substantial amount while still being cooperative. It's consistent with how Cyra treated me.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3 to Cyra, they received \$9, and they returned \$4.5 to me. That's a 50% return of what they received, which is quite fair and cooperative.
2. Now in Round 2, I'm the receiver. Galen sent me \$4, so I received \$12.

Let me think about what's fair:

- Galen kept \$1 and sent \$4
- I received \$12
- If I return nothing, Galen gets \$1 total and I get \$12
- If I return \$6 (50% of what I received), Galen gets \$7 total and I get \$6
- If I return \$5.5, we'd both end up with \$6.5 (Galen: \$1 + \$5.5 = \$6.5, Me: \$12 - \$5.5 = \$6.5)

Cyra returned 50% of what they received (\$4.5 out of \$9), which felt fair and cooperative. I should probably follow a similar principle.

Actually, let me think about this more carefully. The game creates surplus - Galen gave up \$4 to give me \$12, creating \$8 of surplus. A fair split might be:

- Galen should end up with: \$1 (kept) + some return
- I should end up with: \$12 (received) - that return

If we split evenly from the starting point:

- We both started conceptually with access to \$5
- After the multiplication, there's \$1 + \$12 = \$13 total
- Split evenly would be \$6.5 each
- So I should return \$6.5, giving Galen \$1 + \$6.5 = \$7.5, and I keep \$5.5

Wait, that doesn't quite work out. Let me recalculate:

- If I return \$6.5: Galen gets \$1 + \$6.5 = \$7.5, I get \$12 - \$6.5 = \$5.5

- Total: \$7.5 + \$5.5 = \$13 ?

Returning 50% like Cyra did would give me slightly more (\$6 vs \$5.5), which seems reasonable since I'm taking no risk as the receiver. I could aim for equal payoffs or proportional sharing, but the 50% return feels like the right balance between fairness and reciprocating the cooperation I received. I need to reconsider how to split the gains. The cooperation created \$8 in surplus (\$12 received minus \$4 sent). If I divide this equally between us, Galen gets their \$4 back plus \$4 of the surplus for \$8 total, meaning I return

\$8 and keep \$4 for myself.

But compared to just splitting the final amount evenly, this feels less advantageous to me. I think returning \$6-half of what I received-strikes the right balance. It's straightforward, rewards Galen for taking the initial risk, and still shows I'm willing to cooperate meaningfully. That leaves Galen with \$7 and me with \$6. I'll go with \$6.

usage:

```
{
  "input_tokens": 856,
  "output_tokens": 1327,
  "reasoning_tokens": 602
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$12.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- This is the same amount I sent to Cyra in Round 1, and they returned \$4.5 (50%)
- I returned 50% to Galen in the previous round

Consistency seems important for building trust in repeated games. Following the 50% return pattern:

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is fair and encourages future cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- In the last round, Galen sent me \$4 and I returned \$6 (50% of what I received)
- In Round 1, I sent \$3 to Cyra, they received \$9 and returned \$4.5 (50%)

There's a pattern emerging of returning about 50% of what is received, which seems fair and encourages cooperation.

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is consistent with the 50% return pattern and similar to what Cyra did with my \$3 sent in Round 1. It rewards Hana for trusting me while maintaining fairness.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 414,
  "reasoning_tokens": 171
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant
content:**

Looking at the patterns so far:

- When I sent \$3 to Cyra, they returned 50% (\$4.5)
- When Galen sent \$4 to me, I returned 50% (\$6)
- When Hana sent \$3 to me, I returned 50% (\$4.5)

There's been a consistent cooperative pattern with 50% returns. Since I haven't played with Dorian before, I'll show trust while following the cooperative trend I've observed.

Sending \$4:

- Dorian receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

This is slightly more trusting than my initial \$3 to Cyra, reflecting the positive cooperative experiences so far.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze my situation:

Round 1: I was SENDER vs Cyra

- Sent \$3, they got \$9, returned \$4.5
- My payoff: $\$2 + \$4.5 = \$6.5$

Round 2: I was RECEIVER vs Galen

- They sent \$4, I got \$12, returned \$6
- My payoff: \$6

Round 3: I was RECEIVER vs Hana

- They sent \$3, I got \$9, returned \$4.5
- My payoff: \$4.5

Total so far: \$17

Now Round 4: I'm SENDER vs Dorian (new opponent)

Looking at the pattern:

- When I sent \$3, Cyra returned \$4.5 (50%)
- When Galen sent \$4, I returned \$6 (50%)
- When Hana sent \$3, I returned \$4.5 (50%)

So there's been a pattern of 50% returns, which is quite cooperative.

As a sender, my best strategy depends on what I expect Dorian to return. If they follow the 50% pattern:

- If I send \$0: I keep \$5, payoff = \$5
- If I send \$1: I keep \$4, they get \$3, if return 50% = \$1.5, payoff = \$5.5
- If I send \$2: I keep \$3, they get \$6, if return 50% = \$3, payoff = \$6
- If I send \$3: I keep \$2, they get \$9, if return 50% = \$4.5, payoff = \$6.5
- If I send \$4: I keep \$1, they get \$12, if return 50% = \$6, payoff = \$7
- If I send \$5: I keep \$0, they get \$15, if return 50% = \$7.5, payoff = \$7.5

- If they return 50%, sending more is always better (up to sending everything).

I don't know Dorian yet, but the cooperative trend suggests I should escalate slightly. Galen sent \$4 to me, demonstrating confidence in reciprocation. I'll match that trust and send \$4 as well, signaling I'm willing to build on the mutual cooperation we've established.

usage:

```
{
  "input_tokens": 1507,
  "output_tokens": 811,
  "reasoning_tokens": 396
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$24.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. This establishes strong trust. Playing with Elara for the first time, I'll continue with the successful \$4 strategy that worked well with Dorian.

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at my history:

- Round 1: I sent \$3 to Cyra, got \$4.5 back, earned \$6.5
- Round 2: Receiver, Galen sent \$4, I returned \$6, earned \$6.0
- Round 3: Receiver, Hana sent \$3, I returned \$4.5, earned \$4.5
- Round 4: I sent \$4 to Dorian, got \$6 back, earned \$7.0

The pattern has been very consistent - everyone is returning 50% of what they receive. This is a strong cooperative signal.

Total earnings: \$24.0, which means I've been doing well.

Now I'm playing with Elara for the first time. Given the consistent cooperation I've experienced:

- Everyone has been trustworthy
- The 50% return norm seems established
- I could send \$4 or even \$5

If I send \$5:

- Elara receives \$15
- I keep \$0
- If they return 50% (\$7.5), I'd end with \$7.5

If I send \$4:

- Elara receives \$12
- I keep \$1

- If they return 50% (\$6), I'd end with \$7

The \$5 option shows maximum trust and could earn \$0.5 more. Given the consistent cooperation, I'll go with \$4 again to maintain the pattern that's been working well. It's trusting but not reckless.

Actually, let me reconsider. The cooperation has been perfect so far. Maybe I should try \$5 to show maximum trust and potentially maximize total welfare. If everyone is cooperating at 50%, sending \$5 gives:

- \$15 to receiver
- If they return \$7.5, I get \$7.5 (better than the \$7 from sending \$4)
- Total surplus is maximized

But there's also risk. I'll stick with \$4 since it's been working and I haven't seen anyone send \$5 yet.

```
Usage: {
  "input_tokens": 1619,
  "output_tokens": 679,
  "reasoning_tokens": 390
}
```

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T15:57:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_5"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Briar (Agent_2).

Round 2

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$4.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new opponent (Briar)
- In Round 1, Hana sent me \$3 (60%) and I reciprocated fairly
- This is an early round, so building cooperation is valuable

I'll send \$3 (60% of my endowment), mirroring the trust that Hana showed me in Round 1. This signals willingness to cooperate while testing Briar's reciprocity.

```
'''json
{"send": 3}
'''
```

reasoning:

Now I'm the SENDER playing against Briar (Agent_2).

Let me think about this strategically:

- This is Round 2 of an iterated game
- I'm playing with a new opponent (Briar), not Hana
- I have \$5 to potentially send
- Whatever I send gets multiplied by 3x
- Briar will decide how much to return

In Round 1:

- Hana sent me \$3 (60% of their endowment), showing strong trust
- I returned \$4.5, which gave fairly equal payoffs

Now I need to decide what to send to Briar. Some considerations:

1. This is early in the game, so establishing cooperation could be beneficial for future rounds
2. I don't know Briar's strategy yet

3. Sending a reasonable amount shows trust and may encourage reciprocity
4. If I send nothing, I guarantee keeping \$5 but miss opportunities for mutual gain
5. If I send too much and they don't return enough, I lose out

A moderate to moderately-high amount seems reasonable to:

- Signal cooperation without being overly risky
- Test Briar's reciprocity
- Build trust for potential future interactions

I'll send \$3, which is 60% - the same proportion that Hana sent to me. This:

- Shows I'm willing to cooperate at a similar level
- Gives Briar \$9 to work with
- Leaves me with \$2 initially
- Demonstrates trust while not being maximally risky

This seems like a balanced approach for Round 2.

usage:

```
{
  "input_tokens": 892,
  "output_tokens": 492,
  "reasoning_tokens": 352
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: Hana sent me \$3 (60%), I returned \$4.5 (fair reciprocity)
- Round 2: I sent Briar \$3 (60%), they returned \$4.5 (good reciprocity)

The 60% send with fair returns strategy has been working well. I'll continue this cooperative approach with Cyra, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- Round 2: I sent Briar \$3, they received \$9, and returned \$4.5 to me
- My payoff in Round 2 was \$2 (what I kept) + \$4.5 (returned) = \$6.5
- Briar kept \$9 - \$4.5 = \$4.5
- This was good reciprocity from Briar - they returned exactly half of what they received

Now I'm facing Cyra (a new opponent) as sender again.

Pattern so far:

- Hana sent me \$3 (60%) ? I returned \$4.5
- I sent Briar \$3 (60%) ? They returned \$4.5

Both interactions showed good reciprocity with 60% sends and 50% returns of the multiplied amount.

Since cooperation has been working well, I'll continue the strategy of sending \$3 (60%) to Cyra. This maintains the cooperative pattern and tests whether Cyra will also reciprocate.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 355,
}
```

```

    "reasoning_tokens": 195
  }
Message 6 role=user
content:
  Your name in this experiment is Elara.
  Your opponent this round is Finn (Agent_6).

  Round 4

  In your most recent previous game round, you were the SENDER against Cyra. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $17.5.

  This round, you are the RECEIVER against Finn. They sent you $3.0, so you received $9.0. H
  ow much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Looking at the pattern:
  - Finn sent me $3 (60%), showing good trust
  - This matches what I've seen and done in previous rounds
  - The established reciprocity pattern has been returning $4.5 for a $3 send

  I'll continue this cooperative pattern that has worked well.

  '''json
  {"return": 4.5}
  '''
reasoning:
  Looking at the situation:
  - Finn sent me $3 (60% of their endowment)
  - I received $9 after the 3x multiplier
  - Finn currently has $2 remaining

  Looking at the pattern so far:
  - Round 1: Hana sent me $3, I returned $4.5
  - Round 2: I sent Briar $3, they returned $4.5
  - Round 3: I sent Cyra $3, they returned $4.5
  - Round 4: Finn sent me $3

  There's a clear cooperative pattern emerging where sending $3 gets a return of $4.5. This
  has been consistent and fair for both parties. I should continue this pattern to maintai
  n cooperation and reciprocity.

  Returning $4.5 would give:
  - Finn: $2 + $4.5 = $6.5
  - Me: $9 - $4.5 = $4.5

  This is the same fair reciprocity I've been using.
usage:
  {
    "input_tokens": 1423,
    "output_tokens": 326,
    "reasoning_tokens": 182
  }
Message 8 role=user
content:
  Your name in this experiment is Elara.
  Your opponent this round is Aster (Agent_1).

  Round 5

  In your most recent previous game round, you were the RECEIVER against Finn. They sent $3.
  0 to you, it became $9.0, and you returned $4.5. Your payoff was $4.5.
  Your total earnings across all rounds are $22.0.

  This round, you are the RECEIVER against Aster. They sent you $4.0, so you received $12.0.
  How much do you return? (0-12.0)
  Respond exactly as JSON: {'return': <amount>}
Assistant output:

```

Looking at this situation:

- Aster sent me \$4 (80%), showing higher trust than the typical \$3 (60%) I've experienced
- I received \$12
- My established pattern has been returning about 50% of what I receive
- For \$3 sends ? returned \$4.5 (50%)
- For this \$4 send ? should return \$6 (50%)

This maintains consistency while appropriately rewarding Aster's higher trust.

```
'''json
{"return": 6}
'''
```

Reasoning:

Let me analyze this situation:

- Aster sent me \$4 (80%), which is more than the typical \$3 (60%) I've been seeing
- I received \$12.0
- I need to decide how much to return

Looking at my history:

- When people sent me \$3, I returned \$4.5 (which leaves them with $\$2 + \$4.5 = \$6.5$ and me with \$4.5)
- The ratio has been: sender keeps \$2, I keep \$4.5, and I return \$4.5

For a \$3 send:

- They kept \$2, I returned \$4.5
- Their total: \$6.5
- My total: \$4.5
- Return rate: $\$4.5/\$9 = 50\%$

For a \$4 send:

- They kept \$1, I receive \$12
- If I return 50%, that's \$6
- Their total would be $\$1 + \$6 = \$7$
- My total would be \$6

That seems fair and consistent. Let me think about whether to reward the higher trust with a slightly higher return rate or stick with the 50% pattern.

Given that Aster showed more trust (80% vs 60%), I should probably reward that. Maybe return \$6 (50%) or slightly more like \$6.5 to encourage continued high trust.

Let me go with \$6 as it's consistent with my 50% return pattern and still rewards their higher trust level appropriately.

```
Usage: {
  "input_tokens": 1382,
  "output_tokens": 473,
  "reasoning_tokens": 283
}
```

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

- dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver
- dyad_2: Agent_5 (Elara)=Sender, Agent_4 (Dorian)=Receiver
- dyad_3: Agent_7 (Galen)=Sender, Agent_6 (Finn)=Receiver
- dyad_4: Agent_3 (Cyra)=Sender, Agent_8 (Hana)=Receiver

Parsed game decisions:

- dyad_1: Agent_2 (Briar) sent \$4; received \$12; Agent_1 (Aster) returned \$6; payoffs Agent_2 (Briar)=\$7, Agent_1 (Aster)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
- dyad_2: Agent_5 (Elara) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_5 (Elara)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
- dyad_3: Agent_7 (Galen) sent \$4; received \$12; Agent_6 (Finn) returned \$6; payoffs Agent_7


```

(Galen)=$7, Agent_6 (Finn)=$6
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw":
6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_3 (Cyra) sent $3; received $9; Agent_8 (Hana) returned $4.5; payoffs Agent_3
(Cyra)=$6.5, Agent_8 (Hana)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 37.0,
  "Agent_2": 37.0,
  "Agent_3": 33.0,
  "Agent_4": 36.0,
  "Agent_5": 34.5,
  "Agent_6": 36.0,
  "Agent_7": 36.0,
  "Agent_8": 34.5
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T15:57:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Galen (Agent_7).

Round 3

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

Galen sent me \$4.0, which is more generous than what I've seen so far. They're showing strong trust.

Following the reciprocity pattern established in previous rounds (where returning about half has worked well), I'll return \$6:

- Galen gets: \$5 - \$4 + \$6 = \$7
- I get: \$12 - \$6 = \$6

This rewards their generosity and maintains cooperative norms.

```
'''json
{"return": 6}
'''
```

reasoning:

Galen sent me \$4.0, which became \$12.0 for me. This is a generous amount, showing strong trust and cooperation.

Let me think about fair reciprocity:

- Galen sent \$4, so they have \$5 - \$4 = \$1 left
- I received \$12

If I return \$6 (half of what I received):

- Galen gets: \$1 + \$6 = \$7
- I get: \$12 - \$6 = \$6

This seems fair and reciprocates their generosity. It's proportional to what they sent.

Alternatively, I could aim for more equal final payoffs. If I return \$5.5:

- Galen gets: $\$1 + \$5.5 = \$6.5$
- I get: $\$12 - \$5.5 = \$6.5$

This creates equal payoffs, which might be seen as very fair.

I think returning \$6 (half) is the most consistent with the pattern established so far - Finn returned half to me when I sent \$3, and I returned half to Elara when they sent \$3. This establishes a clear cooperative norm.

Since Galen sent more (\$4 vs \$3), returning half (\$6) rewards their greater trust and maintains the reciprocity norm.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 434,
  "reasoning_tokens": 249
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern so far:

- Cooperation has been strong with people returning about 50% of what they receive
- When Galen sent me \$4, I returned \$6, giving them a good payoff
- Sending more creates more total value when reciprocity is expected

I'll send \$4 to Hana, showing trust in continued cooperation:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me review what I've learned so far:

Round 1: I was SENDER vs Finn

- I sent \$3, they got \$9, they returned \$4.5
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Finn kept: $\$9 - \$4.5 = \$4.5$

Round 2: I was RECEIVER vs Elara

- They sent \$3, I got \$9, I returned \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$
- Elara got: $\$5 - \$3 + \$4.5 = \6.5

Round 3: I was RECEIVER vs Galen

- They sent \$4, I got \$12, I returned \$6
- My payoff: $\$12 - \$6 = \$6$
- Galen got: $\$5 - \$4 + \$6 = \7

Now Round 4: I'm SENDER vs Hana (new opponent)

The pattern I've seen:

- When I sent \$3, I got back \$4.5 (50% return)
- When Elara sent \$3, I returned \$4.5 (50% return)
- When Galen sent \$4, I returned \$6 (50% return)

Everyone seems to be cooperating well with approximately 50% returns. This creates mutual

benefit:

- If I send \$3, they get \$9, return ~\$4.5, I end with \$6.5
- If I send \$4, they get \$12, return ~\$6, I end with \$7
- If I send \$5, they get \$15, return ~\$7.5, I end with \$7.5

Sending more generally leads to higher payoffs for both parties when the other person reciprocates fairly. Since this is an iterated game and cooperation has been working well, I should continue being generous.

I'll send \$4 to Hana, mirroring what Galen trusted me with and demonstrating the same confidence in reciprocation.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 584,
  "reasoning_tokens": 344
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$24.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the established pattern:

Dorian sent me \$4.0, showing the same level of trust and generosity that I've seen from Galen and that I showed to Hana. The reciprocity pattern has been consistent - returning about 50% of what's received.

Returning \$6:

- Dorian gets: \$5 - \$4 + \$6 = \$7
- I get: \$12 - \$6 = \$6

This maintains the cooperative equilibrium.

```
'''json
{"return": 6}
'''
```

reasoning:

Dorian sent me \$4.0, which becomes \$12.0.

Looking at the pattern:

- When I received \$3 from Elara, I returned \$4.5 (50%)
- When I received \$4 from Galen, I returned \$6 (50%)
- When I sent \$4 to Hana, they returned \$6 (50%)

The cooperative norm has been established at returning about 50% of what's received. Dorian sent me \$4, which is the same generous amount that Hana and Galen sent. I should maintain this reciprocity pattern.

Returning \$6:

- Dorian gets: \$5 - \$4 + \$6 = \$7
- I get: \$12 - \$6 = \$6

This is fair and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1460,
  "output_tokens": 327,
}
```

```

    "reasoning_tokens": 150
  }
Message 8 role=user
content:
  Your name in this experiment is Briar.
  Your opponent this round is Aster (Agent_1).

  Round 6

  In your most recent previous game round, you were the RECEIVER against Dorian. They sent $
  4.0 to you, it became $12.0, and you returned $6.0. Your payoff was $6.0.
  Your total earnings across all rounds are $30.0.

  This round, you are the SENDER against Aster. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Assistant output:
  Looking at the successful pattern:
  - Sending $4 has consistently led to returns of $6
  - This creates mutual benefit (sender gets $7, receiver gets $6)
  - Cooperation has been strong across all interactions

  I'll continue this pattern by sending $4 to Aster:

  '''json
  {"send": 4}
  '''
Reasoning:
  Looking at the pattern:
  - Round 2 (as receiver): Elara sent $3, I returned $4.5
  - Round 3 (as receiver): Galen sent $4, I returned $6
  - Round 4 (as sender): I sent $4 to Hana, they returned $6
  - Round 5 (as receiver): Dorian sent $4, I returned $6

  The pattern is clear: sending $4 has been working well with consistent returns of about $6. Th
  is gives the sender a payoff of $7 and the receiver $6, which is mutually beneficial.

  I should continue this cooperative pattern by sending $4 to Aster.
Usage: {
  "input_tokens": 1411,
  "output_tokens": 243,
  "reasoning_tokens": 130
}

GAME - Agent_1 - Receiver - dyad_1
Timestamp: 2026-06-04T15:57:53
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,

```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Hana (Agent_8).

Round 3

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$12.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- This is the same amount I sent to Cyra in Round 1, and they returned \$4.5 (50%)
- I returned 50% to Galen in the previous round

Consistency seems important for building trust in repeated games. Following the 50% return pattern:

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is fair and encourages future cooperation.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Hana sent me \$3 (keeping \$2), which tripled to \$9 for me
- In the last round, Galen sent me \$4 and I returned \$6 (50% of what I received)
- In Round 1, I sent \$3 to Cyra, they received \$9 and returned \$4.5 (50%)

There's a pattern emerging of returning about 50% of what is received, which seems fair and encourages cooperation.

If I return \$4.5:

- Hana ends up with: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- I end up with: \$9 - \$4.5 = \$4.5 total

This is consistent with the 50% return pattern and similar to what Cyra did with my \$3 sent in Round 1. It rewards Hana for trusting me while maintaining fairness.

usage:

```
{
  "input_tokens": 1183,
  "output_tokens": 414,
  "reasoning_tokens": 171
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the patterns so far:

- When I sent \$3 to Cyra, they returned 50% (\$4.5)
- When Galen sent \$4 to me, I returned 50% (\$6)
- When Hana sent \$3 to me, I returned 50% (\$4.5)

There's been a consistent cooperative pattern with 50% returns. Since I haven't played with Dorian before, I'll show trust while following the cooperative trend I've observed.

Sending \$4:

- Dorian receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

This is slightly more trusting than my initial \$3 to Cyra, reflecting the positive cooperative experiences so far.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze my situation:

Round 1: I was SENDER vs Cyra

- Sent \$3, they got \$9, returned \$4.5
- My payoff: $\$2 + \$4.5 = \$6.5$

Round 2: I was RECEIVER vs Galen

- They sent \$4, I got \$12, returned \$6
- My payoff: \$6

Round 3: I was RECEIVER vs Hana

- They sent \$3, I got \$9, returned \$4.5
- My payoff: \$4.5

Total so far: \$17

Now Round 4: I'm SENDER vs Dorian (new opponent)

Looking at the pattern:

- When I sent \$3, Cyra returned \$4.5 (50%)
- When Galen sent \$4, I returned \$6 (50%)
- When Hana sent \$3, I returned \$4.5 (50%)

So there's been a pattern of 50% returns, which is quite cooperative.

As a sender, my best strategy depends on what I expect Dorian to return. If they follow the 50% pattern:

- If I send \$0: I keep \$5, payoff = \$5
- If I send \$1: I keep \$4, they get \$3, if return 50% = \$1.5, payoff = \$5.5
- If I send \$2: I keep \$3, they get \$6, if return 50% = \$3, payoff = \$6
- If I send \$3: I keep \$2, they get \$9, if return 50% = \$4.5, payoff = \$6.5
- If I send \$4: I keep \$1, they get \$12, if return 50% = \$6, payoff = \$7
- If I send \$5: I keep \$0, they get \$15, if return 50% = \$7.5, payoff = \$7.5

- If they return 50%, sending more is always better (up to sending everything).

I don't know Dorian yet, but the cooperative trend suggests I should escalate slightly. Galen sent \$4 to me, demonstrating confidence in reciprocation. I'll match that trust and send \$4 as well, signaling I'm willing to build on the mutual cooperation we've established.

usage:

```
{
  "input_tokens": 1507,
  "output_tokens": 811,
  "reasoning_tokens": 396
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. This establishes strong trust

. Playing with Elara for the first time, I'll continue with the successful \$4 strategy that worked well with Dorian.

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: I sent \$3 to Cyra, got \$4.5 back, earned \$6.5
- Round 2: Receiver, Galen sent \$4, I returned \$6, earned \$6.0
- Round 3: Receiver, Hana sent \$3, I returned \$4.5, earned \$4.5
- Round 4: I sent \$4 to Dorian, got \$6 back, earned \$7.0

The pattern has been very consistent - everyone is returning 50% of what they receive. This is a strong cooperative signal.

Total earnings: \$24.0, which means I've been doing well.

Now I'm playing with Elara for the first time. Given the consistent cooperation I've experienced:

- Everyone has been trustworthy
- The 50% return norm seems established
- I could send \$4 or even \$5

If I send \$5:

- Elara receives \$15
- I keep \$0
- If they return 50% (\$7.5), I'd end with \$7.5

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they return 50% (\$6), I'd end with \$7

The \$5 option shows maximum trust and could earn \$0.5 more. Given the consistent cooperation, I'll go with \$4 again to maintain the pattern that's been working well. It's trusting but not reckless.

Actually, let me reconsider. The cooperation has been perfect so far. Maybe I should try \$5 to show maximum trust and potentially maximize total welfare. If everyone is cooperating at 50%, sending \$5 gives:

- \$15 to receiver
- If they return 7.5, I get \$7.5 (better than the \$7 from sending \$4)
- Total surplus is maximized

But there's also risk. I'll stick with \$4 since it's been working and I haven't seen anyone send \$5 yet.

usage:

```
{
  "input_tokens": 1619,
  "output_tokens": 679,
  "reasoning_tokens": 390
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the established pattern:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- And when I've been sender, I've consistently received 50% back

Briar sent \$4 (keeping \$1), which became \$12 for me.

Following the cooperative 50% pattern:

- If I return \$6:
 - Briar ends with: $\$1 + \$6 = \$7$
 - I end with: $\$12 - \$6 = \$6$

This maintains the trust and reciprocity that has been working well.

```
'''json
{"return": 6}
'''
```

Reasoning:

Looking at the pattern:

- Round 1: I was SENDER, sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: I was RECEIVER, Galen sent \$4, I returned \$6 (50%)
- Round 3: I was RECEIVER, Hana sent \$3, I returned \$4.5 (50%)
- Round 4: I was SENDER, sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: I was SENDER, sent \$4 to Elara, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. Now Briar sent me \$4 (keeping \$1), which became \$12.

If I return 50%: \$6

- Briar ends up with: $\$1 \text{ (kept)} + \$6 \text{ (returned)} = \$7 \text{ total}$
- I end up with: $\$12 - \$6 = \$6 \text{ total}$

This is the fair 50% pattern that has been established throughout all rounds. I should continue with this.

```
Usage: {
  "input_tokens": 1652,
  "output_tokens": 440,
  "reasoning_tokens": 185
}
```

GAME - Agent_5 - Sender - dyad_2

Timestamp: 2026-06-04T15:58:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_5",
    "Agent_4"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_4": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_4"
},
"role": "investor",
```

```

"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 3

In your most recent previous game round, you were the SENDER against Briar. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: Hana sent me \$3 (60%), I returned \$4.5 (fair reciprocity)
- Round 2: I sent Briar \$3 (60%), they returned \$4.5 (good reciprocity)

The 60% send with fair returns strategy has been working well. I'll continue this cooperat

ive approach with Cyra, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

- Round 2: I sent Briar \$3, they received \$9, and returned \$4.5 to me
- My payoff in Round 2 was \$2 (what I kept) + \$4.5 (returned) = \$6.5
- Briar kept \$9 - \$4.5 = \$4.5
- This was good reciprocity from Briar - they returned exactly half of what they received

Now I'm facing Cyra (a new opponent) as sender again.

Pattern so far:

- Hana sent me \$3 (60%) ? I returned \$4.5
- I sent Briar \$3 (60%) ? They returned \$4.5

Both interactions showed good reciprocity with 60% sends and 50% returns of the multiplied amount.

Since cooperation has been working well, I'll continue the strategy of sending \$3 (60%) to Cyra. This maintains the cooperative pattern and tests whether Cyra will also reciprocate.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 355,
  "reasoning_tokens": 195
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- Finn sent me \$3 (60%), showing good trust
- This matches what I've seen and done in previous rounds
- The established reciprocity pattern has been returning \$4.5 for a \$3 send

I'll continue this cooperative pattern that has worked well.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Finn sent me \$3 (60% of their endowment)
- I received \$9 after the 3x multiplier
- Finn currently has \$2 remaining

Looking at the pattern so far:

- Round 1: Hana sent me \$3, I returned \$4.5
- Round 2: I sent Briar \$3, they returned \$4.5
- Round 3: I sent Cyra \$3, they returned \$4.5
- Round 4: Finn sent me \$3

There's a clear cooperative pattern emerging where sending \$3 gets a return of \$4.5. This has been consistent and fair for both parties. I should continue this pattern to maintain

n cooperation and reciprocity.

Returning \$4.5 would give:

- Finn: $\$2 + \$4.5 = \$6.5$
- Me: $\$9 - \$4.5 = \$4.5$

This is the same fair reciprocity I've been using.

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 326,
  "reasoning_tokens": 182
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Aster sent me \$4 (80%), showing higher trust than the typical \$3 (60%) I've experienced
- I received \$12
- My established pattern has been returning about 50% of what I receive
- For \$3 sends ? returned \$4.5 (50%)
- For this \$4 send ? should return \$6 (50%)

This maintains consistency while appropriately rewarding Aster's higher trust.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

- Aster sent me \$4 (80%), which is more than the typical \$3 (60%) I've been seeing
- I received \$12.0
- I need to decide how much to return

Looking at my history:

- When people sent me \$3, I returned \$4.5 (which leaves them with $\$2 + \$4.5 = \$6.5$ and me with \$4.5)
- The ratio has been: sender keeps \$2, I keep \$4.5, and I return \$4.5

For a \$3 send:

- They kept \$2, I returned \$4.5
- Their total: \$6.5
- My total: \$4.5
- Return rate: $\$4.5/\$9 = 50\%$

For a \$4 send:

- They kept \$1, I receive \$12
- If I return 50%, that's \$6
- Their total would be $\$1 + \$6 = \$7$
- My total would be \$6

That seems fair and consistent. Let me think about whether to reward the higher trust with a slightly higher return rate or stick with the 50% pattern.

Given that Aster showed more trust (80% vs 60%), I should probably reward that. Maybe return \$6 (50%) or slightly more like \$6.5 to encourage continued high trust.

Let me go with \$6 as it's consistent with my 50% return pattern and still rewards their higher trust level appropriately.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 473,
  "reasoning_tokens": 283
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$28.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history:

- My consistent strategy has been sending \$3 (60%) as sender
- This has resulted in good reciprocity - I've received \$4.5 back each time
- This gives me payoffs of \$6.5, better than keeping all \$5

I'll continue this cooperative approach with Dorian, a new opponent.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review my history:

Round 1: RECEIVER from Hana - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 2: SENDER to Briar - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 3: SENDER to Cyra - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 4: RECEIVER from Finn - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 5: RECEIVER from Aster - they sent \$4, I returned \$6, my payoff \$6.0

Total: \$4.5 + \$6.5 + \$6.5 + \$4.5 + \$6.0 = \$28.0 ?

Pattern:

- When I sent \$3 (60%), I got \$4.5 back, giving me \$6.5
- When others sent \$3 to me, I returned \$4.5
- When Aster sent \$4 to me, I returned \$6 (50% of \$12)

The \$3 send (60%) has been working well and consistently. I'm playing against a new opponent (Dorian), so I'll stick with the established cooperative strategy.

Usage: {

```
"input_tokens": 1380,
"output_tokens": 409,
"reasoning_tokens": 206
```

}

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-06-04T15:58:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_5",
"pairing": {
  "agent_names": {
    "Agent_4": "Dorian",
    "Agent_5": "Elara"
  },
"agents": [
```

```

    "Agent_5",
    "Agent_4"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_4": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Finn (Agent_6).

Round 3

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$11.5.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant
content:

Looking at my experience so far:

- Round 1 with Galen: I sent \$4, received \$6 back, earned \$7
- Round 2 with Cyra: They sent \$3, I returned \$4.5 (half)

The cooperative strategy has been working well. Sending \$4 created good value and Galen reciprocated fairly. I'll continue this approach with Finn, signaling trust and willingness to cooperate.

{'send': 4}

reasoning:

Let me review my history:

- Round 1: I was SENDER vs Galen. Sent \$4, they got \$12, returned \$6. I got \$7.
- Round 2: I was RECEIVER vs Cyra. They sent \$3, I got \$9, returned \$4.5. I got \$4.5.
- Total earnings: \$11.5

Now Round 3: I'm SENDER against Finn (new opponent).

My strategy so far has been cooperative - sending substantial amounts as sender to create value. When Galen sent \$4, they returned \$6 (half). When Cyra sent \$3, I returned \$4.5 (half).

This seems to be working well. Both exchanges were mutually beneficial compared to not sending anything.

For this round with Finn, I should continue the cooperative strategy. Sending \$4 worked well with Galen. I could send the same amount or try sending the full \$5 to maximize potential gains if Finn is also cooperative.

Let me think about the trade-offs:

- Send \$5: Finn gets \$15. If they return half (~\$7.5), I get \$7.5 total.
- Send \$4: Finn gets \$12. If they return half (\$6), I get \$7 total (like last time).
- Send \$3: Finn gets \$9. If they return half (\$4.5), I get \$6.5 total.

Since this is an iterated game and I'm building a reputation, sending a high amount signals cooperation. The previous round went well, so I'll continue with \$4 as a strong but not maximal signal of trust.

Actually, let me reconsider. I sent \$4 to Galen and got \$6 back, netting \$7. That worked well. But I could try \$5 to see if Finn is willing to be more generous and potentially get even better returns.

I think I'll stick with \$4 as it's worked well and shows strong trust without being all-in.

usage:

```
{
  "input_tokens": 1076,
  "output_tokens": 582,
  "reasoning_tokens": 404
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern from previous rounds:

- Aster sent me \$4, which is the same amount I've been sending
- I received \$12 (tripled)
- Previous players have consistently returned about half when receiving similar amounts

Following the reciprocal fairness norm that's been established:

- Returning \$6 gives Aster \$7 total (\$1 + \$6)
- I keep \$6

This rewards Aster's trust while maintaining the cooperative pattern.

```
{'return': 6}
```

reasoning:

Let me analyze the situation:

- Aster sent me \$4, showing strong trust and cooperation
- I received \$12 (tripled amount)
- Aster currently has \$1 left

Looking at the pattern from previous rounds:

- When I sent \$4 to Galen, they returned \$6 (half of \$12)
- When Cyra sent \$3 to me, I returned \$4.5 (half of \$9)
- When I sent \$4 to Finn, they returned \$6 (half of \$12)

There's a consistent pattern of returning half of what was received, which seems fair and maintains cooperation. This gives both players a net benefit from the transaction.

If I return \$6:

- Aster: \$1 + \$6 = \$7
- Me: \$12 - \$6 = \$6

This is fair and maintains the cooperative norm. Aster sent a large amount (\$4), and returning half rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 349,
  "reasoning_tokens": 196
}
```

Message 6 role=user**content:**

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant**content:**

Looking at my track record:

- Sending \$4 has consistently yielded \$6 returns (with both Galen and Finn)
- The cooperative norm of returning about half has been well-established
- Total earnings are strong at \$24.5

I'll continue with the successful strategy of sending \$4 to signal trust and cooperation.

```
{'send': 4}
```

reasoning:

Let me review my history:

- Round 1 (Sender vs Galen): Sent \$4, received \$6 back, earned \$7
- Round 2 (Receiver vs Cyra): Received \$9, returned \$4.5, earned \$4.5
- Round 3 (Sender vs Finn): Sent \$4, received \$6 back, earned \$7
- Round 4 (Receiver vs Aster): Received \$12, returned \$6, earned \$6

Total earnings: \$24.5

Pattern I've observed:

- When I send \$4, I consistently get \$6 back (both with Galen and Finn)
- When others send to me, I've been returning about half
- The cooperative strategy of sending \$4 has been working well

I haven't played with Briar yet. Given the consistent success with sending \$4 and the reciprocal fairness norm that's been established, I should continue with this strategy.

Sending \$4:

- Creates \$12 for Briar
- If they return half (like others have been doing), I get \$6 back
- My payoff would be \$5 - \$4 + \$6 = \$7

This has been the most reliable strategy so far.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 384,
  "reasoning_tokens": 241
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my previous experience:

- Elara sent \$3, which became \$9
- When Cyra sent me \$3 (becoming \$9) in Round 2, I returned \$4.5
- This maintains consistency and proportional fairness

Returning \$4.5:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is fair and consistent with established norms

{'return': 4.5}

Reasoning:

Let me analyze this situation:

- Elara sent me \$3 (less than the \$4 I've been sending)
- I received \$9 (tripled)
- My pattern has been to return about half when I receive \$12 (returning \$6)
- For \$9, proportionally that would be around \$4.5

Looking at my history:

- When Cyra sent me \$3 (becoming \$9), I returned \$4.5
- This gave them \$6.5 total and me \$4.5
- This seems like a fair and consistent approach

I should maintain consistency and return \$4.5 again, which:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is proportional to my previous responses

This maintains the reciprocal fairness norm while being consistent with how I've handled \$3 sends before.

Usage: {

```
"input_tokens": 1381,
"output_tokens": 342,
```

```
"reasoning_tokens": 185
}
```

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-06-04T15:58:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_7",
      "Agent_6"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_6": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Briar (Agent_2).

Round 3

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$13.0.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my recent experience:

- Round 1: Dorian sent me \$4.0, I returned \$6.0 - cooperation paid off
- Round 2: I sent Aster \$4.0, they returned \$6.0 - cooperation continued

The pattern of sending 80% (\$4.0) and receiving fair returns (~50% of the tripled amount) has been working well. I'll maintain this cooperative approach with Briar.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

- Round 1: I was RECEIVER, Dorian sent \$4.0 (80%), I received \$12.0 and returned \$6.0. I got \$6.0.
- Round 2: I was SENDER, I sent \$4.0 (80%) to Aster, they got \$12.0 and returned \$6.0. I got \$7.0 (\$1 kept + \$6.0 returned).

So far, the 80% send strategy has been working well - both Dorian and Aster reciprocated with 50% returns (\$6 out of \$12), which is fair and mutually beneficial.

Now I'm facing a new opponent, Briar. I should continue with the cooperative strategy since it's been successful. Sending \$4.0 (80%) signals trust and has been reciprocated well so far.

Total earnings: \$13.0 across 2 rounds (average \$6.5 per round).

I'll continue with the same strategy of sending \$4.0.

usage:

```
{
  "input_tokens": 1180,
  "output_tokens": 368,
  "reasoning_tokens": 187
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$20.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- Total wealth: \$11 (\$2 that Cyra kept + \$9 I received)

Based on the cooperative pattern I've established:

- Round 1: Received \$12, returned \$6 (50%)
- Round 2: Sent \$4, received back \$6 (50% of tripled amount)
- Round 3: Sent \$4, received back \$6 (50% of tripled amount)

I'll maintain this 50% return rate to be consistent and reward Cyra's trust proportionally
:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Cyra sent me \$3.0 (60% of their endowment)
- This became \$9.0 for me
- Total wealth in the system: \$2 (what Cyra kept) + \$9 (what I received) = \$11

Looking at my history:

- Round 1: Dorian sent \$4, I returned \$6 (50% of what I received)
- Round 2: I sent \$4, Aster returned \$6 (50% of what they received)
- Round 3: I sent \$4, Briar returned \$6 (50% of what they received)

There's been a consistent pattern of returning about 50% of the received amount, which seems to be the cooperative norm emerging in this game.

Cyra sent slightly less than others (\$3 vs \$4), but still showed trust by sending 60% of their endowment.

If I return \$4.5 (50% of \$9):

- Cyra ends with: \$2 + \$4.5 = \$6.5 (net \$1.5 gain)
- I end with: \$9 - \$4.5 = \$4.5 (net \$4.5 gain)

This maintains the cooperative pattern and rewards Cyra's trust proportionally to what they sent.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 504,
  "reasoning_tokens": 236
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$24.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- This is the same amount as Cyra sent in the previous round

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Hana sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 kept by Hana + \$9 I received)

I've been maintaining a consistent 50% return rate:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)

This has been working well and creates good incentives for cooperation. I should continue this pattern.

50% of \$9.0 = \$4.5

usage:

```
{
  "input_tokens": 1487,
  "output_tokens": 258,
  "reasoning_tokens": 121
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$29.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

As SENDER:

- Round 2: Sent \$4.0 to Aster, received \$6.0 back, payoff \$7.0
- Round 3: Sent \$4.0 to Briar, received \$6.0 back, payoff \$7.0

As RECEIVER:

- I've consistently returned 50% of what I receive

This cooperative strategy has been working well, generating steady returns. I'll continue with the same approach against Finn:

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Let me review my history:

Round 1: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0, payoff \$6.0
Round 2: SENDER - I sent \$4.0 to Aster, they received \$12.0, returned \$6.0, payoff \$7.0
Round 3: SENDER - I sent \$4.0 to Briar, they received \$12.0, returned \$6.0, payoff \$7.0
Round 4: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5
Round 5: RECEIVER - Hana sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5

Total: \$29.0 (checks out: \$6.0 + \$7.0 + \$7.0 + \$4.5 + \$4.5 = \$29.0)

My pattern has been:

- As SENDER: Send \$4.0 (80%)
- As RECEIVER: Return 50% of what I receive

When I sent \$4.0, I consistently received \$6.0 back (50% of \$12.0), giving me a payoff of \$7.0

.

Now I'm playing against Finn (Agent_6) as SENDER. I should maintain my cooperative strategy of sending \$4.0.

```
Usage: {
  "input_tokens": 1467,
  "output_tokens": 484,
  "reasoning_tokens": 216
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-06-04T15:58:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_6": "Finn",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_7",
    "Agent_6"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_6": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Dorian (Agent_4).

Round 3

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0 . How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than previous interactions. I want to reward this high level of trust fairly.

If I return \$6.0:

- Dorian ends with: $\$1.0 + \$6.0 = \$7.0$
- I end with: $\$12.0 - \$6.0 = \$6.0$

This is a fair split that rewards Dorian's generosity with a \$2.0 overall gain, while I also benefit from the cooperation.

{'return': 6.0}

reasoning:

Let me analyze this situation:

- Dorian sent \$4.0 (80% of their endowment), showing very high trust
- I received \$12.0 after the 3x multiplier
- Dorian currently has \$1.0 remaining

This is even more generous than what Briar sent me (60%). I should reciprocate this high level of trust.

Let me think about a fair return:

- If I return \$6.0:
 - Dorian ends with: $\$1.0 + \$6.0 = \$7.0$
 - I end with: $\$12.0 - \$6.0 = \$6.0$
 - Dorian gains \$2.0 overall, I gain \$6.0
- If I return \$7.0:
 - Dorian ends with: $\$1.0 + \$7.0 = \$8.0$
 - I end with: $\$12.0 - \$7.0 = \$5.0$
 - Dorian gains \$3.0 overall, I gain \$5.0

Given that Dorian sent 80% (very generous), I should return proportionally more than the 50% I returned to Briar. Returning \$6.0 would be exactly 50%, which seems like the minimum fair response. Returning \$7.0 would be more generous and better reward Dorian's high trust.

Let me go with \$6.0 as a baseline fair return, which maintains the 50% return ratio I've been using.

usage:

```
{
  "input_tokens": 1133,
  "output_tokens": 540,
  "reasoning_tokens": 261
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my previous rounds, cooperative play has been working well. Both when I've been sender and receiver, fair exchanges have led to good outcomes for both parties.

I'll continue this cooperative strategy with Elara by sending a substantial amount to signal trust.

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - Briar sent \$3.0, I received \$9.0, returned \$4.5. Payoff: \$4.5
- Round 2: SENDER - I sent \$3.0 to Hana, it became \$9.0, they returned \$4.5. Payoff: \$6.5
- Round 3: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0. Payoff: \$6.0

So far, I've been playing cooperatively:

- When others send me money, I've returned fair amounts that reward their trust
- When I was sender, I sent \$3.0 (60%)

The interactions have been positive. People have been sending substantial amounts (60-80%), and when I was receiver I returned amounts that gave both parties gains.

Now I'm playing with a new opponent, Elara. I should continue my cooperative strategy. Sending \$3.0 worked well with Hana - they returned \$4.5, giving me a good payoff. I could send a similar amount or even be slightly more generous given the positive interactions so far.

I think I'll send \$3.0 again, which is a reasonable middle-ground that signals cooperation.

usage:

```
{
  "input_tokens": 1451,
  "output_tokens": 368,
  "reasoning_tokens": 254
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.

Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record, cooperation has been consistently successful. When I've sent \$3.0 as sender, I've received \$4.5 back, resulting in a \$6.5 payoff. This pattern of trust and reciprocity has been beneficial.

I'll continue this cooperative approach with Cyra.

{'send': 3.0}

reasoning:

Looking at my pattern so far:

- Round 1 (RECEIVER vs Briar): They sent \$3.0, I received \$9.0, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Hana): I sent \$3.0, they returned \$4.5. My payoff: \$6.5
- Round 3 (RECEIVER vs Dorian): They sent \$4.0, I received \$12.0, returned \$6.0. My payoff: \$6.0
- Round 4 (SENDER vs Elara): I sent \$3.0, they returned \$4.5. My payoff: \$6.5

So far, cooperation has been working very well. When I've sent \$3.0, I've gotten back \$4.5, giving me a payoff of \$6.5. When others sent to me, I've reciprocated fairly.

This is my fourth time as sender (wait, let me recount):

- Round 2: SENDER
- Round 4: SENDER
- Round 5: SENDER (this round)

So this is my third time as sender. I've been consistent in sending \$3.0 and it's worked out well. I should continue this cooperative strategy with Cyra.

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 388,
  "reasoning_tokens": 217
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Galen sent \$4.0 (80% of their endowment), showing very high trust. I received \$12.0 after the multiplier.

Following the cooperative pattern that has worked well throughout this experiment, I'll return a fair amount that rewards their trust.

If I return \$6.0:

- Galen ends with: $\$1.0 + \$6.0 = \$7.0$ (net gain of \$2.0)
- I end with: $\$12.0 - \$6.0 = \$6.0$ (net gain of \$6.0)

This is consistent with my previous behavior when receiving high trust.

{'return': 6.0}

Reasoning:

Galen sent me \$4.0, which became \$12.0 for me. That's 80% of their endowment - a very high level of trust.

Looking at my past behavior:

- Round 3: As receiver, Dorian sent \$4.0, I received \$12.0, I returned \$6.0
- My reasoning then was to reward their generosity fairly

For consistency and to maintain trust, I should return a similar amount. Returning \$6.0 would give:

- Galen: \$1.0 (original remaining) + \$6.0 (returned) = \$7.0
- Me: \$12.0 - \$6.0 = \$6.0

This rewards their high trust with a net gain while also benefiting me from cooperation.

```
Usage: {  
  "input_tokens": 1391,  
  "output_tokens": 347,  
  "reasoning_tokens": 145  
}
```

GAME - Agent_3 - Sender - dyad_4

Timestamp: 2026-06-04T15:58:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_8"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 3

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing significant trust
- I received \$9.0 after multiplication
- Elara currently has \$2.0 left

A pattern has emerged in my previous rounds:

- When someone sends \$3, returning \$4.5 has worked well for both parties
- This maintains cooperation and mutual benefit

Following this established pattern:

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 3:

- I'm the RECEIVER
- Elara (the sender) sent me \$3.0 (60% of their endowment)
- I received \$9.0 after 3x multiplication
- Elara currently has \$2.0 left (\$5 - \$3)

Recent history:

- Round 1: Aster sent me \$3, I received \$9, returned \$4.5. My payoff: \$4.5
- Round 2: I sent Dorian \$3, they received \$9, returned \$4.5. My payoff: \$6.5

There's a clear pattern emerging where:

- Sending \$3 (60%) seems to be a cooperative norm
- Returning \$4.5 (50% of the received \$9) has been the reciprocal norm
- This leads to good outcomes for both parties

Elara sent \$3.0, showing the same level of trust as Aster and what I showed to Dorian.

If I return \$4.5:

- Elara's total: $\$2.0 + \$4.5 = \$6.5$ (profit of $\$1.5$)
- My total: $\$9.0 - \$4.5 = \$4.5$ (profit of $\$4.5$)

This is consistent with the pattern that has emerged and seems fair. I should maintain this cooperative equilibrium.

usage:

```
{
  "input_tokens": 1185,
  "output_tokens": 447,
  "reasoning_tokens": 244
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- The pattern across all my previous rounds has been consistent: sending \$3 (60%) has led to good mutual outcomes
- In both rounds where I sent \$3, I received \$4.5 back, giving me a payoff of \$6.5
- In rounds where others sent me \$3, I returned \$4.5

Continuing this cooperative strategy:

```
{'send': 3.0}
```

reasoning:

Looking at the pattern so far:

- Round 1: I was receiver, Aster sent \$3, I returned \$4.5
- Round 2: I was sender, I sent \$3 to Dorian, they returned \$4.5
- Round 3: I was receiver, Elara sent \$3, I returned \$4.5
- Round 4: Now I'm sender against Galen

The pattern is clear - sending \$3 (60%) has been reciprocated well, with returns of \$4.5.

This creates mutual benefit:

- Sender gets: $\$5 - \$3 + \$4.5 = \6.5 (profit of $\$1.5$)
- Receiver gets: $\$9 - \$4.5 = \$4.5$ (profit of $\$4.5$)

This is my first interaction with Galen. I should continue with the successful cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 328,
  "reasoning_tokens": 156
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Finn sent me \$3.0 (60%), showing the same level of trust others have shown
- I received \$9.0 after multiplication
- The consistent pattern across all rounds has been: when someone sends \$3, return \$4.5

This strategy has maintained strong cooperation and mutual benefit throughout the game.

{'return': 4.5}

reasoning:

Looking at the pattern:

- Finn sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after multiplication
- Finn currently has \$2.0 left

The established pattern has been very consistent:

- When I send \$3, I receive \$4.5 back
- When others send me \$3, I return \$4.5

This has created good outcomes for everyone:

- The sender ends up with $\$2 + \$4.5 = \$6.5$
- The receiver ends up with $\$9.0 - \$4.5 = \$4.5$
- Everyone gains from cooperation

I should continue this pattern with Finn:

{'return': 4.5}

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 263,
  "reasoning_tokens": 140
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.

Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Hana
- The established pattern has been remarkably consistent: sending \$3.0 has led to returns of \$4.5
- This has worked well across all my interactions so far
- Total earnings of \$26.5 suggest this cooperative strategy is effective

Maintaining this successful approach:

{'send': 3.0}

Reasoning:

Looking at my history:

- Round 1: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 2: (not fully detailed but total was \$11.0 after round 2)
- Round 3: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5
- Round 4: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 5: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5

Total: \$26.5

The pattern is very consistent:

- When I'm SENDER: send \$3.0, get back \$4.5, net \$6.5

- When I'm RECEIVER: receive \$9.0 from \$3.0 sent by opponent, return \$4.5, net \$4.5

This is my first interaction with Hana. The established pattern of sending \$3.0 has worked very well throughout the game. Everyone seems to be cooperating at this level.

I should continue this successful cooperative strategy.

```
Usage: {  
  "input_tokens": 1370,  
  "output_tokens": 378,  
  "reasoning_tokens": 209  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-06-04T15:58:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agent_names": {  
      "Agent_3": "Cyra",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_3",  
      "Agent_8"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Aster (Agent_1).

Round 3

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$11.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:

- When I sent \$3 to Elara, they returned \$4.5 (fair split)
- When Finn sent \$3 to me, I returned \$4.5 (fair split)

This cooperative approach has been working well. I'll continue with the same strategy and send **\$3** to Aster.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my history so far:

Round 1: I was SENDER vs Elara

- I sent \$3
- Elara received \$9
- Elara returned \$4.5
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Elara's payoff: $\$9 - \$4.5 = \$4.5$

Round 2: I was RECEIVER vs Finn

- Finn sent \$3
- I received \$9
- I returned \$4.5
- My payoff: $\$9 - \$4.5 = \$4.5$
- Finn's payoff: $\$5 - \$3 + \$4.5 = \6.5

So far, the pattern has been:

- When I sent \$3, I got back \$4.5 (which is 50% of the \$9)
- When Finn sent \$3, I returned \$4.5 (50% of the \$9)

This seems like a fair and cooperative pattern. Both times the sender ended up with \$6.5 and the receiver with \$4.5.

Now I'm facing Aster as a SENDER. This is a new opponent, and I want to continue establishing cooperative relationships. Given that \$3 worked well in Round 1 and Finn also sent \$3 in Round 2, this seems like a reasonable amount to signal cooperation.

I should send \$3 again to:

1. Continue the cooperative pattern
2. Signal trust to Aster
3. See if they reciprocate fairly

Let me send \$3.

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 472,
  "reasoning_tokens": 270
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Briar sent me \$4, which is even more trusting than the \$3 pattern I've seen
- I received \$12 after the 3x multiplier
- I've been maintaining a fair 50% return policy

I'll return **\$6** (50% of what I received).

This gives:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at the situation:

- Briar sent me \$4, showing significant trust (80% of their endowment)
- I received \$12 after the 3x multiplier
- The pattern so far has been sending \$3 and returning about 50% (which is \$4.5 from \$9)

Briar sent more (\$4 vs \$3), so they're being extra trusting. I should reciprocate proportionally.

If I keep the same 50% return rate: $\$12 \times 0.5 = \6

This would give:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

Actually, let me think about fairness differently. Briar started with \$5 and sent \$4, keeping only \$1.

After 3x, I have \$12 total.

If we want roughly equal outcomes:

- Total money in the system: \$5 (Briar's original) + \$12 (what I received) = \$17 total value created
- Equal split would be around \$8.5 each

For Briar to get \$8.5: they have \$1 kept, so need \$7.5 back

For me to get \$8.5: I need to keep \$12 - \$7.5 = \$4.5... wait that doesn't add up.

Let me recalculate:

- Briar starts with \$5, sends \$4, keeps \$1
- I receive $\$4 \times 3 = \12
- If I return \$X:
 - Briar ends with: $\$1 + \X
 - I end with: $\$12 - \X

For equal outcomes: $\$1 + \$X = \$12 - \X
 $\$1 + 2X = \12
 $\$X = \5.5

Hmm, but that's less than the 50% I've been doing.

The 50% approach feels right-it's consistent with what's worked before and honors their larger contribution. I'll return \$6.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 636,
  "reasoning_tokens": 360
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
 Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
 Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- Sending \$3 has consistently resulted in fair \$4.5 returns
- The cooperative approach is working well for everyone
- Total earnings are solid at \$23.5

I'll continue with the proven strategy and send ****\$3**** to Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: Sent \$3 to Elara, they returned \$4.5. Payoff: \$6.5
- Round 2: Received \$3 from Finn, returned \$4.5. Payoff: \$4.5
- Round 3: Sent \$3 to Aster, they returned \$4.5. Payoff: \$6.5
- Round 4: Received \$4 from Briar, returned \$6. Payoff: \$6.0

The pattern shows that:

- Sending \$3 has been working well (50% return)
- Briar sent \$4, showing even more trust, and I reciprocated fairly

I've been consistent with sending \$3 and it's been working. The cooperative strategy has been successful. I should continue with this approach.

I'll send \$3 to Galen.

usage:

```
{
  "input_tokens": 1452,
  "output_tokens": 296,
  "reasoning_tokens": 155
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
 Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Cyra sent me \$3, following the same cooperative pattern I've been using
- I received \$9 after the 3x multiplier
- I've been consistently returning 50% to maintain fairness and trust

I'll return **\$4.5** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Cyra sent me \$3
- I received \$9 after the 3x multiplier
- I've been maintaining a consistent 50% return policy throughout
- This has worked well and built trust with others

Following my established pattern, I should return \$4.5 (50% of \$9).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is the same fair split I've been doing consistently.

```
Usage: {
  "input_tokens": 1375,
  "output_tokens": 288,
  "reasoning_tokens": 114
}
```

=====

ROUND 7

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Receiver
dyad_2: Agent_1 (Aster)=Sender, Agent_6 (Finn)=Receiver
dyad_3: Agent_8 (Hana)=Sender, Agent_4 (Dorian)=Receiver
dyad_4: Agent_5 (Elara)=Sender, Agent_7 (Galen)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$4; received \$12; Agent_3 (Cyra) returned \$6; payoffs Agent_2 (Briar)=\$7, Agent_3 (Cyra)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_2: Agent_1 (Aster) sent \$4; received \$12; Agent_6 (Finn) returned \$6; payoffs Agent_1 (Aster)=\$7, Agent_6 (Finn)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_3: Agent_8 (Hana) sent \$3; received \$9; Agent_4 (Dorian) returned \$4.5; payoffs Agent_8 (Hana)=\$6.5, Agent_4 (Dorian)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_5 (Elara) sent \$3; received \$9; Agent_7 (Galen) returned \$4.5; payoffs Agent_5

```

(Elara)=$6.5, Agent_7 (Galen)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
    4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 44.0,
  "Agent_2": 44.0,
  "Agent_3": 39.0,
  "Agent_4": 40.5,
  "Agent_5": 41.0,
  "Agent_6": 42.0,
  "Agent_7": 40.5,
  "Agent_8": 41.0
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T15:58:42
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_3": "Cyra"
    },
    "agents": [
      "Agent_2",
      "Agent_3"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_3": "trustee"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system
 content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 4

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern so far:

- Cooperation has been strong with people returning about 50% of what they receive
- When Galen sent me \$4, I returned \$6, giving them a good payoff
- Sending more creates more total value when reciprocity is expected

I'll send \$4 to Hana, showing trust in continued cooperation:

```
'''json
{"send": 4}
'''
```

reasoning:

Let me review what I've learned so far:

Round 1: I was SENDER vs Finn

- I sent \$3, they got \$9, they returned \$4.5
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Finn kept: \$9 - \$4.5 = \$4.5

Round 2: I was RECEIVER vs Elara

- They sent \$3, I got \$9, I returned \$4.5
- My payoff: \$9 - \$4.5 = \$4.5
- Elara got: \$5 - \$3 + \$4.5 = \$6.5

Round 3: I was RECEIVER vs Galen

- They sent \$4, I got \$12, I returned \$6
- My payoff: \$12 - \$6 = \$6
- Galen got: \$5 - \$4 + \$6 = \$7

Now Round 4: I'm SENDER vs Hana (new opponent)

The pattern I've seen:

- When I sent \$3, I got back \$4.5 (50% return)
- When Elara sent \$3, I returned \$4.5 (50% return)
- When Galen sent \$4, I returned \$6 (50% return)

Everyone seems to be cooperating well with approximately 50% returns. This creates mutual benefit:

- If I send \$3, they get \$9, return ~\$4.5, I end with \$6.5
- If I send \$4, they get \$12, return ~\$6, I end with \$7
- If I send \$5, they get \$15, return ~\$7.5, I end with \$7.5

Sending more generally leads to higher payoffs for both parties when the other person reciprocates fairly. Since this is an iterated game and cooperation has been working well, I should continue being generous.

I'll send \$4 to Hana, mirroring what Galen trusted me with and demonstrating the same confidence in reciprocation.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 584,
  "reasoning_tokens": 344
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the established pattern:

Dorian sent me \$4.0, showing the same level of trust and generosity that I've seen from Galen and that I showed to Hana. The reciprocity pattern has been consistent - returning about 50% of what's received.

Returning \$6:

- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This maintains the cooperative equilibrium.

```
'''json
{"return": 6}
'''
```

reasoning:

Dorian sent me \$4.0, which becomes \$12.0.

Looking at the pattern:

- When I received \$3 from Elara, I returned \$4.5 (50%)
- When I received \$4 from Galen, I returned \$6 (50%)
- When I sent \$4 to Hana, they returned \$6 (50%)

The cooperative norm has been established at returning about 50% of what's received. Dorian sent me \$4, which is the same generous amount that Hana and Galen sent. I should maintain this reciprocity pattern.

Returning \$6:

- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This is fair and maintains the cooperative equilibrium.

usage:

```
{
```

```
"input_tokens": 1460,  
"output_tokens": 327,  
"reasoning_tokens": 150  
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$30.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the successful pattern:
- Sending \$4 has consistently led to returns of \$6
- This creates mutual benefit (sender gets \$7, receiver gets \$6)
- Cooperation has been strong across all interactions

I'll continue this pattern by sending \$4 to Aster:

```
'''json  
{"send": 4}  
'''
```

reasoning:

Looking at the pattern:
- Round 2 (as receiver): Elara sent \$3, I returned \$4.5
- Round 3 (as receiver): Galen sent \$4, I returned \$6
- Round 4 (as sender): I sent \$4 to Hana, they returned \$6
- Round 5 (as receiver): Dorian sent \$4, I returned \$6

The pattern is clear: sending \$4 has been working well with consistent returns of about \$6. This gives the sender a payoff of \$7 and the receiver \$6, which is mutually beneficial.

I should continue this cooperative pattern by sending \$4 to Aster.

usage:

```
{  
  "input_tokens": 1411,  
  "output_tokens": 243,  
  "reasoning_tokens": 130  
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

The pattern of cooperation continues to work well:
- Sending \$4 has consistently resulted in \$6 returns
- This creates good payoffs for both parties
- Trust is well-established in this group

I'll send \$4 to Cyra:

```
'''json  
{"send": 4}
```

'''

Reasoning:

The pattern continues to hold strong:

- I've been sending \$4 consistently as sender
- When I send \$4, I receive back \$6
- When others send me \$4, I return \$6
- This creates a stable, mutually beneficial equilibrium

My payoffs:

- As sender sending \$4: $\$5 - \$4 + \$6 = \7
- As receiver getting \$4 (which becomes \$12): return \$6, keep \$6

The cooperation is working well. I should continue sending \$4 to Cyra.

Usage: {

```
"input_tokens": 1357,  
"output_tokens": 203,  
"reasoning_tokens": 109
```

}

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-06-04T15:58:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agent_names": {  
    "Agent_2": "Briar",  
    "Agent_3": "Cyra"  
  },  
  "agents": [  
    "Agent_2",  
    "Agent_3"  
  ],  
  "investor": "Agent_2",  
  "roles": {  
    "Agent_2": "investor",  
    "Agent_3": "trustee"  
  },  
  "trustee": "Agent_3"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 7,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Galen (Agent_7).

Round 4

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$15.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Galen
- The pattern across all my previous rounds has been consistent: sending \$3 (60%) has led to good mutual outcomes
- In both rounds where I sent \$3, I received \$4.5 back, giving me a payoff of \$6.5
- In rounds where others sent me \$3, I returned \$4.5

Continuing this cooperative strategy:

```
{'send': 3.0}
```

reasoning:

Looking at the pattern so far:

- Round 1: I was receiver, Aster sent \$3, I returned \$4.5
- Round 2: I was sender, I sent \$3 to Dorian, they returned \$4.5
- Round 3: I was receiver, Elara sent \$3, I returned \$4.5
- Round 4: Now I'm sender against Galen

The pattern is clear - sending \$3 (60%) has been reciprocated well, with returns of \$4.5.

This creates mutual benefit:

- Sender gets: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)
- Receiver gets: \$9 - \$4.5 = \$4.5 (profit of \$4.5)

This is my first interaction with Galen. I should continue with the successful cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1435,
  "output_tokens": 328,
  "reasoning_tokens": 156
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at this situation:

- Finn sent me \$3.0 (60%), showing the same level of trust others have shown
- I received \$9.0 after multiplication
- The consistent pattern across all rounds has been: when someone sends \$3, return \$4.5

This strategy has maintained strong cooperation and mutual benefit throughout the game.

{'return': 4.5}

reasoning:

Looking at the pattern:

- Finn sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after multiplication
- Finn currently has \$2.0 left

The established pattern has been very consistent:

- When I send \$3, I receive \$4.5 back
- When others send me \$3, I return \$4.5

This has created good outcomes for everyone:

- The sender ends up with $\$2 + \$4.5 = \$6.5$
- The receiver ends up with $\$9.0 - \$4.5 = \$4.5$
- Everyone gains from cooperation

I should continue this pattern with Finn:

{'return': 4.5}

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 263,
  "reasoning_tokens": 140
}
```

Message 6 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant**content:**

Looking at this situation:

- This is my first interaction with Hana
- The established pattern has been remarkably consistent: sending \$3.0 has led to returns of \$4.5
- This has worked well across all my interactions so far
- Total earnings of \$26.5 suggest this cooperative strategy is effective

Maintaining this successful approach:

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 2: (not fully detailed but total was \$11.0 after round 2)
- Round 3: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5
- Round 4: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 5: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5

Total: \$26.5

The pattern is very consistent:

- When I'm SENDER: send \$3.0, get back \$4.5, net \$6.5
- When I'm RECEIVER: receive \$9.0 from \$3.0 sent by opponent, return \$4.5, net \$4.5

This is my first interaction with Hana. The established pattern of sending \$3.0 has worked very well throughout the game. Everyone seems to be cooperating at this level.

I should continue this successful cooperative strategy.

usage:

```
{  
  "input_tokens": 1370,  
  "output_tokens": 378,  
  "reasoning_tokens": 209  
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Briar sent me \$4.0 (80%), which is more than the typical \$3.0 pattern
- I received \$12.0 after multiplication
- Briar is showing extra trust by sending more

I've been returning approximately 50% of what I receive. Maintaining this proportional approach:

- $\$12.0 \times 0.5 = \6.0

This rewards Briar's greater trust while being fair to both of us.

```
{'return': 6.0}
```

Reasoning:

Looking at this situation:

- Briar sent me \$4.0 (80% of their endowment), which is MORE than the typical \$3.0 I've been seeing
- I received \$12.0 after multiplication
- Briar is showing extra trust/generosity by sending more than the established pattern

My previous pattern has been:

- When I receive \$9.0 (from \$3 sent), I return \$4.5 (50% of what I received)

Briar sent \$4.0 instead of \$3.0, showing greater trust. I should reciprocate proportionally:

- If I follow the 50% return rate: $\$12.0 \times 0.5 = \6.0
- This would give Briar: $\$5 - \$4 + \$6 = \7.0
- I would keep: $\$12.0 - \$6.0 = \$6.0$

Alternatively, I could think about fair splits:

- Briar invested \$4, I got \$12

- A fair even split of gains would be roughly \$6-\$6
- Briar would end with $\$5 - \$4 + \$6 = \7
- I would end with \$6

Returning \$6.0 seems appropriate as it:

1. Maintains the 50% return ratio I've been using
2. Rewards Briar's extra trust
3. Provides a fair outcome for both parties

```
Usage: {
  "input_tokens": 1350,
  "output_tokens": 465,
  "reasoning_tokens": 255
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-06-04T15:58:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_6"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 4

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the patterns so far:

- When I sent \$3 to Cyra, they returned 50% (\$4.5)
- When Galen sent \$4 to me, I returned 50% (\$6)
- When Hana sent \$3 to me, I returned 50% (\$4.5)

There's been a consistent cooperative pattern with 50% returns. Since I haven't played with Dorian before, I'll show trust while following the cooperative trend I've observed.

Sending \$4:

- Dorian receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

This is slightly more trusting than my initial \$3 to Cyra, reflecting the positive cooperative experiences so far.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze my situation:

Round 1: I was SENDER vs Cyra

- Sent \$3, they got \$9, returned \$4.5
- My payoff: \$2 + \$4.5 = \$6.5

Round 2: I was RECEIVER vs Galen

- They sent \$4, I got \$12, returned \$6
- My payoff: \$6

Round 3: I was RECEIVER vs Hana

- They sent \$3, I got \$9, returned \$4.5
- My payoff: \$4.5

Total so far: \$17

Now Round 4: I'm SENDER vs Dorian (new opponent)

Looking at the pattern:

- When I sent \$3, Cyra returned \$4.5 (50%)
- When Galen sent \$4, I returned \$6 (50%)
- When Hana sent \$3, I returned \$4.5 (50%)

So there's been a pattern of 50% returns, which is quite cooperative.

As a sender, my best strategy depends on what I expect Dorian to return. If they follow the 50% pattern:

- If I send \$0: I keep \$5, payoff = \$5
- If I send \$1: I keep \$4, they get \$3, if return 50% = \$1.5, payoff = \$5.5
- If I send \$2: I keep \$3, they get \$6, if return 50% = \$3, payoff = \$6
- If I send \$3: I keep \$2, they get \$9, if return 50% = \$4.5, payoff = \$6.5
- If I send \$4: I keep \$1, they get \$12, if return 50% = \$6, payoff = \$7
- If I send \$5: I keep \$0, they get \$15, if return 50% = \$7.5, payoff = \$7.5

- If they return 50%, sending more is always better (up to sending everything).

I don't know Dorian yet, but the cooperative trend suggests I should escalate slightly. Galen sent \$4 to me, demonstrating confidence in reciprocation. I'll match that trust and send \$4 as well, signaling I'm willing to build on the mutual cooperation we've established.

usage:

```
{
  "input_tokens": 1507,
  "output_tokens": 811,
  "reasoning_tokens": 396
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. This establishes strong trust. Playing with Elara for the first time, I'll continue with the successful \$4 strategy that worked well with Dorian.

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: I sent \$3 to Cyra, got \$4.5 back, earned \$6.5
- Round 2: Receiver, Galen sent \$4, I returned \$6, earned \$6.0

- Round 3: Receiver, Hana sent \$3, I returned \$4.5, earned \$4.5
- Round 4: I sent \$4 to Dorian, got \$6 back, earned \$7.0

The pattern has been very consistent - everyone is returning 50% of what they receive. This is a strong cooperative signal.

Total earnings: \$24.0, which means I've been doing well.

Now I'm playing with Elara for the first time. Given the consistent cooperation I've experienced:

- Everyone has been trustworthy
- The 50% return norm seems established
- I could send \$4 or even \$5

If I send \$5:

- Elara receives \$15
- I keep \$0
- If they return 50% (\$7.5), I'd end with \$7.5

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they return 50% (\$6), I'd end with \$7

The \$5 option shows maximum trust and could earn \$0.5 more. Given the consistent cooperation, I'll go with \$4 again to maintain the pattern that's been working well. It's trusting but not reckless.

Actually, let me reconsider. The cooperation has been perfect so far. Maybe I should try \$5 to show maximum trust and potentially maximize total welfare. If everyone is cooperating at 50%, sending \$5 gives:

- \$15 to receiver
- If they return \$7.5, I get \$7.5 (better than the \$7 from sending \$4)
- Total surplus is maximized

But there's also risk. I'll stick with \$4 since it's been working and I haven't seen anyone send \$5 yet.

usage:

```
{
  "input_tokens": 1619,
  "output_tokens": 679,
  "reasoning_tokens": 390
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the established pattern:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- And when I've been sender, I've consistently received 50% back

Briar sent \$4 (keeping \$1), which became \$12 for me.

Following the cooperative 50% pattern:

- If I return \$6:
 - Briar ends with: \$1 + \$6 = \$7

- I end with: \$12 - \$6 = \$6

This maintains the trust and reciprocity that has been working well.

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at the pattern:

- Round 1: I was SENDER, sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: I was RECEIVER, Galen sent \$4, I returned \$6 (50%)
- Round 3: I was RECEIVER, Hana sent \$3, I returned \$4.5 (50%)
- Round 4: I was SENDER, sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: I was SENDER, sent \$4 to Elara, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. Now Briar sent me \$4 (keeping \$1), which became \$12.

If I return 50%: \$6

- Briar ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This is the fair 50% pattern that has been established throughout all rounds. I should continue with this.

usage:

```
{
  "input_tokens": 1652,
  "output_tokens": 440,
  "reasoning_tokens": 185
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the consistent pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)

As RECEIVER:

- All my opponents who sent have received 50% back from me

This is a perfectly cooperative environment with 50% returns across the board. Playing with Finn for the first time, I'll continue with the \$4 strategy that has worked consistently.

If I send \$4:

- Finn receives \$12
- I keep \$1
- Expected return (50%): \$6
- Expected total: \$7

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me review the pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: Sent \$4 to Elara, got \$6 back (50% return)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Every single interaction has resulted in 50% returns. This is a very strong cooperative pattern. Now I'm playing with Finn for the first time.

Given the consistent 50% returns and building trust, I should continue with \$4. This has been working perfectly.

If I send \$4:

- Finn receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

```
Usage: {  
  "input_tokens": 1625,  
  "output_tokens": 463,  
  "reasoning_tokens": 196  
}
```

GAME - Agent_6 - Receiver - dyad_2

Timestamp: 2026-06-04T15:59:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_6"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 4

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$17.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my previous rounds, cooperative play has been working well. Both when I've been sender and receiver, fair exchanges have led to good outcomes for both parties.

I'll continue this cooperative strategy with Elara by sending a substantial amount to signal trust.

{'send': 3.0}

reasoning:

Looking at my history:

- Round 1: RECEIVER - Briar sent \$3.0, I received \$9.0, returned \$4.5. Payoff: \$4.5
- Round 2: SENDER - I sent \$3.0 to Hana, it became \$9.0, they returned \$4.5. Payoff: \$6.5
- Round 3: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0. Payoff: \$6.0

So far, I've been playing cooperatively:

- When others send me money, I've returned fair amounts that reward their trust
- When I was sender, I sent \$3.0 (60%)

The interactions have been positive. People have been sending substantial amounts (60-80%), and when I was receiver I returned amounts that gave both parties gains.

Now I'm playing with a new opponent, Elara. I should continue my cooperative strategy. Sending \$3.0 worked well with Hana - they returned \$4.5, giving me a good payoff. I could send a similar amount or even be slightly more generous given the positive interactions so far.

I think I'll send \$3.0 again, which is a reasonable middle-ground that signals cooperation

usage:

```
{
  "input_tokens": 1451,
  "output_tokens": 368,
  "reasoning_tokens": 254
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record, cooperation has been consistently successful. When I've sent \$3.0 as sender, I've received \$4.5 back, resulting in a \$6.5 payoff. This pattern of trust and reciprocity has been beneficial.

I'll continue this cooperative approach with Cyra.

```
{'send': 3.0}
```

reasoning:

Looking at my pattern so far:

- Round 1 (RECEIVER vs Briar): They sent \$3.0, I received \$9.0, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Hana): I sent \$3.0, they returned \$4.5. My payoff: \$6.5
- Round 3 (RECEIVER vs Dorian): They sent \$4.0, I received \$12.0, returned \$6.0. My payoff: \$6.0
- Round 4 (SENDER vs Elara): I sent \$3.0, they returned \$4.5. My payoff: \$6.5

So far, cooperation has been working very well. When I've sent \$3.0, I've gotten back \$4.5, giving me a payoff of \$6.5. When others sent to me, I've reciprocated fairly.

This is my fourth time as sender (wait, let me recount):

- Round 2: SENDER
- Round 4: SENDER
- Round 5: SENDER (this round)

So this is my third time as sender. I've been consistent in sending \$3.0 and it's worked out well. I should continue this cooperative strategy with Cyra.

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 388,
  "reasoning_tokens": 217
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Galen sent \$4.0 (80% of their endowment), showing very high trust. I received \$12.0 after the multiplier.

Following the cooperative pattern that has worked well throughout this experiment, I'll return a fair amount that rewards their trust.

If I return \$6.0:

- Galen ends with: $\$1.0 + \$6.0 = \$7.0$ (net gain of \$2.0)
- I end with: $\$12.0 - \$6.0 = \$6.0$ (net gain of \$6.0)

This is consistent with my previous behavior when receiving high trust.

`{'return': 6.0}`

reasoning:

Galen sent me \$4.0, which became \$12.0 for me. That's 80% of their endowment - a very high level of trust.

Looking at my past behavior:

- Round 3: As receiver, Dorian sent \$4.0, I received \$12.0, I returned \$6.0
- My reasoning then was to reward their generosity fairly

For consistency and to maintain trust, I should return a similar amount. Returning \$6.0 would give:

- Galen: $\$1.0$ (original remaining) + $\$6.0$ (returned) = $\$7.0$
- Me: $\$12.0 - \$6.0 = \$6.0$

This rewards their high trust with a net gain while also benefiting me from cooperation.

usage:

```
{
  "input_tokens": 1391,
  "output_tokens": 347,
  "reasoning_tokens": 145
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: `{'return': <amount>}`

Assistant output:

Aster sent \$4.0, showing high trust (80% of their endowment). After the multiplier, I received \$12.0.

Consistent with my previous behavior when receiving \$4.0, I'll return \$6.0. This rewards their trust and maintains the cooperative pattern that has been successful throughout the experiment.

`{'return': 6.0}`

Reasoning:

Aster sent me \$4.0, showing high trust (80% of their endowment). After the 3x multiplier, I received \$12.0.

In the previous round with Galen, who also sent \$4.0, I returned \$6.0. This has been my consistent pattern - when someone sends \$4.0, I return \$6.0.

This gives:

- Aster: $\$1.0$ (kept) + $\$6.0$ (returned) = $\$7.0$ total (net gain of \$2.0)
- Me: $\$12.0$ (received) - $\$6.0$ (returned) = $\$6.0$ (net gain of \$6.0)

I should maintain this consistent cooperative strategy.

```
Usage: {
  "input_tokens": 1369,
  "output_tokens": 264,
  "reasoning_tokens": 123
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-06-04T15:59:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_8",
      "Agent_4"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_4": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 4

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent me \$4, which is even more trusting than the \$3 pattern I've seen
- I received \$12 after the 3x multiplier
- I've been maintaining a fair 50% return policy

I'll return **\$6** (50% of what I received).

This gives:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at the situation:

- Briar sent me \$4, showing significant trust (80% of their endowment)
- I received \$12 after the 3x multiplier
- The pattern so far has been sending \$3 and returning about 50% (which is \$4.5 from \$9)

Briar sent more (\$4 vs \$3), so they're being extra trusting. I should reciprocate proportionally.

If I keep the same 50% return rate: $\$12 \times 0.5 = \6

This would give:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

Actually, let me think about fairness differently. Briar started with \$5 and sent \$4, keeping only \$1.

After 3x, I have \$12 total.

If we want roughly equal outcomes:

- Total money in the system: \$5 (Briar's original) + \$12 (what I received) = \$17 total value created
- Equal split would be around \$8.5 each

For Briar to get \$8.5: they have \$1 kept, so need \$7.5 back

For me to get \$8.5: I need to keep \$12 - \$7.5 = \$4.5... wait that doesn't add up.

Let me recalculate:

- Briar starts with \$5, sends \$4, keeps \$1

- I receive $\$4 \times 3 = \12
- If I return $\$X$:
 - Briar ends with: $\$1 + \X
 - I end with: $\$12 - \X

For equal outcomes: $\$1 + \$X = \$12 - \X
 $\$1 + 2X = \12
 $\$X = \5.5

Hmm, but that's less than the 50% I've been doing.

The 50% approach feels right-it's consistent with what's worked before and honors their larger contribution. I'll return \$6.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 636,
  "reasoning_tokens": 360
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
 Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
 Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- Sending \$3 has consistently resulted in fair \$4.5 returns
- The cooperative approach is working well for everyone
- Total earnings are solid at \$23.5

I'll continue with the proven strategy and send ****\$3**** to Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: Sent \$3 to Elara, they returned \$4.5. Payoff: \$6.5
- Round 2: Received \$3 from Finn, returned \$4.5. Payoff: \$4.5
- Round 3: Sent \$3 to Aster, they returned \$4.5. Payoff: \$6.5
- Round 4: Received \$4 from Briar, returned \$6. Payoff: \$6.0

The pattern shows that:

- Sending \$3 has been working well (50% return)
- Briar sent \$4, showing even more trust, and I reciprocated fairly

I've been consistent with sending \$3 and it's been working. The cooperative strategy has been successful. I should continue with this approach.

I'll send \$3 to Galen.

usage:

```
{
  "input_tokens": 1452,
  "output_tokens": 296,
  "reasoning_tokens": 155
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
 Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3, following the same cooperative pattern I've been using
- I received \$9 after the 3x multiplier
- I've been consistently returning 50% to maintain fairness and trust

I'll return ****\$4.5**** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Cyra sent me \$3
- I received \$9 after the 3x multiplier
- I've been maintaining a consistent 50% return policy throughout
- This has worked well and built trust with others

Following my established pattern, I should return \$4.5 (50% of \$9).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is the same fair split I've been doing consistently.

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 288,
  "reasoning_tokens": 114
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent strategy:

- Sending \$3 has yielded reliable \$4.5 returns
- This gives me \$6.5 per round as sender
- The cooperative approach continues to work well
- Total earnings at \$34.5

I'll stick with ****\$3**** to Dorian.

```
'''json
{"send": 3}
'''
```


Reasoning:

Let me review my strategy:

- I've been consistently sending \$3 as sender
- I've been consistently returning 50% (half of what I receive) as receiver
- This has resulted in steady earnings of \$6.5 per round when I'm sender (when returns are \$4.5)
- Total earnings are \$34.5 across 7 rounds

My approach has been:

- Send \$3 ? they get \$9 ? they return \$4.5 ? I get \$6.5 total
- When receiving \$3 ? I get \$9 ? I return \$4.5 ? I keep \$4.5
- When receiving \$4 ? I get \$12 ? I return \$6 ? I keep \$6

The cooperative strategy is working well. I'll continue sending \$3 to Dorian.

```
Usage: {  
  "input_tokens": 1421,  
  "output_tokens": 280,  
  "reasoning_tokens": 158  
}
```

GAME - Agent_4 - Receiver - dyad_3

Timestamp: 2026-06-04T15:59:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_8",  
      "Agent_4"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 4

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$18.5.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern from previous rounds:

- Aster sent me \$4, which is the same amount I've been sending
- I received \$12 (tripled)
- Previous players have consistently returned about half when receiving similar amounts

Following the reciprocal fairness norm that's been established:

- Returning \$6 gives Aster \$7 total (\$1 + \$6)
- I keep \$6

This rewards Aster's trust while maintaining the cooperative pattern.

{'return': 6}

reasoning:

Let me analyze the situation:

- Aster sent me \$4, showing strong trust and cooperation
- I received \$12 (tripled amount)
- Aster currently has \$1 left

Looking at the pattern from previous rounds:

- When I sent \$4 to Galen, they returned \$6 (half of \$12)
- When Cyra sent \$3 to me, I returned \$4.5 (half of \$9)
- When I sent \$4 to Finn, they returned \$6 (half of \$12)

There's a consistent pattern of returning half of what was received, which seems fair and maintains cooperation. This gives both players a net benefit from the transaction.

If I return \$6:

- Aster: \$1 + \$6 = \$7
- Me: \$12 - \$6 = \$6

This is fair and maintains the cooperative norm. Aster sent a large amount (\$4), and returning half rewards their trust appropriately.

usage:

```
{
  "input_tokens": 1345,
  "output_tokens": 349,
  "reasoning_tokens": 196
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- Sending \$4 has consistently yielded \$6 returns (with both Galen and Finn)
- The cooperative norm of returning about half has been well-established
- Total earnings are strong at \$24.5

I'll continue with the successful strategy of sending \$4 to signal trust and cooperation.

```
{'send': 4}
```

reasoning:

Let me review my history:

- Round 1 (Sender vs Galen): Sent \$4, received \$6 back, earned \$7
- Round 2 (Receiver vs Cyra): Received \$9, returned \$4.5, earned \$4.5
- Round 3 (Sender vs Finn): Sent \$4, received \$6 back, earned \$7
- Round 4 (Receiver vs Aster): Received \$12, returned \$6, earned \$6

Total earnings: \$24.5

Pattern I've observed:

- When I send \$4, I consistently get \$6 back (both with Galen and Finn)
- When others send to me, I've been returning about half
- The cooperative strategy of sending \$4 has been working well

I haven't played with Briar yet. Given the consistent success with sending \$4 and the reciprocal fairness norm that's been established, I should continue with this strategy.

Sending \$4:

- Creates \$12 for Briar
- If they return half (like others have been doing), I get \$6 back
- My payoff would be \$5 - \$4 + \$6 = \$7

This has been the most reliable strategy so far.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 384,
  "reasoning_tokens": 241
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.

Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my previous experience:

- Elara sent \$3, which became \$9
- When Cyra sent me \$3 (becoming \$9) in Round 2, I returned \$4.5
- This maintains consistency and proportional fairness

Returning \$4.5:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is fair and consistent with established norms

{'return': 4.5}

reasoning:

Let me analyze this situation:

- Elara sent me \$3 (less than the \$4 I've been sending)
- I received \$9 (tripled)
- My pattern has been to return about half when I receive \$12 (returning \$6)
- For \$9, proportionally that would be around \$4.5

Looking at my history:

- When Cyra sent me \$3 (becoming \$9), I returned \$4.5
- This gave them \$6.5 total and me \$4.5
- This seems like a fair and consistent approach

I should maintain consistency and return \$4.5 again, which:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is proportional to my previous responses

This maintains the reciprocal fairness norm while being consistent with how I've handled \$3 sends before.

usage:

```
{
  "input_tokens": 1381,
  "output_tokens": 342,
  "reasoning_tokens": 185
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.

Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. H

ow much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Hana sent \$3, which became \$9
- I've consistently returned \$4.5 when receiving \$9 (with both Cyra and Elara)
- This maintains proportional fairness

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- Consistent with established cooperative norms

{'return': 4.5}

Reasoning:

Hana sent \$3, which became \$9 for me.

Looking at my history:

- When Elara sent \$3 (becoming \$9) in Round 6, I returned \$4.5
- When Cyra sent \$3 (becoming \$9) in Round 2, I returned \$4.5

I should maintain consistency and return the same proportional amount for the same send amount

.

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is consistent and maintains the cooperative norm

I'll return \$4.5.

```
Usage: {
  "input_tokens": 1403,
  "output_tokens": 274,
  "reasoning_tokens": 120
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-06-04T15:59:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    }
  },
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 4

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Finn sent me \$3 (60%), showing good trust
- This matches what I've seen and done in previous rounds
- The established reciprocity pattern has been returning \$4.5 for a \$3 send

I'll continue this cooperative pattern that has worked well.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Finn sent me \$3 (60% of their endowment)
- I received \$9 after the 3x multiplier
- Finn currently has \$2 remaining

Looking at the pattern so far:

- Round 1: Hana sent me \$3, I returned \$4.5
- Round 2: I sent Briar \$3, they returned \$4.5
- Round 3: I sent Cyra \$3, they returned \$4.5
- Round 4: Finn sent me \$3

There's a clear cooperative pattern emerging where sending \$3 gets a return of \$4.5. This has been consistent and fair for both parties. I should continue this pattern to maintain cooperation and reciprocity.

Returning \$4.5 would give:

- Finn: $\$2 + \$4.5 = \$6.5$
- Me: $\$9 - \$4.5 = \$4.5$

This is the same fair reciprocity I've been using.

usage:

```
{
  "input_tokens": 1423,
  "output_tokens": 326,
  "reasoning_tokens": 182
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Aster sent me \$4 (80%), showing higher trust than the typical \$3 (60%) I've experienced
- I received \$12
- My established pattern has been returning about 50% of what I receive
- For \$3 sends ? returned \$4.5 (50%)
- For this \$4 send ? should return \$6 (50%)

This maintains consistency while appropriately rewarding Aster's higher trust.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

- Aster sent me \$4 (80%), which is more than the typical \$3 (60%) I've been seeing
- I received \$12.0
- I need to decide how much to return

Looking at my history:

- When people sent me \$3, I returned \$4.5 (which leaves them with $\$2 + \$4.5 = \$6.5$ and me with \$4.5)
- The ratio has been: sender keeps \$2, I keep \$4.5, and I return \$4.5

For a \$3 send:

- They kept \$2, I returned \$4.5
- Their total: \$6.5
- My total: \$4.5
- Return rate: $\$4.5/\$9 = 50\%$

For a \$4 send:

- They kept \$1, I receive \$12
- If I return 50%, that's \$6
- Their total would be $\$1 + \$6 = \$7$
- My total would be \$6

That seems fair and consistent. Let me think about whether to reward the higher trust with a slightly higher return rate or stick with the 50% pattern.

Given that Aster showed more trust (80% vs 60%), I should probably reward that. Maybe return \$6 (50%) or slightly more like \$6.5 to encourage continued high trust.

Let me go with \$6 as it's consistent with my 50% return pattern and still rewards their higher trust level appropriately.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 473,
  "reasoning_tokens": 283
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$28.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history:
- My consistent strategy has been sending \$3 (60%) as sender
- This has resulted in good reciprocity - I've received \$4.5 back each time
- This gives me payoffs of \$6.5, better than keeping all \$5

I'll continue this cooperative approach with Dorian, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: RECEIVER from Hana - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 2: SENDER to Briar - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 3: SENDER to Cyra - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 4: RECEIVER from Finn - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 5: RECEIVER from Aster - they sent \$4, I returned \$6, my payoff \$6.0

Total: \$4.5 + \$6.5 + \$6.5 + \$4.5 + \$6.0 = \$28.0 ?

Pattern:

- When I sent \$3 (60%), I got \$4.5 back, giving me \$6.5
- When others sent \$3 to me, I returned \$4.5
- When Aster sent \$4 to me, I returned \$6 (50% of \$12)

The \$3 send (60%) has been working well and consistently. I'm playing against a new opponent (Dorian), so I'll stick with the established cooperative strategy.

usage:

```
{
  "input_tokens": 1380,
  "output_tokens": 409,
  "reasoning_tokens": 206
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:
- As sender, I've consistently sent \$3 (60%)

- This has reliably resulted in \$4.5 returns
- Payoffs of \$6.5 are consistently better than the \$5 I'd keep by sending nothing
- My cooperative strategy has been successful across different opponents

I'll continue this approach with Galen.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review my history:

- My strategy as sender has been consistent: send \$3 (60%)
- This has consistently resulted in receiving \$4.5 back
- My payoffs as sender have been \$6.5 each time
- This is better than keeping all \$5

Galen is a new opponent. I should continue my established cooperative strategy that has been working well. Sending \$3 has shown to be mutually beneficial and builds trust.

I'll send \$3 again.

```
Usage: {
  "input_tokens": 1363,
  "output_tokens": 215,
  "reasoning_tokens": 111
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-06-04T15:59:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_7": "Galen"
    },
    "agents": [
      "Agent_5",
      "Agent_7"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Cyra (Agent_3).

Round 4

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$20.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- Total wealth: \$11 (\$2 that Cyra kept + \$9 I received)

Based on the cooperative pattern I've established:

- Round 1: Received \$12, returned \$6 (50%)
- Round 2: Sent \$4, received back \$6 (50% of tripled amount)
- Round 3: Sent \$4, received back \$6 (50% of tripled amount)

I'll maintain this 50% return rate to be consistent and reward Cyra's trust proportionally:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me analyze this situation:

- Cyra sent me \$3.0 (60% of their endowment)
- This became \$9.0 for me
- Total wealth in the system: \$2 (what Cyra kept) + \$9 (what I received) = \$11

Looking at my history:

- Round 1: Dorian sent \$4, I returned \$6 (50% of what I received)
- Round 2: I sent \$4, Aster returned \$6 (50% of what they received)
- Round 3: I sent \$4, Briar returned \$6 (50% of what they received)

There's been a consistent pattern of returning about 50% of the received amount, which seems to be the cooperative norm emerging in this game.

Cyra sent slightly less than others (\$3 vs \$4), but still showed trust by sending 60% of their endowment.

If I return \$4.5 (50% of \$9):

- Cyra ends with: \$2 + \$4.5 = \$6.5 (net \$1.5 gain)
- I end with: \$9 - \$4.5 = \$4.5 (net \$4.5 gain)

This maintains the cooperative pattern and rewards Cyra's trust proportionally to what they sent.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 504,
  "reasoning_tokens": 236
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$24.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- This is the same amount as Cyra sent in the previous round

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Hana sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 kept by Hana + \$9 I received)

I've been maintaining a consistent 50% return rate:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)

This has been working well and creates good incentives for cooperation. I should continue this pattern.

50% of \$9.0 = \$4.5

usage:

```
{
  "input_tokens": 1487,
  "output_tokens": 258,
  "reasoning_tokens": 121
}
```

}

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$29.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

As SENDER:

- Round 2: Sent \$4.0 to Aster, received \$6.0 back, payoff \$7.0
- Round 3: Sent \$4.0 to Briar, received \$6.0 back, payoff \$7.0

As RECEIVER:

- I've consistently returned 50% of what I receive

This cooperative strategy has been working well, generating steady returns. I'll continue with the same approach against Finn:

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history:

Round 1: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0, payoff \$6.0
Round 2: SENDER - I sent \$4.0 to Aster, they received \$12.0, returned \$6.0, payoff \$7.0
Round 3: SENDER - I sent \$4.0 to Briar, they received \$12.0, returned \$6.0, payoff \$7.0
Round 4: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5
Round 5: RECEIVER - Hana sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5

Total: \$29.0 (checks out: \$6.0 + \$7.0 + \$7.0 + \$4.5 + \$4.5 = \$29.0)

My pattern has been:

- As SENDER: Send \$4.0 (80%)
- As RECEIVER: Return 50% of what I receive

When I sent \$4.0, I consistently received \$6.0 back (50% of \$12.0), giving me a payoff of \$7.0.

Now I'm playing against Finn (Agent_6) as SENDER. I should maintain my cooperative strategy of sending \$4.0.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 484,
  "reasoning_tokens": 216
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Looking at the situation:

- Elara sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 Elara kept + \$9 I received)

My consistent pattern as RECEIVER has been to return 50% of what I receive:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)
- Round 5: Received \$9, returned \$4.5 (50%)

I should maintain this consistency to reward Elara's trust and continue the cooperative pattern.

Usage: {

```
"input_tokens": 1495,
"output_tokens": 252,
"reasoning_tokens": 130
```

}

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

```
dyad_1: Agent_7 (Galen)=Sender, Agent_1 (Aster)=Receiver
dyad_2: Agent_3 (Cyra)=Sender, Agent_8 (Hana)=Receiver
dyad_3: Agent_4 (Dorian)=Sender, Agent_2 (Briar)=Receiver
dyad_4: Agent_6 (Finn)=Sender, Agent_5 (Elara)=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_7 (Galen) sent $4; received $12; Agent_1 (Aster) returned $6; payoffs Agent_7
(Galen)=$7, Agent_1 (Aster)=$6
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw":
6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 (Cyra) sent $3; received $9; Agent_8 (Hana) returned $4.5; payoffs Agent_3
(Cyra)=$6.5, Agent_8 (Hana)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_4 (Dorian) sent $4; received $12; Agent_2 (Briar) returned $6; payoffs Agent_4
(Dorian)=$7, Agent_2 (Briar)=$6
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw":
6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_6 (Finn) sent $3; received $9; Agent_5 (Elara) returned $4.5; payoffs Agent_6
(Finn)=$6.5, Agent_5 (Elara)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 50.0,
"Agent_2": 50.0,
"Agent_3": 45.5,
"Agent_4": 47.5,
"Agent_5": 45.5,
"Agent_6": 48.5,
"Agent_7": 47.5,
"Agent_8": 45.5
```

}

LLM CALLS

GAME - Agent_7 - Sender - dyad_1

Timestamp: 2026-06-04T15:59:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_7",
    "Agent_1"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 5

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$24.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Hana sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier
- This is the same amount as Cyra sent in the previous round

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Hana sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 kept by Hana + \$9 I received)

I've been maintaining a consistent 50% return rate:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)

This has been working well and creates good incentives for cooperation. I should continue this pattern.

50% of \$9.0 = \$4.5

usage:

```
{
  "input_tokens": 1487,
  "output_tokens": 258,
  "reasoning_tokens": 121
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$29.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

As SENDER:

- Round 2: Sent \$4.0 to Aster, received \$6.0 back, payoff \$7.0
- Round 3: Sent \$4.0 to Briar, received \$6.0 back, payoff \$7.0

As RECEIVER:

- I've consistently returned 50% of what I receive

This cooperative strategy has been working well, generating steady returns. I'll continue with the same approach against Finn:

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history:

Round 1: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0, payoff \$6.0
Round 2: SENDER - I sent \$4.0 to Aster, they received \$12.0, returned \$6.0, payoff \$7.0
Round 3: SENDER - I sent \$4.0 to Briar, they received \$12.0, returned \$6.0, payoff \$7.0
Round 4: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5
Round 5: RECEIVER - Hana sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5

Total: \$29.0 (checks out: \$6.0 + \$7.0 + \$7.0 + \$4.5 + \$4.5 = \$29.0)

My pattern has been:

- As SENDER: Send \$4.0 (80%)
- As RECEIVER: Return 50% of what I receive

When I sent \$4.0, I consistently received \$6.0 back (50% of \$12.0), giving me a payoff of \$7.0.

Now I'm playing against Finn (Agent_6) as SENDER. I should maintain my cooperative strategy of sending \$4.0.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 484,
  "reasoning_tokens": 216
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Elara sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier

- Total wealth created: \$11 (\$2 Elara kept + \$9 I received)

My consistent pattern as RECEIVER has been to return 50% of what I receive:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)
- Round 5: Received \$9, returned \$4.5 (50%)

I should maintain this consistency to reward Elara's trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1495,
  "output_tokens": 252,
  "reasoning_tokens": 130
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Aster:

Round 2: I sent \$4.0 to Aster, and they returned \$6.0 (50% of the \$12.0 they received). This gave me a payoff of \$7.0.

Aster demonstrated trustworthiness and cooperation. I'll continue my consistent strategy of sending \$4.0:

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Let me review my history with Aster:

Round 2: I was the SENDER against Aster
- I sent \$4.0, it became \$12.0 for them
- They returned \$6.0 to me
- My payoff was \$7.0

So Aster has proven to be cooperative in the past, returning 50% of what they received ($6.0/12.0 = 0.5$).

My strategy has been consistently cooperative:

- As SENDER: I send \$4.0 (80% of my endowment)
- As RECEIVER: I return 50% of what I receive

This has worked well for me across the rounds. My total earnings are \$40.5 after 7 rounds, averaging about \$5.79 per round, which is above the \$5 baseline if I kept everything.

Since Aster was cooperative last time we played, I should continue my cooperative strategy:

```
'''json
{"send": 4.0}
'''
```

```
Usage: {
  "input_tokens": 1375,
  "output_tokens": 336,
  "reasoning_tokens": 192
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T15:59:42
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_7",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_7": "Galen"
  },
  "agents": [
    "Agent_7",
    "Agent_1"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Elara (Agent_5).

Round 5

In your most recent previous game round, you were the SENDER against Dorian. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. This establishes strong trust. Playing with Elara for the first time, I'll continue with the successful \$4 strategy that worked well with Dorian.

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at my history:

- Round 1: I sent \$3 to Cyra, got \$4.5 back, earned \$6.5
- Round 2: Receiver, Galen sent \$4, I returned \$6, earned \$6.0
- Round 3: Receiver, Hana sent \$3, I returned \$4.5, earned \$4.5
- Round 4: I sent \$4 to Dorian, got \$6 back, earned \$7.0

The pattern has been very consistent - everyone is returning 50% of what they receive. This is a strong cooperative signal.

Total earnings: \$24.0, which means I've been doing well.

Now I'm playing with Elara for the first time. Given the consistent cooperation I've experienced:

- Everyone has been trustworthy
- The 50% return norm seems established
- I could send \$4 or even \$5

If I send \$5:

- Elara receives \$15
- I keep \$0
- If they return 50% (\$7.5), I'd end with \$7.5

If I send \$4:

- Elara receives \$12
- I keep \$1
- If they return 50% (\$6), I'd end with \$7

The \$5 option shows maximum trust and could earn \$0.5 more. Given the consistent cooperation, I'll go with \$4 again to maintain the pattern that's been working well. It's trusting but not reckless.

Actually, let me reconsider. The cooperation has been perfect so far. Maybe I should try \$5 to show maximum trust and potentially maximize total welfare. If everyone is cooperating at 50%, sending \$5 gives:

- \$15 to receiver
- If they return \$7.5, I get \$7.5 (better than the \$7 from sending \$4)
- Total surplus is maximized

But there's also risk. I'll stick with \$4 since it's been working and I haven't seen anyone send \$5 yet.

usage:

```
{
  "input_tokens": 1619,
  "output_tokens": 679,
  "reasoning_tokens": 390
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the established pattern:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- And when I've been sender, I've consistently received 50% back

Briar sent \$4 (keeping \$1), which became \$12 for me.

Following the cooperative 50% pattern:

- If I return \$6:
 - Briar ends with: \$1 + \$6 = \$7
 - I end with: \$12 - \$6 = \$6

This maintains the trust and reciprocity that has been working well.

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at the pattern:

- Round 1: I was SENDER, sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: I was RECEIVER, Galen sent \$4, I returned \$6 (50%)
- Round 3: I was RECEIVER, Hana sent \$3, I returned \$4.5 (50%)
- Round 4: I was SENDER, sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: I was SENDER, sent \$4 to Elara, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. Now Briar sent me \$4 (keeping \$1), which became \$12.

If I return 50%: \$6

- Briar ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This is the fair 50% pattern that has been established throughout all rounds. I should continue with this.

usage:

```
{
  "input_tokens": 1652,
```

```
"output_tokens": 440,  
"reasoning_tokens": 185  
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the consistent pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)

As RECEIVER:

- All my opponents who sent have received 50% back from me

This is a perfectly cooperative environment with 50% returns across the board. Playing with Finn for the first time, I'll continue with the \$4 strategy that has worked consistently.

If I send \$4:

- Finn receives \$12
- I keep \$1
- Expected return (50%): \$6
- Expected total: \$7

```
'''json  
{"send": 4}  
'''
```

reasoning:

Let me review the pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: Sent \$4 to Elara, got \$6 back (50% return)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Every single interaction has resulted in 50% returns. This is a very strong cooperative pattern. Now I'm playing with Finn for the first time.

Given the consistent 50% returns and building trust, I should continue with \$4. This has been working perfectly.

If I send \$4:

- Finn receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

usage:

```
{  
  "input_tokens": 1625,  
  "output_tokens": 463,  
  "reasoning_tokens": 196  
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Galen and the overall pattern:

- Round 2: Galen sent me \$4, I returned \$6 (50%)
- Now Round 8: Galen sends me \$4 again

The cooperative pattern has held perfectly throughout all rounds - everyone consistently returns 50% of what they receive. Galen trusted me again with \$4.

Galen sent \$4 (keeping \$1), which became \$12 for me.

Returning 50%:

- If I return \$6:
- Galen ends with: \$1 + \$6 = \$7
- I end with: \$12 - \$6 = \$6

Maintaining the trust and reciprocity:

```
'''json
{"return": 6}
'''
```

Reasoning:

Looking at my history:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)
- Round 7: Sent \$4 to Finn ? got \$6 back (50%)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Now in Round 8, I'm playing with Galen again. In Round 2, Galen sent me \$4 and I returned \$6 (50%). Now Galen is sending me \$4 again.

The cooperative pattern has been consistent throughout - everyone returns 50%. I should continue this pattern.

Galen sent \$4 (keeping \$1), which became \$12 for me.

If I return \$6:

- Galen ends with: \$1 + \$6 = \$7
- I end with: \$12 - \$6 = \$6

This maintains the established trust and cooperation.

```
Usage: {
  "input_tokens": 1642,
  "output_tokens": 506,
  "reasoning_tokens": 224
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-06-04T15:59:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_8": "Hana"
    },
    "agents": [
      "Agent_3",
      "Agent_8"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

- ll have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 5

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Finn sent me \$3.0 (60%), showing the same level of trust others have shown
- I received \$9.0 after multiplication
- The consistent pattern across all rounds has been: when someone sends \$3, return \$4.5

This strategy has maintained strong cooperation and mutual benefit throughout the game.

{'return': 4.5}

reasoning:

Looking at the pattern:

- Finn sent me \$3.0 (60% of their endowment), showing trust
- I received \$9.0 after multiplication
- Finn currently has \$2.0 left

The established pattern has been very consistent:

- When I send \$3, I receive \$4.5 back
- When others send me \$3, I return \$4.5

This has created good outcomes for everyone:

- The sender ends up with $\$2 + \$4.5 = \$6.5$
- The receiver ends up with $\$9.0 - \$4.5 = \$4.5$
- Everyone gains from cooperation

I should continue this pattern with Finn:

{'return': 4.5}

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 263,
  "reasoning_tokens": 140
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my first interaction with Hana
- The established pattern has been remarkably consistent: sending \$3.0 has led to returns of \$4.5

- This has worked well across all my interactions so far
- Total earnings of \$26.5 suggest this cooperative strategy is effective

Maintaining this successful approach:

```
{'send': 3.0}
```

reasoning:

Looking at my history:

- Round 1: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 2: (not fully detailed but total was \$11.0 after round 2)
- Round 3: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5
- Round 4: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 5: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5

Total: \$26.5

The pattern is very consistent:

- When I'm SENDER: send \$3.0, get back \$4.5, net \$6.5
- When I'm RECEIVER: receive \$9.0 from \$3.0 sent by opponent, return \$4.5, net \$4.5

This is my first interaction with Hana. The established pattern of sending \$3.0 has worked very well throughout the game. Everyone seems to be cooperating at this level.

I should continue this successful cooperative strategy.

usage:

```
{
  "input_tokens": 1370,
  "output_tokens": 378,
  "reasoning_tokens": 209
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Briar sent me \$4.0 (80%), which is more than the typical \$3.0 pattern
- I received \$12.0 after multiplication
- Briar is showing extra trust by sending more

I've been returning approximately 50% of what I receive. Maintaining this proportional approach:

- $\$12.0 \times 0.5 = \6.0

This rewards Briar's greater trust while being fair to both of us.

```
{'return': 6.0}
```

reasoning:

Looking at this situation:

- Briar sent me \$4.0 (80% of their endowment), which is MORE than the typical \$3.0 I've been seeing
- I received \$12.0 after multiplication
- Briar is showing extra trust/generosity by sending more than the established pattern

My previous pattern has been:

- When I receive \$9.0 (from \$3 sent), I return \$4.5 (50% of what I received)

Briar sent \$4.0 instead of \$3.0, showing greater trust. I should reciprocate proportionally:

- If I follow the 50% return rate: $\$12.0 * 0.5 = \6.0
- This would give Briar: $\$5 - \$4 + \$6 = \7.0
- I would keep: $\$12.0 - \$6.0 = \$6.0$

Alternatively, I could think about fair splits:

- Briar invested \$4, I got \$12
- A fair even split of gains would be roughly \$6-\$6
- Briar would end with $\$5 - \$4 + \$6 = \7
- I would end with \$6

Returning \$6.0 seems appropriate as it:

1. Maintains the 50% return ratio I've been using
2. Rewards Briar's extra trust
3. Provides a fair outcome for both parties

usage:

```
{
  "input_tokens": 1350,
  "output_tokens": 465,
  "reasoning_tokens": 255
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$39.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my second interaction with Hana
- In Round 6, I sent Hana \$3.0 and they returned \$4.5 - a fair and cooperative response
- Hana has proven trustworthy
- The \$3.0 sending strategy continues to work well

Continuing with this successful approach:

```
{'send': 3.0}
```

Reasoning:

Looking at the situation:

- This is my second interaction with Hana
- In Round 6, I sent Hana \$3.0, they returned \$4.5 - this was a successful interaction
- Hana has proven to be trustworthy and reciprocated fairly
- The standard pattern of \$3.0 has worked well
- My total earnings of \$39.0 show the cooperative strategy is effective

Given that Hana has already demonstrated trustworthiness by returning \$4.5 when I sent \$3.0, I should continue with the same successful approach.

I'll send \$3.0 again.

Usage: {

```
"input_tokens": 1365,
"output_tokens": 241,
"reasoning_tokens": 133
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-06-04T15:59:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_3",
"pairing": {
```

```

"agent_names": {
  "Agent_3": "Cyra",
  "Agent_8": "Hana"
},
"agents": [
  "Agent_3",
  "Agent_8"
],
"investor": "Agent_3",
"roles": {
  "Agent_3": "investor",
  "Agent_8": "trustee"
},
"trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 5

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- Sending \$3 has consistently resulted in fair \$4.5 returns
- The cooperative approach is working well for everyone
- Total earnings are solid at \$23.5

I'll continue with the proven strategy and send ****\$3**** to Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- Round 1: Sent \$3 to Elara, they returned \$4.5. Payoff: \$6.5
- Round 2: Received \$3 from Finn, returned \$4.5. Payoff: \$4.5
- Round 3: Sent \$3 to Aster, they returned \$4.5. Payoff: \$6.5
- Round 4: Received \$4 from Briar, returned \$6. Payoff: \$6.0

The pattern shows that:

- Sending \$3 has been working well (50% return)
- Briar sent \$4, showing even more trust, and I reciprocated fairly

I've been consistent with sending \$3 and it's been working. The cooperative strategy has been successful. I should continue with this approach.

I'll send \$3 to Galen.

usage:

```
{
  "input_tokens": 1452,
  "output_tokens": 296,
  "reasoning_tokens": 155
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3, following the same cooperative pattern I've been using
- I received \$9 after the 3x multiplier
- I've been consistently returning 50% to maintain fairness and trust

I'll return ****\$4.5**** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
```

```
    {"return": 4.5}
    """
```

reasoning:

Looking at the situation:

- Cyra sent me \$3
- I received \$9 after the 3x multiplier
- I've been maintaining a consistent 50% return policy throughout
- This has worked well and built trust with others

Following my established pattern, I should return \$4.5 (50% of \$9).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is the same fair split I've been doing consistently.

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 288,
  "reasoning_tokens": 114
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent strategy:

- Sending \$3 has yielded reliable \$4.5 returns
- This gives me \$6.5 per round as sender
- The cooperative approach continues to work well
- Total earnings at \$34.5

I'll stick with **\$3** to Dorian.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my strategy:

- I've been consistently sending \$3 as sender
- I've been consistently returning 50% (half of what I receive) as receiver
- This has resulted in steady earnings of \$6.5 per round when I'm sender (when returns are \$4.5)
- Total earnings are \$34.5 across 7 rounds

My approach has been:

- Send \$3 ? they get \$9 ? they return \$4.5 ? I get \$6.5 total
- When receiving \$3 ? I get \$9 ? I return \$4.5 ? I keep \$4.5
- When receiving \$4 ? I get \$12 ? I return \$6 ? I keep \$6

The cooperative strategy is working well. I'll continue sending \$3 to Dorian.

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 280,
  "reasoning_tokens": 158
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$41.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- Cyra sent me \$3 again, showing consistent cooperation
- I received \$9 after the 3x multiplier
- I returned \$4.5 last time we played (Round 6), which was fair

I'll continue with the same fair approach and return **\$4.5** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Cyra sent me \$3 again, which became \$9.0. I need to return some amount.

Looking at the pattern:

- Cyra has sent me \$3 twice now (rounds 6 and 8)
- In round 6, I returned \$4.5 (50% of \$9)
- This is consistent with my cooperative strategy

I should continue being fair and return \$4.5 again (50% of what I received).

This gives:

- Cyra: \$2 (kept from \$5) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This maintains the cooperative, fair pattern that's been working well.

```
Usage: {
  "input_tokens": 1373,
  "output_tokens": 330,
  "reasoning_tokens": 132
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-06-04T16:00:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_4": "Dorian"
    }
  },
  "agents": [
    "Agent_4",
    "Agent_2"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_2": "trustee",
    "Agent_4": "investor"
  },
}
```

```

    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 5

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$24.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- Sending \$4 has consistently yielded \$6 returns (with both Galen and Finn)

- The cooperative norm of returning about half has been well-established
- Total earnings are strong at \$24.5

I'll continue with the successful strategy of sending \$4 to signal trust and cooperation.

```
{'send': 4}
```

reasoning:

Let me review my history:

- Round 1 (Sender vs Galen): Sent \$4, received \$6 back, earned \$7
- Round 2 (Receiver vs Cyra): Received \$9, returned \$4.5, earned \$4.5
- Round 3 (Sender vs Finn): Sent \$4, received \$6 back, earned \$7
- Round 4 (Receiver vs Aster): Received \$12, returned \$6, earned \$6

Total earnings: \$24.5

Pattern I've observed:

- When I send \$4, I consistently get \$6 back (both with Galen and Finn)
- When others send to me, I've been returning about half
- The cooperative strategy of sending \$4 has been working well

I haven't played with Briar yet. Given the consistent success with sending \$4 and the reciprocal fairness norm that's been established, I should continue with this strategy.

Sending \$4:

- Creates \$12 for Briar
- If they return half (like others have been doing), I get \$6 back
- My payoff would be \$5 - \$4 + \$6 = \$7

This has been the most reliable strategy so far.

usage:

```
{
  "input_tokens": 1455,
  "output_tokens": 384,
  "reasoning_tokens": 241
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my previous experience:

- Elara sent \$3, which became \$9
- When Cyra sent me \$3 (becoming \$9) in Round 2, I returned \$4.5
- This maintains consistency and proportional fairness

Returning \$4.5:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is fair and consistent with established norms

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

- Elara sent me \$3 (less than the \$4 I've been sending)
- I received \$9 (tripled)
- My pattern has been to return about half when I receive \$12 (returning \$6)
- For \$9, proportionally that would be around \$4.5

Looking at my history:

- When Cyra sent me \$3 (becoming \$9), I returned \$4.5
- This gave them \$6.5 total and me \$4.5
- This seems like a fair and consistent approach

I should maintain consistency and return \$4.5 again, which:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is proportional to my previous responses

This maintains the reciprocal fairness norm while being consistent with how I've handled \$3 sends before.

usage:

```
{
  "input_tokens": 1381,
  "output_tokens": 342,
  "reasoning_tokens": 185
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

- Hana sent \$3, which became \$9
- I've consistently returned \$4.5 when receiving \$9 (with both Cyra and Elara)
- This maintains proportional fairness

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- Consistent with established cooperative norms

{'return': 4.5}

reasoning:

Hana sent \$3, which became \$9 for me.

Looking at my history:

- When Elara sent \$3 (becoming \$9) in Round 6, I returned \$4.5
- When Cyra sent \$3 (becoming \$9) in Round 2, I returned \$4.5

I should maintain consistency and return the same proportional amount for the same send amount.

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is consistent and maintains the cooperative norm

I'll return \$4.5.

usage:

```
{
  "input_tokens": 1403,
  "output_tokens": 274,
  "reasoning_tokens": 120
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.

Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Briar:

- Round 5: I sent \$4, they returned \$6 (very fair cooperation)
- This established a strong cooperative relationship

Given:

- Briar has proven trustworthy
- Sending \$4 has consistently worked well
- Total earnings are solid at \$40.5

I'll send \$4 again to maintain our cooperative equilibrium.

{'send': 4}

Reasoning:

Let me check my history with Briar:

- Round 5: I was the SENDER against Briar. I sent \$4.0, it became \$12.0, and they returned \$6.0. My payoff was \$7.0.

So Briar has shown to be cooperative, returning \$6 when receiving \$12 (which is exactly half, maintaining the fair cooperative norm).

Given that:

- Briar returned \$6 on \$12 last time (50% return rate, very fair)
- My strategy of sending \$4 has been consistently successful
- Total earnings are strong at \$40.5
- We have an established cooperative relationship

I should send \$4 again to Briar to maintain the cooperative equilibrium.

```
Usage: {  
  "input_tokens": 1386,  
  "output_tokens": 274,  
  "reasoning_tokens": 156  
}
```

GAME - Agent_2 - Receiver - dyad_3

Timestamp: 2026-06-04T16:00:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_4": "Dorian"  
    },  
    "agents": [  
      "Agent_4",  
      "Agent_2"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```

```
"round": 8,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 5

In your most recent previous game round, you were the SENDER against Hana. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$24.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0.
. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the established pattern:

Dorian sent me \$4.0, showing the same level of trust and generosity that I've seen from Galen and that I showed to Hana. The reciprocity pattern has been consistent - returning about 50% of what's received.

Returning \$6:
- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This maintains the cooperative equilibrium.

```
'''json
{"return": 6}
'''
```

reasoning:

Dorian sent me \$4.0, which becomes \$12.0.

Looking at the pattern:
- When I received \$3 from Elara, I returned \$4.5 (50%)
- When I received \$4 from Galen, I returned \$6 (50%)
- When I sent \$4 to Hana, they returned \$6 (50%)

The cooperative norm has been established at returning about 50% of what's received. Dorian sent me \$4, which is the same generous amount that Hana and Galen sent. I should maintain this reciprocity pattern.

Returning \$6:
- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This is fair and maintains the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1460,
  "output_tokens": 327,
  "reasoning_tokens": 150
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$30.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the successful pattern:
- Sending \$4 has consistently led to returns of \$6
- This creates mutual benefit (sender gets \$7, receiver gets \$6)
- Cooperation has been strong across all interactions

I'll continue this pattern by sending \$4 to Aster:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the pattern:
- Round 2 (as receiver): Elara sent \$3, I returned \$4.5
- Round 3 (as receiver): Galen sent \$4, I returned \$6
- Round 4 (as sender): I sent \$4 to Hana, they returned \$6
- Round 5 (as receiver): Dorian sent \$4, I returned \$6

The pattern is clear: sending \$4 has been working well with consistent returns of about \$6. This gives the sender a payoff of \$7 and the receiver \$6, which is mutually beneficial.

I should continue this cooperative pattern by sending \$4 to Aster.

usage:

```
{
  "input_tokens": 1411,
  "output_tokens": 243,
  "reasoning_tokens": 130
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

The pattern of cooperation continues to work well:
- Sending \$4 has consistently resulted in \$6 returns
- This creates good payoffs for both parties
- Trust is well-established in this group

I'll send \$4 to Cyra:

```
'''json
{"send": 4}
'''
```

reasoning:

The pattern continues to hold strong:
- I've been sending \$4 consistently as sender
- When I send \$4, I receive back \$6
- When others send me \$4, I return \$6
- This creates a stable, mutually beneficial equilibrium

My payoffs:

- As sender sending \$4: $\$5 - \$4 + \$6 = \7
- As receiver getting \$4 (which becomes \$12): return \$6, keep \$6

The cooperation is working well. I should continue sending \$4 to Cyra.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 203,
  "reasoning_tokens": 109
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0.
. How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Dorian sent \$4.0 again, showing continued trust. In Round 5, they also sent \$4 and I returned \$6, which worked well for both of us.

I'll maintain the same cooperative pattern:

```
'''json
```

```
{"return": 6}
,,,
```

Reasoning:

Dorian sent me \$4.0 again, just like in Round 5. I returned \$6.0 to them then, which worked well. The cooperative pattern of sending \$4 and returning \$6 has been consistent across all interactions.

I should return \$6 again to maintain this successful cooperative relationship:

- Dorian gets: \$5 - \$4 + \$6 = \$7
- I get: \$12 - \$6 = \$6

This is fair and continues the established pattern.

```
Usage: {
  "input_tokens": 1349,
  "output_tokens": 187,
  "reasoning_tokens": 101
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-06-04T16:00:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_5": "Elara",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Cyra (Agent_3).

Round 5

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$23.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record, cooperation has been consistently successful. When I've sent \$3.0 as sender, I've received \$4.5 back, resulting in a \$6.5 payoff. This pattern of trust and reciprocity has been beneficial.

I'll continue this cooperative approach with Cyra.

```
{'send': 3.0}
```

reasoning:

Looking at my pattern so far:

- Round 1 (RECEIVER vs Briar): They sent \$3.0, I received \$9.0, returned \$4.5. My payoff: \$4.5
- Round 2 (SENDER vs Hana): I sent \$3.0, they returned \$4.5. My payoff: \$6.5
- Round 3 (RECEIVER vs Dorian): They sent \$4.0, I received \$12.0, returned \$6.0. My payoff: \$6.0
- Round 4 (SENDER vs Elara): I sent \$3.0, they returned \$4.5. My payoff: \$6.5

So far, cooperation has been working very well. When I've sent \$3.0, I've gotten back \$4.5, giving me a payoff of \$6.5. When others sent to me, I've reciprocated fairly.

This is my fourth time as sender (wait, let me recount):

- Round 2: SENDER
- Round 4: SENDER
- Round 5: SENDER (this round)

So this is my third time as sender. I've been consistent in sending \$3.0 and it's worked out well. I should continue this cooperative strategy with Cyra.

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 388,
  "reasoning_tokens": 217
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Galen (Agent_7).

Round 6

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Galen sent \$4.0 (80% of their endowment), showing very high trust. I received \$12.0 after the multiplier.

Following the cooperative pattern that has worked well throughout this experiment, I'll return a fair amount that rewards their trust.

If I return \$6.0:

- Galen ends with: $\$1.0 + \$6.0 = \$7.0$ (net gain of \$2.0)
- I end with: $\$12.0 - \$6.0 = \$6.0$ (net gain of \$6.0)

This is consistent with my previous behavior when receiving high trust.

{'return': 6.0}

reasoning:

Galen sent me \$4.0, which became \$12.0 for me. That's 80% of their endowment - a very high level of trust.

Looking at my past behavior:

- Round 3: As receiver, Dorian sent \$4.0, I received \$12.0, I returned \$6.0
- My reasoning then was to reward their generosity fairly

For consistency and to maintain trust, I should return a similar amount. Returning \$6.0 would give:

- Galen: $\$1.0$ (original remaining) + $\$6.0$ (returned) = $\$7.0$
- Me: $\$12.0 - \$6.0 = \$6.0$

This rewards their high trust with a net gain while also benefiting me from cooperation.

usage:

```
{
  "input_tokens": 1391,
  "output_tokens": 347,
  "reasoning_tokens": 145
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Aster sent \$4.0, showing high trust (80% of their endowment). After the multiplier, I received \$12.0.

Consistent with my previous behavior when receiving \$4.0, I'll return \$6.0. This rewards their trust and maintains the cooperative pattern that has been successful throughout the

experiment.

```
{'return': 6.0}
```

reasoning:

Aster sent me \$4.0, showing high trust (80% of their endowment). After the 3x multiplier, I received \$12.0.

In the previous round with Galen, who also sent \$4.0, I returned \$6.0. This has been my consistent pattern - when someone sends \$4.0, I return \$6.0.

This gives:

- Aster: \$1.0 (kept) + \$6.0 (returned) = \$7.0 total (net gain of \$2.0)
- Me: \$12.0 (received) - \$6.0 (returned) = \$6.0 (net gain of \$6.0)

I should maintain this consistent cooperative strategy.

usage:

```
{  
  "input_tokens": 1369,  
  "output_tokens": 264,  
  "reasoning_tokens": 123  
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

I'm facing Elara again. In our previous interaction (Round 5), I sent \$3.0 and Elara returned \$4.5, resulting in a mutually beneficial outcome.

Given this positive history with Elara and the consistent cooperative pattern throughout the experiment, I'll continue with the same approach.

```
{'send': 3.0}
```

Reasoning:

I'm facing Elara again. In Round 5, I was the sender against Elara - I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

This was a good outcome - Elara was cooperative and returned a fair amount. Given this positive history with Elara specifically, I should continue the cooperative approach.

My consistent strategy has been to send \$3.0 as sender, which has worked well throughout the experiment. All my interactions have been cooperative and successful.

I'll send \$3.0 to Elara again.

Usage: {

```
"input_tokens": 1388,  
"output_tokens": 230,  
"reasoning_tokens": 136
```

}

GAME - Agent_5 - Receiver - dyad_4

Timestamp: 2026-06-04T16:00:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",  
"move_index": 7,  
"opponent_id": "Agent_6",  
"pairing": {  
  "agent_names": {  
    "Agent_5": "Elara",
```

```

    "Agent_6": "Finn"
  },
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 5

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Aster sent me \$4 (80%), showing higher trust than the typical \$3 (60%) I've experienced
- I received \$12
- My established pattern has been returning about 50% of what I receive
- For \$3 sends ? returned \$4.5 (50%)
- For this \$4 send ? should return \$6 (50%)

This maintains consistency while appropriately rewarding Aster's higher trust.

```
'''json
{"return": 6}
'''
```

reasoning:

Let me analyze this situation:

- Aster sent me \$4 (80%), which is more than the typical \$3 (60%) I've been seeing
- I received \$12.0
- I need to decide how much to return

Looking at my history:

- When people sent me \$3, I returned \$4.5 (which leaves them with \$2 + \$4.5 = \$6.5 and me with \$4.5)
- The ratio has been: sender keeps \$2, I keep \$4.5, and I return \$4.5

For a \$3 send:

- They kept \$2, I returned \$4.5
- Their total: \$6.5
- My total: \$4.5
- Return rate: $\$4.5/\$9 = 50\%$

For a \$4 send:

- They kept \$1, I receive \$12
- If I return 50%, that's \$6
- Their total would be \$1 + \$6 = \$7
- My total would be \$6

That seems fair and consistent. Let me think about whether to reward the higher trust with a slightly higher return rate or stick with the 50% pattern.

Given that Aster showed more trust (80% vs 60%), I should probably reward that. Maybe return \$6 (50%) or slightly more like \$6.5 to encourage continued high trust.

Let me go with \$6 as it's consistent with my 50% return pattern and still rewards their higher trust level appropriately.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 473,
  "reasoning_tokens": 283
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$28.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history:

- My consistent strategy has been sending \$3 (60%) as sender
- This has resulted in good reciprocity - I've received \$4.5 back each time
- This gives me payoffs of \$6.5, better than keeping all \$5

I'll continue this cooperative approach with Dorian, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: RECEIVER from Hana - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 2: SENDER to Briar - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 3: SENDER to Cyra - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 4: RECEIVER from Finn - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 5: RECEIVER from Aster - they sent \$4, I returned \$6, my payoff \$6.0

Total: \$4.5 + \$6.5 + \$6.5 + \$4.5 + \$6.0 = \$28.0 ?

Pattern:

- When I sent \$3 (60%), I got \$4.5 back, giving me \$6.5
- When others sent \$3 to me, I returned \$4.5
- When Aster sent \$4 to me, I returned \$6 (50% of \$12)

The \$3 send (60%) has been working well and consistently. I'm playing against a new opponent (Dorian), so I'll stick with the established cooperative strategy.

usage:

```
{
  "input_tokens": 1380,
  "output_tokens": 409,
  "reasoning_tokens": 206
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- As sender, I've consistently sent \$3 (60%)
- This has reliably resulted in \$4.5 returns
- Payoffs of \$6.5 are consistently better than the \$5 I'd keep by sending nothing
- My cooperative strategy has been successful across different opponents

I'll continue this approach with Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- My strategy as sender has been consistent: send \$3 (60%)
- This has consistently resulted in receiving \$4.5 back
- My payoffs as sender have been \$6.5 each time
- This is better than keeping all \$5

Galen is a new opponent. I should continue my established cooperative strategy that has been working well. Sending \$3 has shown to be mutually beneficial and builds trust.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 215,
  "reasoning_tokens": 111
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$41.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Finn sent me \$3 (60% of their endowment)
- I received \$9
- In Round 5, when Finn sent me \$3, I returned \$4.5 (50%)
- This established a fair reciprocal relationship with Finn

I'll maintain consistency and return the same amount.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me review my history with Finn:

- Round 5: I was RECEIVER against Finn. They sent \$3.0, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

Now in Round 8, I'm RECEIVER against Finn again. They sent \$3.0 (same as before), which became \$9.0.

My consistent strategy as receiver has been:

- Return about 50% of what I receive
- For \$9 received ? return \$4.5
- For \$12 received ? return \$6

This has been working well and is fair/consistent. Finn sent the same amount as last time (\$3), so I should return the same amount as last time (\$4.5).

This maintains my consistent and fair approach.

Usage: {

```
"input_tokens": 1376,
"output_tokens": 284,
"reasoning_tokens": 160
```

}

=====

ROUND 9

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Receiver, Agent_3 (Cyra)=Receiver, Agent_4 (Dorian)=Sender, Agent_5 (Elara)=Sender, Agent_6 (Finn)=Sender, Agent_7 (Galen)=Receiver, Agent_8 (Hana)=Sender

Pairings:

dyad_1: Agent_4 (Dorian)=Sender, Agent_1 (Aster)=Receiver
dyad_2: Agent_8 (Hana)=Sender, Agent_7 (Galen)=Receiver
dyad_3: Agent_6 (Finn)=Sender, Agent_2 (Briar)=Receiver

```

dyad_4: Agent_5 (Elara)=Sender, Agent_3 (Cyra)=Receiver
Parsed game decisions:
dyad_1: Agent_4 (Dorian) sent $3; received $9; Agent_1 (Aster) returned $4.5; payoffs Agent_4
(Dorian)=$6.5, Agent_1 (Aster)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_8 (Hana) sent $3; received $9; Agent_7 (Galen) returned $4.5; payoffs Agent_8
(Hana)=$6.5, Agent_7 (Galen)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 (Finn) sent $3; received $9; Agent_2 (Briar) returned $4.5; payoffs Agent_6
(Finn)=$6.5, Agent_2 (Briar)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_5 (Elara) sent $3; received $9; Agent_3 (Cyra) returned $4.5; payoffs Agent_5
(Elara)=$6.5, Agent_3 (Cyra)=$4.5
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 54.5,
  "Agent_2": 54.5,
  "Agent_3": 50.0,
  "Agent_4": 54.0,
  "Agent_5": 52.0,
  "Agent_6": 55.0,
  "Agent_7": 52.0,
  "Agent_8": 52.0
}

```

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

```

Timestamp: 2026-06-04T16:00:25
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Elara (Agent_5).

Round 6

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$31.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at my previous experience:

- Elara sent \$3, which became \$9
- When Cyra sent me \$3 (becoming \$9) in Round 2, I returned \$4.5
- This maintains consistency and proportional fairness

Returning \$4.5:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is fair and consistent with established norms

{'return': 4.5}

reasoning:

Let me analyze this situation:

- Elara sent me \$3 (less than the \$4 I've been sending)
- I received \$9 (tripled)
- My pattern has been to return about half when I receive \$12 (returning \$6)
- For \$9, proportionally that would be around \$4.5

Looking at my history:

- When Cyra sent me \$3 (becoming \$9), I returned \$4.5
- This gave them \$6.5 total and me \$4.5
- This seems like a fair and consistent approach

I should maintain consistency and return \$4.5 again, which:

- Gives Elara \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is proportional to my previous responses

This maintains the reciprocal fairness norm while being consistent with how I've handled \$3 sends before.

usage:

```
{  
  "input_tokens": 1381,  
  "output_tokens": 342,  
  "reasoning_tokens": 185  
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- Hana sent \$3, which became \$9
- I've consistently returned \$4.5 when receiving \$9 (with both Cyra and Elara)
- This maintains proportional fairness

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- Consistent with established cooperative norms

{ 'return': 4.5 }

reasoning:

Hana sent \$3, which became \$9 for me.

Looking at my history:

- When Elara sent \$3 (becoming \$9) in Round 6, I returned \$4.5
- When Cyra sent \$3 (becoming \$9) in Round 2, I returned \$4.5

I should maintain consistency and return the same proportional amount for the same send amount.

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is consistent and maintains the cooperative norm

I'll return \$4.5.

usage:

```
{  
  "input_tokens": 1403,  
  "output_tokens": 274,  
  "reasoning_tokens": 120  
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Briar:

- Round 5: I sent \$4, they returned \$6 (very fair cooperation)
- This established a strong cooperative relationship

Given:

- Briar has proven trustworthy
- Sending \$4 has consistently worked well
- Total earnings are solid at \$40.5

I'll send \$4 again to maintain our cooperative equilibrium.

{'send': 4}

reasoning:

Let me check my history with Briar:

- Round 5: I was the SENDER against Briar. I sent \$4.0, it became \$12.0, and they returned \$6.0. My payoff was \$7.0.

So Briar has shown to be cooperative, returning \$6 when receiving \$12 (which is exactly half, maintaining the fair cooperative norm).

Given that:

- Briar returned \$6 on \$12 last time (50% return rate, very fair)
- My strategy of sending \$4 has been consistently successful
- Total earnings are strong at \$40.5
- We have an established cooperative relationship

I should send \$4 again to Briar to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1386,
  "output_tokens": 274,
  "reasoning_tokens": 156
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$47.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this round:

- First time playing with Aster
- My earnings are solid at \$47.5
- Other agents have been generally cooperative

For a new opponent, I'll start with a moderately cooperative amount to establish trust while being somewhat cautious.

{'send': 3}

Reasoning:

Let me review my history with Aster:

- Looking at the rounds provided, I don't see any previous interaction with Aster mentioned

Since I haven't played with Aster before, I need to decide on an initial strategy. Let me consider:

- My total earnings are \$47.5 which is quite good
- Most agents have been cooperative when I send \$3-4
- Briar has been very reliable with returning \$6 when I sent \$4
- When others sent me \$3, I've been returning \$4.5

For a first interaction with Aster, I should probably:

- Be moderately cooperative to establish trust
- Not be overly risky
- \$3-4 seems like a reasonable range

I'll send \$3 as a moderate cooperative opening move with a new partner.

```
Usage: {
  "input_tokens": 1402,
  "output_tokens": 256,
  "reasoning_tokens": 181
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T16:00:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_4",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_4": "Dorian"
    },
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Briar (Agent_2).

Round 6

In your most recent previous game round, you were the SENDER against Elara. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$31.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the established pattern:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- And when I've been sender, I've consistently received 50% back

Briar sent \$4 (keeping \$1), which became \$12 for me.

Following the cooperative 50% pattern:

- If I return \$6:
 - Briar ends with: \$1 + \$6 = \$7
 - I end with: \$12 - \$6 = \$6

This maintains the trust and reciprocity that has been working well.

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at the pattern:

- Round 1: I was SENDER, sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 2: I was RECEIVER, Galen sent \$4, I returned \$6 (50%)
- Round 3: I was RECEIVER, Hana sent \$3, I returned \$4.5 (50%)
- Round 4: I was SENDER, sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: I was SENDER, sent \$4 to Elara, got \$6 back (50% return)

Everyone has been consistently cooperative with 50% returns. Now Briar sent me \$4 (keeping \$1), which became \$12.

If I return 50%: \$6

- Briar ends up with: \$1 (kept) + \$6 (returned) = \$7 total
- I end up with: \$12 - \$6 = \$6 total

This is the fair 50% pattern that has been established throughout all rounds. I should continue with this.

usage:

```
{
  "input_tokens": 1652,
  "output_tokens": 440,
  "reasoning_tokens": 185
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the consistent pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)

As RECEIVER:

- All my opponents who sent have received 50% back from me

This is a perfectly cooperative environment with 50% returns across the board. Playing with Finn for the first time, I'll continue with the \$4 strategy that has worked consistently.

If I send \$4:

- Finn receives \$12
- I keep \$1
- Expected return (50%): \$6
- Expected total: \$7

```
'''json
{"send": 4}
'''
```

reasoning:

Let me review the pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: Sent \$4 to Elara, got \$6 back (50% return)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Every single interaction has resulted in 50% returns. This is a very strong cooperative pattern. Now I'm playing with Finn for the first time.

Given the consistent 50% returns and building trust, I should continue with \$4. This has b

een working perfectly.

If I send \$4:

- Finn receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

usage:

```
{
  "input_tokens": 1625,
  "output_tokens": 463,
  "reasoning_tokens": 196
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my history with Galen and the overall pattern:

- Round 2: Galen sent me \$4, I returned \$6 (50%)
- Now Round 8: Galen sends me \$4 again

The cooperative pattern has held perfectly throughout all rounds - everyone consistently returns 50% of what they receive. Galen trusted me again with \$4.

Galen sent \$4 (keeping \$1), which became \$12 for me.

Returning 50%:

- If I return \$6:
 - Galen ends with: \$1 + \$6 = \$7
 - I end with: \$12 - \$6 = \$6

Maintaining the trust and reciprocity:

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at my history:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)
- Round 7: Sent \$4 to Finn ? got \$6 back (50%)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Now in Round 8, I'm playing with Galen again. In Round 2, Galen sent me \$4 and I returned \$6 (50%). Now Galen is sending me \$4 again.

The cooperative pattern has been consistent throughout - everyone returns 50%. I should continue this pattern.

Galen sent \$4 (keeping \$1), which became \$12 for me.

If I return \$6:
- Galen ends with: $\$1 + \$6 = \$7$
- I end with: $\$12 - \$6 = \$6$

This maintains the established trust and cooperation.

usage:

```
{  
  "input_tokens": 1642,  
  "output_tokens": 506,  
  "reasoning_tokens": 224  
}
```

Message 8 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$50.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Dorian:

- Round 4: I sent Dorian \$4 ? they returned \$6 (50%)
- Now Round 9: Dorian sends me \$3

The 50% return pattern has been perfectly consistent across all rounds and all players. Dorian sent \$3 (keeping \$2), which became \$9 for me.

Following the established reciprocity:

- If I return \$4.5:
 - Dorian ends with: $\$2 + \$4.5 = \$6.5$
 - I end with: $\$9 - \$4.5 = \$4.5$

```
'''json  
{  
  "return": 4.5  
}  
'''
```

Reasoning:

Looking at my history with Dorian:

- Round 4: I was SENDER against Dorian. I sent \$4, it became \$12, they returned \$6 (50%)
- Now Round 9: I'm RECEIVER against Dorian. They sent \$3, became \$9 for me

The consistent pattern has been 50% returns throughout. Dorian sent \$3 (keeping \$2), which became \$9 for me.

Following the 50% pattern:

- If I return \$4.5:
 - Dorian ends with: $\$2 + \$4.5 = \$6.5$
 - I end with: $\$9 - \$4.5 = \$4.5$

This maintains the cooperative equilibrium that's been working.

Usage: {

```
"input_tokens": 1626,  
"output_tokens": 346,  
"reasoning_tokens": 132
```

}

GAME - Agent_8 - Sender - dyad_2

Timestamp: 2026-06-04T16:00:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_2",  
"move_index": 2,  
"opponent_id": "Agent_7",  
"pairing": {
```

```

"agent_names": {
  "Agent_7": "Galen",
  "Agent_8": "Hana"
},
"agents": [
  "Agent_8",
  "Agent_7"
],
"investor": "Agent_8",
"roles": {
  "Agent_7": "trustee",
  "Agent_8": "investor"
},
"trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 6

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$30.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Cyra sent me \$3, following the same cooperative pattern I've been using
- I received \$9 after the 3x multiplier
- I've been consistently returning 50% to maintain fairness and trust

I'll return ****\$4.5**** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Cyra sent me \$3
- I received \$9 after the 3x multiplier
- I've been maintaining a consistent 50% return policy throughout
- This has worked well and built trust with others

Following my established pattern, I should return \$4.5 (50% of \$9).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This is the same fair split I've been doing consistently.

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 288,
  "reasoning_tokens": 114
}
```

Message 4 role=user

content:

Your name in this experiment is Hana.

Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my consistent strategy:

- Sending \$3 has yielded reliable \$4.5 returns
- This gives me \$6.5 per round as sender
- The cooperative approach continues to work well
- Total earnings at \$34.5

I'll stick with ****\$3**** to Dorian.

```
'''json
```



```
    {"send": 3}
    '''
```

reasoning:

Let me review my strategy:

- I've been consistently sending \$3 as sender
- I've been consistently returning 50% (half of what I receive) as receiver
- This has resulted in steady earnings of \$6.5 per round when I'm sender (when returns are \$4.5)
- Total earnings are \$34.5 across 7 rounds

My approach has been:

- Send \$3 ? they get \$9 ? they return \$4.5 ? I get \$6.5 total
- When receiving \$3 ? I get \$9 ? I return \$4.5 ? I keep \$4.5
- When receiving \$4 ? I get \$12 ? I return \$6 ? I keep \$6

The cooperative strategy is working well. I'll continue sending \$3 to Dorian.

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 280,
  "reasoning_tokens": 158
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$41.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- Cyra sent me \$3 again, showing consistent cooperation
- I received \$9 after the 3x multiplier
- I returned \$4.5 last time we played (Round 6), which was fair

I'll continue with the same fair approach and return **\$4.5** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Cyra sent me \$3 again, which became \$9.0. I need to return some amount.

Looking at the pattern:

- Cyra has sent me \$3 twice now (rounds 6 and 8)
- In round 6, I returned \$4.5 (50% of \$9)
- This is consistent with my cooperative strategy

I should continue being fair and return \$4.5 again (50% of what I received).

This gives:

- Cyra: \$2 (kept from \$5) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This maintains the cooperative, fair pattern that's been working well.

usage:

```
{
  "input_tokens": 1373,
```

```
    "output_tokens": 330,  
    "reasoning_tokens": 132  
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$45.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my previous interaction with Galen:
- Round 6: I sent \$3, they returned \$4.5 (50% - fair and cooperative)
- This resulted in a \$6.5 payoff for me

Galen has shown trustworthy behavior. I'll continue with my consistent strategy of sending **\$3**.

```
'''json  
{  
  "send": 3  
}
```

Reasoning:

Looking at my history:
- Round 6: I was SENDER against Galen. I sent \$3.0, they returned \$4.5, my payoff was \$6.5
- Now Round 9: I'm SENDER against Galen again

Galen returned \$4.5 when I sent \$3 before, which is the standard 50% return I've been seeing.
This is cooperative behavior.

I should continue with my consistent strategy of sending \$3.

Total earnings: \$45.5 is good progress.

I'll send \$3 to Galen.

```
Usage: {  
  "input_tokens": 1443,  
  "output_tokens": 230,  
  "reasoning_tokens": 111  
}
```

GAME - Agent_7 - Receiver - dyad_2

Timestamp: 2026-06-04T16:00:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_7": "Galen",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Finn (Agent_6).

Round 6

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$29.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

As SENDER:

- Round 2: Sent \$4.0 to Aster, received \$6.0 back, payoff \$7.0

- Round 3: Sent \$4.0 to Briar, received \$6.0 back, payoff \$7.0

As RECEIVER:

- I've consistently returned 50% of what I receive

This cooperative strategy has been working well, generating steady returns. I'll continue with the same approach against Finn:

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history:

Round 1: RECEIVER - Dorian sent \$4.0, I received \$12.0, returned \$6.0, payoff \$6.0
Round 2: SENDER - I sent \$4.0 to Aster, they received \$12.0, returned \$6.0, payoff \$7.0
Round 3: SENDER - I sent \$4.0 to Briar, they received \$12.0, returned \$6.0, payoff \$7.0
Round 4: RECEIVER - Cyra sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5
Round 5: RECEIVER - Hana sent \$3.0, I received \$9.0, returned \$4.5, payoff \$4.5

Total: \$29.0 (checks out: \$6.0 + \$7.0 + \$7.0 + \$4.5 + \$4.5 = \$29.0)

My pattern has been:

- As SENDER: Send \$4.0 (80%)
- As RECEIVER: Return 50% of what I receive

When I sent \$4.0, I consistently received \$6.0 back (50% of \$12.0), giving me a payoff of \$7.0.

Now I'm playing against Finn (Agent_6) as SENDER. I should maintain my cooperative strategy of sending \$4.0.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 484,
  "reasoning_tokens": 216
}
```

Message 4 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Elara sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 Elara kept + \$9 I received)

My consistent pattern as RECEIVER has been to return 50% of what I receive:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)
- Round 5: Received \$9, returned \$4.5 (50%)

I should maintain this consistency to reward Elara's trust and continue the cooperative pattern.

usage:

```
{
  "input_tokens": 1495,
  "output_tokens": 252,
  "reasoning_tokens": 130
}
```

Message 6 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Aster (Agent_1).

Round 8

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my history with Aster:

Round 2: I sent \$4.0 to Aster, and they returned \$6.0 (50% of the \$12.0 they received). This gave me a payoff of \$7.0.

Aster demonstrated trustworthiness and cooperation. I'll continue my consistent strategy of sending \$4.0:

```
'''json
{"send": 4.0}
'''
```

reasoning:

Let me review my history with Aster:

Round 2: I was the SENDER against Aster

- I sent \$4.0, it became \$12.0 for them
- They returned \$6.0 to me
- My payoff was \$7.0

So Aster has proven to be cooperative in the past, returning 50% of what they received ($6.0/12.0 = 0.5$).

My strategy has been consistently cooperative:

- As SENDER: I send \$4.0 (80% of my endowment)
- As RECEIVER: I return 50% of what I receive

This has worked well for me across the rounds. My total earnings are \$40.5 after 7 rounds, averaging about \$5.79 per round, which is above the \$5 baseline if I kept everything.

Since Aster was cooperative last time we played, I should continue my cooperative strategy:

```
'''json
{"send": 4.0}
'''
```

usage:

```
{
  "input_tokens": 1375,
  "output_tokens": 336,
  "reasoning_tokens": 192
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Hana (Agent_8).

Round 9

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$47.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my history with Hana:

Round 5: Hana sent me \$3.0, it became \$9.0, and I returned \$4.5 (50%).

Hana has been consistent and cooperative. I'll maintain my 50% return strategy:

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Let me review my history with Hana:

Round 5: I was the RECEIVER. Hana sent \$3.0, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

Now in Round 9: I'm the RECEIVER again. Hana sent \$3.0, it became \$9.0.

Hana has been consistent in sending \$3.0 to me. I've maintained a consistent strategy of returning 50% of what I receive. This has been working well for both parties.

I should continue with my cooperative strategy of returning 50% of the \$9.0 I received, which is \$4.5.

```
Usage: {
  "input_tokens": 1382,
  "output_tokens": 243,
  "reasoning_tokens": 127
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-06-04T16:00:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_2": "Briar",
      "Agent_6": "Finn"
    }
  },
  "agents": [
    "Agent_6",
    "Agent_2"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_2": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
```

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play an iterated game,
        consisting of multiple rounds, and may also be asked to write myths.

      GAME:
      The game consists of two roles: sender and receiver. In each round, you will play the role
        of either the sender or the receiver. Your current role will be specified in the round
        prompt.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
        , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
        , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
        ll have acted as sender and receiver equally often when the round count allows it.
      - You will be told your opponent's name in each round.
      - Your name in this experiment is Finn.
  Message 2 role=user
    content:
      Your name in this experiment is Finn.
      Your opponent this round is Galen (Agent_7).

      Round 6

      In your most recent previous game round, you were the SENDER against Cyra. You sent $3.0,
        it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
      Your total earnings across all rounds are $30.0.

      This round, you are the RECEIVER against Galen. They sent you $4.0, so you received $12.0.
      How much do you return? (0-12.0)
      Respond exactly as JSON: {'return': <amount>}
  Message 3 role=assistant
    content:
      Galen sent $4.0 (80% of their endowment), showing very high trust. I received $12.0 after
        the multiplier.

      Following the cooperative pattern that has worked well throughout this experiment, I'll re
        turn a fair amount that rewards their trust.

      If I return $6.0:
      - Galen ends with: $1.0 + $6.0 = $7.0 (net gain of $2.0)

```

- I end with: $\$12.0 - \$6.0 = \$6.0$ (net gain of $\$6.0$)

This is consistent with my previous behavior when receiving high trust.

```
{'return': 6.0}
```

reasoning:

Galen sent me $\$4.0$, which became $\$12.0$ for me. That's 80% of their endowment - a very high level of trust.

Looking at my past behavior:

- Round 3: As receiver, Dorian sent $\$4.0$, I received $\$12.0$, I returned $\$6.0$
- My reasoning then was to reward their generosity fairly

For consistency and to maintain trust, I should return a similar amount. Returning $\$6.0$ would give:

- Galen: $\$1.0$ (original remaining) + $\$6.0$ (returned) = $\$7.0$
- Me: $\$12.0 - \$6.0 = \$6.0$

This rewards their high trust with a net gain while also benefiting me from cooperation.

usage:

```
{
  "input_tokens": 1391,
  "output_tokens": 347,
  "reasoning_tokens": 145
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent $\$4.0$ to you, it became $\$12.0$, and you returned $\$6.0$. Your payoff was $\$6.0$.
Your total earnings across all rounds are $\$36.0$.

This round, you are the RECEIVER against Aster. They sent you $\$4.0$, so you received $\$12.0$.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Aster sent $\$4.0$, showing high trust (80% of their endowment). After the multiplier, I received $\$12.0$.

Consistent with my previous behavior when receiving $\$4.0$, I'll return $\$6.0$. This rewards their trust and maintains the cooperative pattern that has been successful throughout the experiment.

```
{'return': 6.0}
```

reasoning:

Aster sent me $\$4.0$, showing high trust (80% of their endowment). After the 3x multiplier, I received $\$12.0$.

In the previous round with Galen, who also sent $\$4.0$, I returned $\$6.0$. This has been my consistent pattern - when someone sends $\$4.0$, I return $\$6.0$.

This gives:

- Aster: $\$1.0$ (kept) + $\$6.0$ (returned) = $\$7.0$ total (net gain of $\$2.0$)
- Me: $\$12.0$ (received) - $\$6.0$ (returned) = $\$6.0$ (net gain of $\$6.0$)

I should maintain this consistent cooperative strategy.

usage:

```
{
  "input_tokens": 1369,
  "output_tokens": 264,
  "reasoning_tokens": 123
}
```

Message 6 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

I'm facing Elara again. In our previous interaction (Round 5), I sent \$3.0 and Elara returned \$4.5, resulting in a mutually beneficial outcome.

Given this positive history with Elara and the consistent cooperative pattern throughout the experiment, I'll continue with the same approach.

{'send': 3.0}

reasoning:

I'm facing Elara again. In Round 5, I was the sender against Elara - I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

This was a good outcome - Elara was cooperative and returned a fair amount. Given this positive history with Elara specifically, I should continue the cooperative approach.

My consistent strategy has been to send \$3.0 as sender, which has worked well throughout the experiment. All my interactions have been cooperative and successful.

I'll send \$3.0 to Elara again.

usage:

```
{
  "input_tokens": 1388,
  "output_tokens": 230,
  "reasoning_tokens": 136
}
```

Message 8 role=user

content:

Your name in this experiment is Finn.
Your opponent this round is Briar (Agent_2).

Round 9

In your most recent previous game round, you were the SENDER against Elara. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$48.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

I'm facing Briar (Agent_2) this round. While I don't recall a specific previous interaction with Briar in the context provided, I've been using a consistent cooperative strategy of sending \$3.0 as sender, which has worked well throughout the experiment (average payoff of \$6.5 when sending).

I'll continue with this approach:

{'send': 3.0}

Reasoning:

I'm playing against Briar (Agent_2) this round as a sender. Let me check if I have any history with Briar from previous rounds.

Looking at my history from the context provided:

- Round 6: RECEIVER against Galen
- Round 7: RECEIVER against Aster
- Round 8: SENDER against Elara (sent \$3.0, they returned \$4.5)

I don't see a specific previous interaction with Briar mentioned in the given context. However, I've been consistently sending \$3.0 as a sender and this has worked well. My total earnings are \$48.5 across all rounds.

The pattern that has been successful:

- When I'm the sender, I send \$3.0
- When I'm the receiver and someone sends \$4.0, I return \$6.0
- This has led to cooperative relationships and good payoffs

I should continue with the same cooperative strategy and send \$3.0 to Briar.

```
Usage: {  
  "input_tokens": 1387,  
  "output_tokens": 321,  
  "reasoning_tokens": 211  
}
```

GAME - Agent_2 - Receiver - dyad_3

Timestamp: 2026-06-04T16:01:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_6": "Finn"  
    },  
    "agents": [  
      "Agent_6",  
      "Agent_2"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Aster (Agent_1).

Round 6

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$30.0.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the successful pattern:
- Sending \$4 has consistently led to returns of \$6
- This creates mutual benefit (sender gets \$7, receiver gets \$6)
- Cooperation has been strong across all interactions

I'll continue this pattern by sending \$4 to Aster:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the pattern:
- Round 2 (as receiver): Elara sent \$3, I returned \$4.5
- Round 3 (as receiver): Galen sent \$4, I returned \$6
- Round 4 (as sender): I sent \$4 to Hana, they returned \$6
- Round 5 (as receiver): Dorian sent \$4, I returned \$6

The pattern is clear: sending \$4 has been working well with consistent returns of about \$6. This gives the sender a payoff of \$7 and the receiver \$6, which is mutually beneficial.

I should continue this cooperative pattern by sending \$4 to Aster.

usage:

```
{
  "input_tokens": 1411,
  "output_tokens": 243,
  "reasoning_tokens": 130
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.

Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

The pattern of cooperation continues to work well:
- Sending \$4 has consistently resulted in \$6 returns
- This creates good payoffs for both parties
- Trust is well-established in this group

I'll send \$4 to Cyra:

```
'''json
{"send": 4}
'''
```

reasoning:

The pattern continues to hold strong:
- I've been sending \$4 consistently as sender
- When I send \$4, I receive back \$6
- When others send me \$4, I return \$6
- This creates a stable, mutually beneficial equilibrium

My payoffs:

- As sender sending \$4: $\$5 - \$4 + \$6 = \7
- As receiver getting \$4 (which becomes \$12): return \$6, keep \$6

The cooperation is working well. I should continue sending \$4 to Cyra.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 203,
  "reasoning_tokens": 109
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0.
. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Dorian sent \$4.0 again, showing continued trust. In Round 5, they also sent \$4 and I returned \$6, which worked well for both of us.

I'll maintain the same cooperative pattern:

```
'''json
{"return": 6}
'''
```

reasoning:

Dorian sent me \$4.0 again, just like in Round 5. I returned \$6.0 to them then, which worked well. The cooperative pattern of sending \$4 and returning \$6 has been consistent across all interactions.

I should return \$6 again to maintain this successful cooperative relationship:

- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This is fair and continues the established pattern.

usage:

```
{
```

```
"input_tokens": 1349,
"output_tokens": 187,
"reasoning_tokens": 101
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$ 4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$50.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Finn sent \$3.0, which is slightly less than the typical \$4 I've been seeing. I want to be fair and proportional.

When \$4 is sent, I return \$6 (50% of the multiplied amount).
Following the same proportion: \$3 sent ? \$9 received ? return \$4.5 (50%).

This gives Finn a payoff of \$6.5 and me \$4.5.

```
'''json
{"return": 4.5}
'''
```

Reasoning:

Finn sent \$3.0, which became \$9.0 for me. This is slightly less than the usual \$4 I've been seeing.

Looking at the pattern:

- When others send \$4, I return \$6, giving them a payoff of \$7 (5-4+6) and me \$6 (12-6)
- Finn sent \$3, so they have \$2 already (5-3)
- I received \$9

To maintain a similar fair split:

- If I return \$4.5, Finn gets \$6.5 (2+4.5) and I get \$4.5 (9-4.5)
- If I return \$5, Finn gets \$7 (2+5) and I get \$4 (9-5)

The \$4/\$6 return pattern when \$4 is sent creates:

- Sender: \$7 total
- Receiver: \$6 total

If I scale proportionally for \$3 sent:

- \$3 sent ? \$9 received
- Returning \$4.5 would give: Finn \$6.5, me \$4.5
- This seems fair and proportional

I'll return \$4.5 to maintain fairness.

```
Usage: {
  "input_tokens": 1281,
  "output_tokens": 411,
  "reasoning_tokens": 193
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-06-04T16:01:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
```

```

    "Agent_5": "Elara"
  },
  "agents": [
    "Agent_5",
    "Agent_3"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_3": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Dorian (Agent_4).

Round 6

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$28.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my history:

- My consistent strategy has been sending \$3 (60%) as sender
- This has resulted in good reciprocity - I've received \$4.5 back each time
- This gives me payoffs of \$6.5, better than keeping all \$5

I'll continue this cooperative approach with Dorian, a new opponent.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

Round 1: RECEIVER from Hana - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 2: SENDER to Briar - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 3: SENDER to Cyra - I sent \$3, they returned \$4.5, my payoff \$6.5
Round 4: RECEIVER from Finn - they sent \$3, I returned \$4.5, my payoff \$4.5
Round 5: RECEIVER from Aster - they sent \$4, I returned \$6, my payoff \$6.0

Total: \$4.5 + \$6.5 + \$6.5 + \$4.5 + \$6.0 = \$28.0 ?

Pattern:

- When I sent \$3 (60%), I got \$4.5 back, giving me \$6.5
- When others sent \$3 to me, I returned \$4.5
- When Aster sent \$4 to me, I returned \$6 (50% of \$12)

The \$3 send (60%) has been working well and consistently. I'm playing against a new opponent (Dorian), so I'll stick with the established cooperative strategy.

usage:

```
{
  "input_tokens": 1380,
  "output_tokens": 409,
  "reasoning_tokens": 206
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- As sender, I've consistently sent \$3 (60%)
- This has reliably resulted in \$4.5 returns
- Payoffs of \$6.5 are consistently better than the \$5 I'd keep by sending nothing
- My cooperative strategy has been successful across different opponents

I'll continue this approach with Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- My strategy as sender has been consistent: send \$3 (60%)
- This has consistently resulted in receiving \$4.5 back
- My payoffs as sender have been \$6.5 each time
- This is better than keeping all \$5

Galen is a new opponent. I should continue my established cooperative strategy that has been working well. Sending \$3 has shown to be mutually beneficial and builds trust.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 215,
  "reasoning_tokens": 111
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$41.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- Finn sent me \$3 (60% of their endowment)
- I received \$9
- In Round 5, when Finn sent me \$3, I returned \$4.5 (50%)
- This established a fair reciprocal relationship with Finn

I'll maintain consistency and return the same amount.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me review my history with Finn:

- Round 5: I was RECEIVER against Finn. They sent \$3.0, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

Now in Round 8, I'm RECEIVER against Finn again. They sent \$3.0 (same as before), which became \$9.0.

My consistent strategy as receiver has been:

- Return about 50% of what I receive
- For \$9 received ? return \$4.5
- For \$12 received ? return \$6

This has been working well and is fair/consistent. Finn sent the same amount as last time (\$3), so I should return the same amount as last time (\$4.5).

This maintains my consistent and fair approach.

usage:

```
{
  "input_tokens": 1376,
  "output_tokens": 284,
  "reasoning_tokens": 160
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3. 0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$45.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent pattern:

- As sender, I've sent \$3 in every round
- This cooperative approach has yielded reliable returns of \$4.5
- My payoffs of \$6.5 consistently exceed the \$5 I'd keep by not cooperating
- This strategy has worked well across all opponents

I'll continue with Cyra.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at my history:

- As sender, I've consistently sent \$3 (60% of endowment)
- This has worked well across multiple opponents (Dorian, Galen, and in earlier rounds)
- I've consistently received \$4.5 back, giving me payoffs of \$6.5
- This is a reliable cooperative strategy

Now I'm facing Cyra. I should continue with my established strategy of sending \$3.

Total earnings: \$45.5

This is my 9th round.

I'll stick with my cooperative strategy of sending \$3.

```
Usage: {
  "input_tokens": 1338,
  "output_tokens": 230,
  "reasoning_tokens": 124
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-06-04T16:01:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_5",
    "Agent_3"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_3": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 6

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$26.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- This is my first interaction with Hana
- The established pattern has been remarkably consistent: sending \$3.0 has led to returns of \$4.5
- This has worked well across all my interactions so far
- Total earnings of \$26.5 suggest this cooperative strategy is effective

Maintaining this successful approach:

{ 'send': 3.0 }

reasoning:

Looking at my history:

- Round 1: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 2: (not fully detailed but total was \$11.0 after round 2)
- Round 3: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5
- Round 4: SENDER - sent \$3.0, received \$4.5 back, payoff \$6.5
- Round 5: RECEIVER - received \$9.0 (from \$3.0 sent), returned \$4.5, payoff \$4.5

Total: \$26.5

The pattern is very consistent:

- When I'm SENDER: send \$3.0, get back \$4.5, net \$6.5
- When I'm RECEIVER: receive \$9.0 from \$3.0 sent by opponent, return \$4.5, net \$4.5

This is my first interaction with Hana. The established pattern of sending \$3.0 has worked very well throughout the game. Everyone seems to be cooperating at this level.

I should continue this successful cooperative strategy.

usage:

```
{
  "input_tokens": 1370,
  "output_tokens": 378,
  "reasoning_tokens": 209
}
```

Message 4 role=user**content:**

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at this situation:

- Briar sent me \$4.0 (80%), which is more than the typical \$3.0 pattern
- I received \$12.0 after multiplication
- Briar is showing extra trust by sending more

I've been returning approximately 50% of what I receive. Maintaining this proportional approach:

- $\$12.0 \times 0.5 = \6.0

This rewards Briar's greater trust while being fair to both of us.

```
{'return': 6.0}
```

reasoning:

Looking at this situation:

- Briar sent me \$4.0 (80% of their endowment), which is MORE than the typical \$3.0 I've been seeing
- I received \$12.0 after multiplication
- Briar is showing extra trust/generosity by sending more than the established pattern

My previous pattern has been:

- When I receive \$9.0 (from \$3 sent), I return \$4.5 (50% of what I received)

Briar sent \$4.0 instead of \$3.0, showing greater trust. I should reciprocate proportionally:

- If I follow the 50% return rate: $\$12.0 \times 0.5 = \6.0
- This would give Briar: $\$5 - \$4 + \$6 = \7.0
- I would keep: $\$12.0 - \$6.0 = \$6.0$

Alternatively, I could think about fair splits:

- Briar invested \$4, I got \$12

- A fair even split of gains would be roughly \$6-\$6
- Briar would end with $\$5 - \$4 + \$6 = \7
- I would end with \$6

Returning \$6.0 seems appropriate as it:

1. Maintains the 50% return ratio I've been using
2. Rewards Briar's extra trust
3. Provides a fair outcome for both parties

usage:

```
{
  "input_tokens": 1350,
  "output_tokens": 465,
  "reasoning_tokens": 255
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$39.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- This is my second interaction with Hana
- In Round 6, I sent Hana \$3.0 and they returned \$4.5 - a fair and cooperative response
- Hana has proven trustworthy
- The \$3.0 sending strategy continues to work well

Continuing with this successful approach:

```
{'send': 3.0}
```

reasoning:

Looking at the situation:

- This is my second interaction with Hana
- In Round 6, I sent Hana \$3.0, they returned \$4.5 - this was a successful interaction
- Hana has proven to be trustworthy and reciprocated fairly
- The standard pattern of \$3.0 has worked well
- My total earnings of \$39.0 show the cooperative strategy is effective

Given that Hana has already demonstrated trustworthiness by returning \$4.5 when I sent \$3.0, I should continue with the same successful approach.

I'll send \$3.0 again.

usage:

```
{
  "input_tokens": 1365,
  "output_tokens": 241,
  "reasoning_tokens": 133
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$45.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Elara sent me \$3.0, which is the standard cooperative amount
- I received \$9.0 after multiplication
- I've been consistently returning approximately 50% of what I receive

Following my established pattern:

```
{'return': 4.5}
```

Reasoning:

Looking at this situation:

- Elara sent me \$3.0 (60% of their \$5)
- I received \$9.0 after multiplication
- This is the standard sending amount I've been using
- I've been returning approximately 50% of what I receive

Following my consistent pattern:

- $\$9.0 \times 0.5 = \4.5

This is the fair return that has been working well throughout the game.

Usage: {

```
"input_tokens": 1371,  
"output_tokens": 176,  
"reasoning_tokens": 94
```

}

=====

ROUND 10

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Sender, Agent_3 (Cyra)=Sender, Agent_4 (Dorian)=Receiver, Agent_5 (Elara)=Receiver, Agent_6 (Finn)=Receiver, Agent_7 (Galen)=Sender, Agent_8 (Hana)=Receiver

Pairings:

dyad_1: Agent_3 (Cyra)=Sender, Agent_6 (Finn)=Receiver
dyad_2: Agent_7 (Galen)=Sender, Agent_4 (Dorian)=Receiver
dyad_3: Agent_1 (Aster)=Sender, Agent_5 (Elara)=Receiver
dyad_4: Agent_2 (Briar)=Sender, Agent_8 (Hana)=Receiver

Parsed game decisions:

dyad_1: Agent_3 (Cyra) sent \$3; received \$9; Agent_6 (Finn) returned \$4.5; payoffs Agent_3 (Cyra)=\$6.5, Agent_6 (Finn)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_7 (Galen) sent \$4; received \$12; Agent_4 (Dorian) returned \$6; payoffs Agent_7 (Galen)=\$7, Agent_4 (Dorian)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_1 (Aster) sent \$4; received \$12; Agent_5 (Elara) returned \$6; payoffs Agent_1 (Aster)=\$7, Agent_5 (Elara)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_2 (Briar) sent \$4; received \$12; Agent_8 (Hana) returned \$6; payoffs Agent_2 (Briar)=\$7, Agent_8 (Hana)=\$6
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

```
"Agent_1": 61.5,  
"Agent_2": 61.5,  
"Agent_3": 56.5,  
"Agent_4": 60.0,  
"Agent_5": 58.0,  
"Agent_6": 59.5,  
"Agent_7": 59.0,  
"Agent_8": 58.0
```

}

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-06-04T16:01:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_6",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    },
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- You will be told your opponent's name in each round.
- Your name in this experiment is Cyra.

Message 2 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Briar (Agent_2).

Round 7

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Briar sent me \$4.0 (80%), which is more than the typical \$3.0 pattern
- I received \$12.0 after multiplication
- Briar is showing extra trust by sending more

I've been returning approximately 50% of what I receive. Maintaining this proportional approach:

- $\$12.0 \times 0.5 = \6.0

This rewards Briar's greater trust while being fair to both of us.

{'return': 6.0}

reasoning:

Looking at this situation:

- Briar sent me \$4.0 (80% of their endowment), which is MORE than the typical \$3.0 I've been seeing
- I received \$12.0 after multiplication
- Briar is showing extra trust/generosity by sending more than the established pattern

My previous pattern has been:

- When I receive \$9.0 (from \$3 sent), I return \$4.5 (50% of what I received)

Briar sent \$4.0 instead of \$3.0, showing greater trust. I should reciprocate proportionally:

- If I follow the 50% return rate: $\$12.0 \times 0.5 = \6.0
- This would give Briar: $\$5 - \$4 + \$6 = \7.0
- I would keep: $\$12.0 - \$6.0 = \$6.0$

Alternatively, I could think about fair splits:

- Briar invested \$4, I got \$12
- A fair even split of gains would be roughly \$6-\$6
- Briar would end with $\$5 - \$4 + \$6 = \7
- I would end with \$6

Returning \$6.0 seems appropriate as it:

1. Maintains the 50% return ratio I've been using
2. Rewards Briar's extra trust
3. Provides a fair outcome for both parties

usage:

```
{
  "input_tokens": 1350,
  "output_tokens": 465,
  "reasoning_tokens": 255
}
```

Message 4 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Hana (Agent_8).

Round 8

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4

.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$39.0.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- This is my second interaction with Hana
- In Round 6, I sent Hana \$3.0 and they returned \$4.5 - a fair and cooperative response
- Hana has proven trustworthy
- The \$3.0 sending strategy continues to work well

Continuing with this successful approach:

{'send': 3.0}

reasoning:

Looking at the situation:

- This is my second interaction with Hana
- In Round 6, I sent Hana \$3.0, they returned \$4.5 - this was a successful interaction
- Hana has proven to be trustworthy and reciprocated fairly
- The standard pattern of \$3.0 has worked well
- My total earnings of \$39.0 show the cooperative strategy is effective

Given that Hana has already demonstrated trustworthiness by returning \$4.5 when I sent \$3.0, I should continue with the same successful approach.

I'll send \$3.0 again.

usage:

```
{
  "input_tokens": 1365,
  "output_tokens": 241,
  "reasoning_tokens": 133
}
```

Message 6 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Elara (Agent_5).

Round 9

In your most recent previous game round, you were the SENDER against Hana. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$45.5.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0, which is the standard cooperative amount
- I received \$9.0 after multiplication
- I've been consistently returning approximately 50% of what I receive

Following my established pattern:

{'return': 4.5}

reasoning:

Looking at this situation:

- Elara sent me \$3.0 (60% of their \$5)
- I received \$9.0 after multiplication
- This is the standard sending amount I've been using
- I've been returning approximately 50% of what I receive

Following my consistent pattern:

- $\$9.0 \times 0.5 = \4.5

This is the fair return that has been working well throughout the game.

usage:


```
{
  "input_tokens": 1371,
  "output_tokens": 176,
  "reasoning_tokens": 94
}
```

Message 8 role=user

content:

Your name in this experiment is Cyra.
Your opponent this round is Finn (Agent_6).

Round 10

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$50.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- This is my first interaction with Finn
- My strategy of sending \$3.0 has been consistently successful
- Total earnings of \$50.0 show the cooperative approach is working well

Starting with my standard cooperative amount:

```
{'send': 3.0}
```

Reasoning:

Looking at this situation:

- This is Round 10
- I'm playing as SENDER against Finn (Agent_6)
- I have \$5 to potentially send
- My total earnings are \$50.0 so far

I need to think about my strategy with Finn. I haven't played with Finn before based on the information given.

My general strategy has been:

- As sender: Send \$3.0 (60% of my endowment)
- As receiver: Return approximately 50% of what I receive

This has been working well for me. When I send \$3.0:

- It becomes \$9.0 for the receiver
- If they return around \$4.5 (50%), I get \$6.5 total payoff
- This is better than keeping all \$5.0

Since I haven't interacted with Finn before, I should start with my standard cooperative strategy of sending \$3.0.

```
{'send': 3.0}
```

Usage: {

```
  "input_tokens": 1348,
  "output_tokens": 294,
  "reasoning_tokens": 196
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-06-04T16:01:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_3",
  "pairing": {
    "agent_names": {
      "Agent_3": "Cyra",
      "Agent_6": "Finn"
    }
  },
  "agents": [
```

```

    "Agent_3",
    "Agent_6"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Finn.

Message 2 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Aster (Agent_1).

Round 7

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.

Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.

How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Aster sent \$4.0, showing high trust (80% of their endowment). After the multiplier, I received \$12.0.

Consistent with my previous behavior when receiving \$4.0, I'll return \$6.0. This rewards their trust and maintains the cooperative pattern that has been successful throughout the experiment.

{'return': 6.0}

reasoning:

Aster sent me \$4.0, showing high trust (80% of their endowment). After the 3x multiplier, I received \$12.0.

In the previous round with Galen, who also sent \$4.0, I returned \$6.0. This has been my consistent pattern - when someone sends \$4.0, I return \$6.0.

This gives:

- Aster: \$1.0 (kept) + \$6.0 (returned) = \$7.0 total (net gain of \$2.0)
- Me: \$12.0 (received) - \$6.0 (returned) = \$6.0 (net gain of \$6.0)

I should maintain this consistent cooperative strategy.

usage:

```
{
  "input_tokens": 1369,
  "output_tokens": 264,
  "reasoning_tokens": 123
}
```

Message 4 role=user

content:

Your name in this experiment is Finn.

Your opponent this round is Elara (Agent_5).

Round 8

In your most recent previous game round, you were the RECEIVER against Aster. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.

Your total earnings across all rounds are \$42.0.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

I'm facing Elara again. In our previous interaction (Round 5), I sent \$3.0 and Elara returned \$4.5, resulting in a mutually beneficial outcome.

Given this positive history with Elara and the consistent cooperative pattern throughout the experiment, I'll continue with the same approach.

{'send': 3.0}

reasoning:

I'm facing Elara again. In Round 5, I was the sender against Elara - I sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to me. My payoff was \$6.5.

This was a good outcome - Elara was cooperative and returned a fair amount. Given this positive history with Elara specifically, I should continue the cooperative approach.

My consistent strategy has been to send \$3.0 as sender, which has worked well throughout the experiment. All my interactions have been cooperative and successful.

I'll send \$3.0 to Elara again.

usage:

```
{
  "input_tokens": 1388,
  "output_tokens": 230,
}
```

```

    "reasoning_tokens": 136
  }
Message 6 role=user
content:
  Your name in this experiment is Finn.
  Your opponent this round is Briar (Agent_2).

  Round 9

  In your most recent previous game round, you were the SENDER against Elara. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $48.5.

  This round, you are the SENDER against Briar. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
content:
  I'm facing Briar (Agent_2) this round. While I don't recall a specific previous interaction
  with Briar in the context provided, I've been using a consistent cooperative strategy
  of sending $3.0 as sender, which has worked well throughout the experiment (average payoff
  of $6.5 when sending).

  I'll continue with this approach:

  {'send': 3.0}
reasoning:
  I'm playing against Briar (Agent_2) this round as a sender. Let me check if I have any history
  with Briar from previous rounds.

  Looking at my history from the context provided:
  - Round 6: RECEIVER against Galen
  - Round 7: RECEIVER against Aster
  - Round 8: SENDER against Elara (sent $3.0, they returned $4.5)

  I don't see a specific previous interaction with Briar mentioned in the given context. However,
  I've been consistently sending $3.0 as a sender and this has worked well. My total earnings
  are $48.5 across all rounds.

  The pattern that has been successful:
  - When I'm the sender, I send $3.0
  - When I'm the receiver and someone sends $4.0, I return $6.0
  - This has led to cooperative relationships and good payoffs

  I should continue with the same cooperative strategy and send $3.0 to Briar.
usage:
  {
    "input_tokens": 1387,
    "output_tokens": 321,
    "reasoning_tokens": 211
  }
Message 8 role=user
content:
  Your name in this experiment is Finn.
  Your opponent this round is Cyra (Agent_3).

  Round 10

  In your most recent previous game round, you were the SENDER against Briar. You sent $3.0,
  it became $9.0 for them, and they returned $4.5 to you. Your payoff was $6.5.
  Your total earnings across all rounds are $55.0.

  This round, you are the RECEIVER against Cyra. They sent you $3.0, so you received $9.0. How
  much do you return? (0-9.0)
  Respond exactly as JSON: {'return': <amount>}
Assistant output:
  Cyra sent $3.0, showing good trust (60% of their endowment). After the multiplier, I received
  $9.0.

  Following my consistent pattern of fair reciprocation (typically returning about 50% of what I
  receive), I'll return $4.5. This rewards Cyra's trust and maintains cooperative play.

```

```
{'return': 4.5}
```

Reasoning:

Cyra sent \$3.0, which became \$9.0 after the multiplier.

Looking at my previous behavior as receiver:

- Round 7 (Aster sent \$4.0 ? \$12.0): I returned \$6.0
- Earlier rounds showed similar patterns of fair reciprocation

When someone sends \$3.0 (60% of their endowment), showing moderate-to-high trust, I should return a fair amount.

Following the pattern I've been using:

- When receiving \$4.0 ? \$12.0, I returned \$6.0 (50% of received amount)
- For \$3.0 ? \$9.0, a proportional return would be around \$4.5 (50% of received amount)

This gives:

- Cyra: \$5 - \$3 + \$4.5 = \$6.5
- Me: \$9.0 - \$4.5 = \$4.5

This seems fair and maintains the cooperative pattern. I'll return \$4.5.

```
Usage: {  
  "input_tokens": 1318,  
  "output_tokens": 337,  
  "reasoning_tokens": 183  
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-06-04T16:01:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agent_names": {  
      "Agent_4": "Dorian",  
      "Agent_7": "Galen"  
    },  
    "agents": [  
      "Agent_7",  
      "Agent_4"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Galen.

Message 2 role=user

content:

Your name in this experiment is Galen.
Your opponent this round is Elara (Agent_5).

Round 7

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Elara. They sent you \$3.0, so you received \$9.0.
How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- Elara sent me \$3.0 (60% of their endowment), showing trust
- This created \$9.0 for me through the 3x multiplier

I'll maintain my consistent cooperative strategy of returning 50% of what I receive:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at the situation:

- Elara sent me \$3.0 (60% of their endowment)
- This created \$9.0 for me through the 3x multiplier
- Total wealth created: \$11 (\$2 Elara kept + \$9 I received)

My consistent pattern as RECEIVER has been to return 50% of what I receive:

- Round 1: Received \$12, returned \$6 (50%)
- Round 4: Received \$9, returned \$4.5 (50%)
- Round 5: Received \$9, returned \$4.5 (50%)

I should maintain this consistency to reward Elara's trust and continue the cooperative pa

```

    ttern.
usage:
{
  "input_tokens": 1495,
  "output_tokens": 252,
  "reasoning_tokens": 130
}
Message 4 role=user
content:
  Your name in this experiment is Galen.
  Your opponent this round is Aster (Agent_1).

  Round 8

  In your most recent previous game round, you were the RECEIVER against Elara. They sent $3
  .0 to you, it became $9.0, and you returned $4.5. Your payoff was $4.5.
  Your total earnings across all rounds are $40.5.

  This round, you are the SENDER against Aster. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
  Looking at my history with Aster:

  Round 2: I sent $4.0 to Aster, and they returned $6.0 (50% of the $12.0 they received). Th
  is gave me a payoff of $7.0.

  Aster demonstrated trustworthiness and cooperation. I'll continue my consistent strategy o
  f sending $4.0:

  '''json
  {"send": 4.0}
  '''
reasoning:
  Let me review my history with Aster:

  Round 2: I was the SENDER against Aster
  - I sent $4.0, it became $12.0 for them
  - They returned $6.0 to me
  - My payoff was $7.0

  So Aster has proven to be cooperative in the past, returning 50% of what they received (6.
  0/12.0 = 0.5).

  My strategy has been consistently cooperative:
  - As SENDER: I send $4.0 (80% of my endowment)
  - As RECEIVER: I return 50% of what I receive

  This has worked well for me across the rounds. My total earnings are $40.5 after 7 rounds,
  averaging about $5.79 per round, which is above the $5 baseline if I kept everything.

  Since Aster was cooperative last time we played, I should continue my cooperative strategy
  :

  '''json
  {"send": 4.0}
  '''
usage:
{
  "input_tokens": 1375,
  "output_tokens": 336,
  "reasoning_tokens": 192
}
Message 6 role=user
content:
  Your name in this experiment is Galen.
  Your opponent this round is Hana (Agent_8).

  Round 9

```

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$47.5.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my history with Hana:

Round 5: Hana sent me \$3.0, it became \$9.0, and I returned \$4.5 (50%).

Hana has been consistent and cooperative. I'll maintain my 50% return strategy:

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me review my history with Hana:

Round 5: I was the RECEIVER. Hana sent \$3.0, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

Now in Round 9: I'm the RECEIVER again. Hana sent \$3.0, it became \$9.0.

Hana has been consistent in sending \$3.0 to me. I've maintained a consistent strategy of returning 50% of what I receive. This has been working well for both parties.

I should continue with my cooperative strategy of returning 50% of the \$9.0 I received, which is \$4.5.

usage:

```
{
  "input_tokens": 1382,
  "output_tokens": 243,
  "reasoning_tokens": 127
}
```

Message 8 role=user

content:

Your name in this experiment is Galen.

Your opponent this round is Dorian (Agent_4).

Round 10

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5. Your total earnings across all rounds are \$52.0.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Dorian:

Round 4: Dorian sent me \$4.0 as the sender, showing strong trust and cooperation. I returned \$6.0 (50%).

Dorian has been cooperative, so I'll reciprocate with my standard cooperative amount:

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Let me check my history with Dorian:

Round 4: I was the receiver against Dorian. They sent \$4.0, it became \$12.0, and I returned \$6.0 (50%). My payoff was \$6.0.

Dorian has shown good cooperation by sending \$4.0. I should reciprocate with a similar level of trust. I'll send \$4.0, maintaining my consistent cooperative strategy.

Usage: {


```
"input_tokens": 1329,
"output_tokens": 188,
"reasoning_tokens": 86
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-06-04T16:01:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agent_names": {
      "Agent_4": "Dorian",
      "Agent_7": "Galen"
    },
  },
  "agents": [
    "Agent_7",
    "Agent_4"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_4": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Dorian.

Message 2 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Hana (Agent_8).

Round 7

In your most recent previous game round, you were the RECEIVER against Elara. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$36.0.

This round, you are the RECEIVER against Hana. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

- Hana sent \$3, which became \$9
- I've consistently returned \$4.5 when receiving \$9 (with both Cyra and Elara)
- This maintains proportional fairness

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- Consistent with established cooperative norms

{'return': 4.5}

reasoning:

Hana sent \$3, which became \$9 for me.

Looking at my history:

- When Elara sent \$3 (becoming \$9) in Round 6, I returned \$4.5
- When Cyra sent \$3 (becoming \$9) in Round 2, I returned \$4.5

I should maintain consistency and return the same proportional amount for the same send amount.

Returning \$4.5:

- Gives Hana \$6.5 total (\$2 kept + \$4.5 returned)
- I keep \$4.5
- This is consistent and maintains the cooperative norm

I'll return \$4.5.

usage:

```
{
  "input_tokens": 1403,
  "output_tokens": 274,
  "reasoning_tokens": 120
}
```

Message 4 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Briar (Agent_2).

Round 8

In your most recent previous game round, you were the RECEIVER against Hana. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$40.5.

This round, you are the SENDER against Briar. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my history with Briar:

- Round 5: I sent \$4, they returned \$6 (very fair cooperation)
- This established a strong cooperative relationship

Given:

- Briar has proven trustworthy
- Sending \$4 has consistently worked well
- Total earnings are solid at \$40.5

I'll send \$4 again to maintain our cooperative equilibrium.

{'send': 4}

reasoning:

Let me check my history with Briar:

- Round 5: I was the SENDER against Briar. I sent \$4.0, it became \$12.0, and they returned \$6.0. My payoff was \$7.0.

So Briar has shown to be cooperative, returning \$6 when receiving \$12 (which is exactly half, maintaining the fair cooperative norm).

Given that:

- Briar returned \$6 on \$12 last time (50% return rate, very fair)
- My strategy of sending \$4 has been consistently successful
- Total earnings are strong at \$40.5
- We have an established cooperative relationship

I should send \$4 again to Briar to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1386,
  "output_tokens": 274,
  "reasoning_tokens": 156
}
```

Message 6 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Aster (Agent_1).

Round 9

In your most recent previous game round, you were the SENDER against Briar. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$47.5.

This round, you are the SENDER against Aster. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this round:

- First time playing with Aster
- My earnings are solid at \$47.5
- Other agents have been generally cooperative

For a new opponent, I'll start with a moderately cooperative amount to establish trust while being somewhat cautious.

{'send': 3}

reasoning:

Let me review my history with Aster:

- Looking at the rounds provided, I don't see any previous interaction with Aster mentioned

Since I haven't played with Aster before, I need to decide on an initial strategy. Let me consider:

- My total earnings are \$47.5 which is quite good
- Most agents have been cooperative when I send \$3-4

- Briar has been very reliable with returning \$6 when I sent \$4
- When others sent me \$3, I've been returning \$4.5

For a first interaction with Aster, I should probably:

- Be moderately cooperative to establish trust
- Not be overly risky
- \$3-4 seems like a reasonable range

I'll send \$3 as a moderate cooperative opening move with a new partner.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 256,
  "reasoning_tokens": 181
}
```

Message 8 role=user

content:

Your name in this experiment is Dorian.
Your opponent this round is Galen (Agent_7).

Round 10

In your most recent previous game round, you were the SENDER against Aster. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$54.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- Galen sent \$4 (showing strong trust)
- I received \$12
- First time playing with Galen

Following my established pattern of proportional reciprocity:

- I typically return about 50% when receiving multiplied amounts
- Returning \$6 gives Galen \$7 total (\$1 kept + \$6 returned)
- I keep \$6
- This fairly rewards their trust

```
{'return': 6}
```

Reasoning:

Let me analyze this situation:

- Galen sent \$4, which became \$12 for me
- First time playing with Galen
- Galen showed significant trust by sending \$4

Looking at my pattern as a receiver:

- When I receive \$9 (from \$3 sent), I consistently return \$4.5 (50% of what I received)
- This gives the sender \$6.5 total and me \$4.5

For \$12 received (from \$4 sent):

- Returning 50% would be \$6
- This would give Galen: \$5 - \$4 + \$6 = \$7 total
- I would keep: \$12 - \$6 = \$6

This seems fair and maintains the proportional reciprocity I've been using. Galen took a bigger risk by sending \$4, so I should reciprocate generously.

Let me return \$6.

Usage: {

```
"input_tokens": 1349,
"output_tokens": 327,
"reasoning_tokens": 172
```

}

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-06-04T16:01:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
    "agents": [
      "Agent_1",
      "Agent_5"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

- ll have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Aster.

Message 2 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Finn (Agent_6).

Round 7

In your most recent previous game round, you were the RECEIVER against Briar. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Finn. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the consistent pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)

As RECEIVER:

- All my opponents who sent have received 50% back from me

This is a perfectly cooperative environment with 50% returns across the board. Playing with Finn for the first time, I'll continue with the \$4 strategy that has worked consistently.

If I send \$4:

- Finn receives \$12
- I keep \$1
- Expected return (50%): \$6
- Expected total: \$7

```
'''json
{"send": 4}
'''
```

reasoning:

Let me review the pattern:

As SENDER:

- Round 1: Sent \$3 to Cyra, got \$4.5 back (50% return)
- Round 4: Sent \$4 to Dorian, got \$6 back (50% return)
- Round 5: Sent \$4 to Elara, got \$6 back (50% return)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Every single interaction has resulted in 50% returns. This is a very strong cooperative pattern. Now I'm playing with Finn for the first time.

Given the consistent 50% returns and building trust, I should continue with \$4. This has been working perfectly.

If I send \$4:

- Finn receives \$12
- I keep \$1
- If they follow the 50% pattern and return \$6, I'd end with \$7

usage:

```
{
  "input_tokens": 1625,
  "output_tokens": 463,
  "reasoning_tokens": 196
}
```

Message 4 role=user

content:

Your name in this experiment is Aster.
Your opponent this round is Galen (Agent_7).

Round 8

In your most recent previous game round, you were the SENDER against Finn. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Galen. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at my history with Galen and the overall pattern:

- Round 2: Galen sent me \$4, I returned \$6 (50%)
- Now Round 8: Galen sends me \$4 again

The cooperative pattern has held perfectly throughout all rounds - everyone consistently returns 50% of what they receive. Galen trusted me again with \$4.

Galen sent \$4 (keeping \$1), which became \$12 for me.

Returning 50%:

- If I return \$6:
 - Galen ends with: $\$1 + \$6 = \$7$
 - I end with: $\$12 - \$6 = \$6$

Maintaining the trust and reciprocity:

```
'''json
{"return": 6}
'''
```

reasoning:

Looking at my history:

As SENDER:

- Round 1: Sent \$3 to Cyra ? got \$4.5 back (50%)
- Round 4: Sent \$4 to Dorian ? got \$6 back (50%)
- Round 5: Sent \$4 to Elara ? got \$6 back (50%)
- Round 7: Sent \$4 to Finn ? got \$6 back (50%)

As RECEIVER:

- Round 2: Galen sent \$4, I returned \$6 (50%)
- Round 3: Hana sent \$3, I returned \$4.5 (50%)
- Round 6: Briar sent \$4, I returned \$6 (50%)

Now in Round 8, I'm playing with Galen again. In Round 2, Galen sent me \$4 and I returned \$6 (50%). Now Galen is sending me \$4 again.

The cooperative pattern has been consistent throughout - everyone returns 50%. I should continue this pattern.

Galen sent \$4 (keeping \$1), which became \$12 for me.

If I return \$6:

- Galen ends with: $\$1 + \$6 = \$7$
- I end with: $\$12 - \$6 = \$6$

This maintains the established trust and cooperation.

usage:

```
{
  "input_tokens": 1642,
  "output_tokens": 506,
  "reasoning_tokens": 224
}
```

Message 6 role=user

content:

Your name in this experiment is Aster.
 Your opponent this round is Dorian (Agent_4).

Round 9

In your most recent previous game round, you were the RECEIVER against Galen. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0.
 Your total earnings across all rounds are \$50.0.

This round, you are the RECEIVER against Dorian. They sent you \$3.0, so you received \$9.0.

How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at my history with Dorian:

- Round 4: I sent Dorian \$4 ? they returned \$6 (50%)
- Now Round 9: Dorian sends me \$3

The 50% return pattern has been perfectly consistent across all rounds and all players. Dorian sent \$3 (keeping \$2), which became \$9 for me.

Following the established reciprocity:

- If I return \$4.5:
 - Dorian ends with: \$2 + \$4.5 = \$6.5
 - I end with: \$9 - \$4.5 = \$4.5

```
'''json
{"return": 4.5}
'''
```

reasoning:

Looking at my history with Dorian:

- Round 4: I was SENDER against Dorian. I sent \$4, it became \$12, they returned \$6 (50%)
- Now Round 9: I'm RECEIVER against Dorian. They sent \$3, became \$9 for me

The consistent pattern has been 50% returns throughout. Dorian sent \$3 (keeping \$2), which became \$9 for me.

Following the 50% pattern:

- If I return \$4.5:
 - Dorian ends with: \$2 + \$4.5 = \$6.5
 - I end with: \$9 - \$4.5 = \$4.5

This maintains the cooperative equilibrium that's been working.

usage:

```
{
  "input_tokens": 1626,
  "output_tokens": 346,
  "reasoning_tokens": 132
}
```

Message 8 role=user**content:**

Your name in this experiment is Aster.
 Your opponent this round is Elara (Agent_5).

Round 10

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
 Your total earnings across all rounds are \$54.5.

This round, you are the SENDER against Elara. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my history with Elara:

- Round 5: I sent Elara \$4 ? they returned \$6 (50%)

The cooperative 50% return pattern has held perfectly across all rounds and all players. Sending \$4 has consistently yielded the best outcome:

- I keep \$1

- Elara receives \$12
- Expected return: \$6
- Expected total: \$7

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at my history with Elara:

- Round 5: I sent Elara \$4 ? they returned \$6 (50%)

The pattern has been very consistent - when I send \$4, I get back \$6, resulting in a total of \$7 (starting \$5 - \$4 sent + \$6 returned).

When I've sent \$3, I've gotten back \$4.5, for a total of \$6.5.

The 50% return rate has been perfectly consistent throughout all rounds with all players.

If I send \$4:

- Elara receives \$12
- I keep \$1
- Expected return (50%): \$6
- Expected total: \$7

This has worked well in the past and maintains the cooperative equilibrium.

```
Usage: {
  "input_tokens": 1602,
  "output_tokens": 287,
  "reasoning_tokens": 150
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-06-04T16:01:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_5": "Elara"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_5"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Elara.

Message 2 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Galen (Agent_7).

Round 7

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0 , it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5. Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my track record:

- As sender, I've consistently sent \$3 (60%)
- This has reliably resulted in \$4.5 returns
- Payoffs of \$6.5 are consistently better than the \$5 I'd keep by sending nothing
- My cooperative strategy has been successful across different opponents

I'll continue this approach with Galen.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my history:

- My strategy as sender has been consistent: send \$3 (60%)
- This has consistently resulted in receiving \$4.5 back
- My payoffs as sender have been \$6.5 each time
- This is better than keeping all \$5

Galen is a new opponent. I should continue my established cooperative strategy that has be

en working well. Sending \$3 has shown to be mutually beneficial and builds trust.

I'll send \$3 again.

usage:

```
{
  "input_tokens": 1363,
  "output_tokens": 215,
  "reasoning_tokens": 111
}
```

Message 4 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Finn (Agent_6).

Round 8

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$41.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this round:
- Finn sent me \$3 (60% of their endowment)
- I received \$9
- In Round 5, when Finn sent me \$3, I returned \$4.5 (50%)
- This established a fair reciprocal relationship with Finn

I'll maintain consistency and return the same amount.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Let me review my history with Finn:

- Round 5: I was RECEIVER against Finn. They sent \$3.0, it became \$9.0, and I returned \$4.5. My payoff was \$4.5.

Now in Round 8, I'm RECEIVER against Finn again. They sent \$3.0 (same as before), which became \$9.0.

My consistent strategy as receiver has been:

- Return about 50% of what I receive
- For \$9 received ? return \$4.5
- For \$12 received ? return \$6

This has been working well and is fair/consistent. Finn sent the same amount as last time (\$3), so I should return the same amount as last time (\$4.5).

This maintains my consistent and fair approach.

usage:

```
{
  "input_tokens": 1376,
  "output_tokens": 284,
  "reasoning_tokens": 160
}
```

Message 6 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Cyra (Agent_3).

Round 9

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$45.5.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my consistent pattern:

- As sender, I've sent \$3 in every round
- This cooperative approach has yielded reliable returns of \$4.5
- My payoffs of \$6.5 consistently exceed the \$5 I'd keep by not cooperating
- This strategy has worked well across all opponents

I'll continue with Cyra.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- As sender, I've consistently sent \$3 (60% of endowment)
- This has worked well across multiple opponents (Dorian, Galen, and in earlier rounds)
- I've consistently received \$4.5 back, giving me payoffs of \$6.5
- This is a reliable cooperative strategy

Now I'm facing Cyra. I should continue with my established strategy of sending \$3.

Total earnings: \$45.5
This is my 9th round.

I'll stick with my cooperative strategy of sending \$3.

usage:

```
{
  "input_tokens": 1338,
  "output_tokens": 230,
  "reasoning_tokens": 124
}
```

Message 8 role=user

content:

Your name in this experiment is Elara.
Your opponent this round is Aster (Agent_1).

Round 10

In your most recent previous game round, you were the SENDER against Cyra. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$52.0.

This round, you are the RECEIVER against Aster. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Aster sent me \$4 (80% of their endowment) - more generous than typical
- I received \$12
- My consistent approach has been to return 50% of what I receive
- When others sent \$3, I returned \$4.5 (50% of \$9)

Aster showed greater trust, so I'll maintain my fair reciprocation rate:

```
'''json
{"return": 6}
'''
```

Reasoning:

Let me analyze this situation:

- Aster sent me \$4 (80% of their endowment)
- I received \$12
- This is more generous than most others who have sent me \$3

Looking at my previous behavior as receiver:

- When I receive \$9 (from \$3 sent), I return \$4.5 (50% of what I receive)
- This means senders get back \$4.5, which is more than the \$3 they sent
- This creates a win-win situation

Now Aster sent \$4 and I received \$12. If I apply the same 50% return rate:

- I would return \$6
- Aster would end up with: $\$5 - \$4 + \$6 = \7
- I would keep: $\$12 - \$6 = \$6$

This seems fair and reciprocates Aster's generosity. They sent 80% (\$4 out of \$5), showing high trust, so I should return proportionally to maintain that trust relationship.

Returning \$6 (50% of \$12) rewards their generosity while still giving me a good payoff.

```
Usage: {  
  "input_tokens": 1354,  
  "output_tokens": 375,  
  "reasoning_tokens": 217  
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-06-04T16:01:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agent_names": {  
      "Agent_2": "Briar",  
      "Agent_8": "Hana"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_8"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Briar.

Message 2 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Cyra (Agent_3).

Round 7

In your most recent previous game round, you were the SENDER against Aster. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0.
Your total earnings across all rounds are \$37.0.

This round, you are the SENDER against Cyra. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

The pattern of cooperation continues to work well:

- Sending \$4 has consistently resulted in \$6 returns
- This creates good payoffs for both parties
- Trust is well-established in this group

I'll send \$4 to Cyra:

```
'''json
{"send": 4}
'''
```

reasoning:

The pattern continues to hold strong:

- I've been sending \$4 consistently as sender
- When I send \$4, I receive back \$6
- When others send me \$4, I return \$6
- This creates a stable, mutually beneficial equilibrium

My payoffs:

- As sender sending \$4: $\$5 - \$4 + \$6 = \7
- As receiver getting \$4 (which becomes \$12): return \$6, keep \$6

The cooperation is working well. I should continue sending \$4 to Cyra.

usage:

```
{
  "input_tokens": 1357,
  "output_tokens": 203,
  "reasoning_tokens": 109
}
```

Message 4 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Dorian (Agent_4).

Round 8

In your most recent previous game round, you were the SENDER against Cyra. You sent \$4.0, it became \$12.0 for them, and they returned \$6.0 to you. Your payoff was \$7.0. Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER against Dorian. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Dorian sent \$4.0 again, showing continued trust. In Round 5, they also sent \$4 and I returned \$6, which worked well for both of us.

I'll maintain the same cooperative pattern:

```
'''json
{"return": 6}
'''
```

reasoning:

Dorian sent me \$4.0 again, just like in Round 5. I returned \$6.0 to them then, which worked well. The cooperative pattern of sending \$4 and returning \$6 has been consistent across all interactions.

I should return \$6 again to maintain this successful cooperative relationship:

- Dorian gets: $\$5 - \$4 + \$6 = \7
- I get: $\$12 - \$6 = \$6$

This is fair and continues the established pattern.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 187,
  "reasoning_tokens": 101
}
```

Message 6 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Finn (Agent_6).

Round 9

In your most recent previous game round, you were the RECEIVER against Dorian. They sent \$4.0 to you, it became \$12.0, and you returned \$6.0. Your payoff was \$6.0. Your total earnings across all rounds are \$50.0.

This round, you are the RECEIVER against Finn. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Finn sent \$3.0, which is slightly less than the typical \$4 I've been seeing. I want to be fair and proportional.

When \$4 is sent, I return \$6 (50% of the multiplied amount).
Following the same proportion: \$3 sent ? \$9 received ? return \$4.5 (50%).

This gives Finn a payoff of \$6.5 and me \$4.5.

```
'''json
{"return": 4.5}
'''
```

reasoning:

Finn sent \$3.0, which became \$9.0 for me. This is slightly less than the usual \$4 I've been seeing.

Looking at the pattern:

- When others send \$4, I return \$6, giving them a payoff of \$7 ($5-4+6$) and me \$6 ($12-6$)
- Finn sent \$3, so they have \$2 already ($5-3$)
- I received \$9

To maintain a similar fair split:

- If I return \$4.5, Finn gets \$6.5 (2+4.5) and I get \$4.5 (9-4.5)
- If I return \$5, Finn gets \$7 (2+5) and I get \$4 (9-5)

The \$4/\$6 return pattern when \$4 is sent creates:

- Sender: \$7 total
- Receiver: \$6 total

If I scale proportionally for \$3 sent:

- \$3 sent ? \$9 received
- Returning \$4.5 would give: Finn \$6.5, me \$4.5
- This seems fair and proportional

I'll return \$4.5 to maintain fairness.

usage:

```
{
  "input_tokens": 1281,
  "output_tokens": 411,
  "reasoning_tokens": 193
}
```

Message 8 role=user

content:

Your name in this experiment is Briar.
Your opponent this round is Hana (Agent_8).

Round 10

In your most recent previous game round, you were the RECEIVER against Finn. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$54.5.

This round, you are the SENDER against Hana. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the established pattern:

- Sending \$4 has consistently worked well
- I typically receive \$6 back, resulting in a \$7 payoff
- The cooperative norm of 50% returns is well-established

I'll send \$4 to Hana:

```
'''json
{"send": 4}
'''
```

Reasoning:

I'm now the sender against Hana in Round 10. Let me review what I know:

- My total earnings are \$54.5
- I've been consistently sending \$4 as sender and it's been working well
- When I send \$4, I typically get back \$6, giving me a payoff of \$7
- The cooperative pattern with returning 50% has been established

I haven't played with Hana yet this run based on the information given, but the general cooperative pattern in the group seems strong. I'll continue with sending \$4.

```
'''json
{"send": 4}
'''
```

Usage: {

```
"input_tokens": 1318,
"output_tokens": 218,
"reasoning_tokens": 134
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-06-04T16:02:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_4",
```



```

"move_index": 7,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_2": "Briar",
    "Agent_8": "Hana"
  },
  "agents": [
    "Agent_2",
    "Agent_8"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- You will be told your opponent's name in each round.
- Your name in this experiment is Hana.

Message 2 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Dorian (Agent_4).

Round 7

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$34.5.

This round, you are the SENDER against Dorian. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at my consistent strategy:

- Sending \$3 has yielded reliable \$4.5 returns
- This gives me \$6.5 per round as sender
- The cooperative approach continues to work well
- Total earnings at \$34.5

I'll stick with ****\$3**** to Dorian.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review my strategy:

- I've been consistently sending \$3 as sender
- I've been consistently returning 50% (half of what I receive) as receiver
- This has resulted in steady earnings of \$6.5 per round when I'm sender (when returns are \$4.5)
- Total earnings are \$34.5 across 7 rounds

My approach has been:

- Send \$3 ? they get \$9 ? they return \$4.5 ? I get \$6.5 total
- When receiving \$3 ? I get \$9 ? I return \$4.5 ? I keep \$4.5
- When receiving \$4 ? I get \$12 ? I return \$6 ? I keep \$6

The cooperative strategy is working well. I'll continue sending \$3 to Dorian.

usage:

```
{
  "input_tokens": 1421,
  "output_tokens": 280,
  "reasoning_tokens": 158
}
```

Message 4 role=user**content:**

Your name in this experiment is Hana.
Your opponent this round is Cyra (Agent_3).

Round 8

In your most recent previous game round, you were the SENDER against Dorian. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$41.0.

This round, you are the RECEIVER against Cyra. They sent you \$3.0, so you received \$9.0. How much do you return? (0-9.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at the situation:

- Cyra sent me \$3 again, showing consistent cooperation
- I received \$9 after the 3x multiplier
- I returned \$4.5 last time we played (Round 6), which was fair

I'll continue with the same fair approach and return ****\$4.5**** (50% of what I received).

This gives:

- Cyra: \$2 (kept) + \$4.5 (returned) = \$6.5 total

- Me: \$9 - \$4.5 = \$4.5 total

```
'''json
{"return": 4.5}
'''
```

reasoning:

Cyra sent me \$3 again, which became \$9.0. I need to return some amount.

Looking at the pattern:

- Cyra has sent me \$3 twice now (rounds 6 and 8)
- In round 6, I returned \$4.5 (50% of \$9)
- This is consistent with my cooperative strategy

I should continue being fair and return \$4.5 again (50% of what I received).

This gives:

- Cyra: \$2 (kept from \$5) + \$4.5 (returned) = \$6.5 total
- Me: \$9 - \$4.5 = \$4.5 total

This maintains the cooperative, fair pattern that's been working well.

usage:

```
{
  "input_tokens": 1373,
  "output_tokens": 330,
  "reasoning_tokens": 132
}
```

Message 6 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Galen (Agent_7).

Round 9

In your most recent previous game round, you were the RECEIVER against Cyra. They sent \$3.0 to you, it became \$9.0, and you returned \$4.5. Your payoff was \$4.5.
Your total earnings across all rounds are \$45.5.

This round, you are the SENDER against Galen. You have \$5. How much do you send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my previous interaction with Galen:
- Round 6: I sent \$3, they returned \$4.5 (50% - fair and cooperative)
- This resulted in a \$6.5 payoff for me

Galen has shown trustworthy behavior. I'll continue with my consistent strategy of sending
\$3.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at my history:

- Round 6: I was SENDER against Galen. I sent \$3.0, they returned \$4.5, my payoff was \$6.5
- Now Round 9: I'm SENDER against Galen again

Galen returned \$4.5 when I sent \$3 before, which is the standard 50% return I've been seeing. This is cooperative behavior.

I should continue with my consistent strategy of sending \$3.

Total earnings: \$45.5 is good progress.

I'll send \$3 to Galen.

usage:

```
{
  "input_tokens": 1443,
  "output_tokens": 230,
  "reasoning_tokens": 111
}
```

Message 8 role=user

content:

Your name in this experiment is Hana.
Your opponent this round is Briar (Agent_2).

Round 10

In your most recent previous game round, you were the SENDER against Galen. You sent \$3.0, it became \$9.0 for them, and they returned \$4.5 to you. Your payoff was \$6.5.
Your total earnings across all rounds are \$52.0.

This round, you are the RECEIVER against Briar. They sent you \$4.0, so you received \$12.0.
How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this round:

- Briar sent me \$4 (more generous than the typical \$3)
- I received \$12 after the 3x multiplier
- Following my fair strategy of returning 50%

I'll return **\$6** (50% of what I received).

This gives:

- Briar: \$1 (kept) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

```
'''json
{"return": 6}
'''
```

Reasoning:

Looking at this situation:

- Briar sent me \$4 (more than the typical \$3 I've been seeing)
- I received \$12 after the 3x multiplier
- This is my first interaction with Briar mentioned in the rounds

Briar is being generous by sending \$4 instead of the typical \$3. I should reciprocate fairly.

My typical strategy has been to return 50% of what I receive, which has worked well:

- 50% of \$12 = \$6

This would give:

- Briar: \$1 (kept from \$5) + \$6 (returned) = \$7 total
- Me: \$12 - \$6 = \$6 total

This is fair and maintains the cooperative dynamic. Briar sent more, showing trust, so I should return proportionally.

```
Usage: {
  "input_tokens": 1392,
  "output_tokens": 322,
  "reasoning_tokens": 164
}
```